

# ClaudeForth

*An ANS Forth System for the Motorola 6809*

*Target: the Minimalist Eurocard Board (MECB), 6809 version, configured with 16kB of ROM.*

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# Table of Contents

|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>1. Introduction.....</b>                            | <b>6</b>  |
| 1.1 Overview.....                                      | 6         |
| 1.2 Inner-Interpreter Model: Subroutine Threading..... | 6         |
| 1.3 Register Assignment.....                           | 6         |
| 1.4 Cell Size and Character Set.....                   | 6         |
| 1.5 The Three-Region Memory Split.....                 | 6         |
| 1.6 Exception Handling.....                            | 7         |
| 1.7 What This Document Covers.....                     | 7         |
| <b>2. Memory Map.....</b>                              | <b>8</b>  |
| <b>3. Glossary.....</b>                                | <b>11</b> |
| 3.1 System / Console I/O.....                          | 11        |
| 3.2 Stack Manipulation — Data Stack.....               | 12        |
| 3.3 Stack Manipulation — Return Stack.....             | 13        |
| 3.4 Arithmetic — Single Cell.....                      | 14        |
| 3.5 Arithmetic — Mixed & Double Precision.....         | 16        |
| 3.6 Logic, Shifts & Address Arithmetic.....            | 17        |
| 3.7 Comparison.....                                    | 18        |
| 3.8 Control Flow — Conditionals & Loops.....           | 19        |
| 3.9 Defining Words.....                                | 21        |
| 3.10 Compiling Words.....                              | 23        |
| 3.11 Memory.....                                       | 24        |
| 3.12 Strings & Parsing.....                            | 27        |
| 3.13 Numeric Output.....                               | 29        |
| 3.14 Base / Radix Control.....                         | 30        |
| 3.15 Exception Handling.....                           | 30        |
| 3.16 Comments.....                                     | 31        |
| 3.17 Environmental & System Queries.....               | 31        |
| 3.18 Tools.....                                        | 32        |
| <b>4. Forth Development Notes.....</b>                 | <b>33</b> |
| 4.1 Serial I/O and Handshaking.....                    | 33        |
| 4.2 Forth Syntax and Stack Operations.....             | 35        |
| 4.3 Tools Word Set: Explanation and Examples.....      | 36        |
| 4.4 Dictionary Management with MARKER.....             | 37        |
| 4.5 The Search-Order Word Set Is Not Included.....     | 38        |
| 4.6 Exception Handling.....                            | 38        |
| <b>5. Open Items Checklist.....</b>                    | <b>40</b> |

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| 5.1 Known bugs — resolved since the original checklist.....                  | 40        |
| 5.2 Structural duplication (identified, some resolved, some not).....        | 50        |
| 5.3 Real, load-bearing gaps.....                                             | 51        |
| 5.4 Never resolved after being explicitly raised.....                        | 52        |
| 5.5 Open design questions (not defects — deliberate unresolved choices)..... | 53        |
| 5.6 What's solid (for reference, not action).....                            | 53        |
| <b>6. Build Instructions.....</b>                                            | <b>54</b> |
| <b>7. Architecture.....</b>                                                  | <b>56</b> |
| 7.1 (Outer) Interpreter.....                                                 | 56        |
| 7.2 Dictionary Operations.....                                               | 57        |
| <b>8. Assembler Source.....</b>                                              | <b>57</b> |
| 8.1 Memory Map, Constants & GLOBALS Layout.....                              | 58        |
| 8.2 Hardware Vector Table.....                                               | 67        |
| 8.3 Init Code (COLDSTRT / WARM).....                                         | 67        |
| 8.4 ACIA Interrupt Handler.....                                              | 68        |
| 8.5 COLD / ABORT / QUIT.....                                                 | 70        |
| 8.6 Inner-Interpreter Support.....                                           | 72        |
| 8.7 Comma Family.....                                                        | 73        |
| 8.8 HEADER.....                                                              | 74        |
| 8.9 Defining Words.....                                                      | 75        |
| 8.10 Outer Interpreter.....                                                  | 80        |
| 8.11 QUERY / ACCEPT / EXPECT / KEY / KEY? / EMIT.....                        | 87        |
| 8.12 CATCH / THROW.....                                                      | 89        |
| 8.13 Control Flow.....                                                       | 90        |
| 8.14 Compiling Words.....                                                    | 98        |
| 8.15 Stack Manipulation.....                                                 | 101       |
| 8.16 Arithmetic.....                                                         | 104       |
| 8.17 Logic, Shifts & Address Arithmetic.....                                 | 113       |
| 8.18 Comparison.....                                                         | 114       |
| 8.19 Memory.....                                                             | 117       |
| 8.20 String Words.....                                                       | 119       |
| 8.21 Numeric Output.....                                                     | 129       |
| 8.22 Base / Radix Control.....                                               | 134       |
| 8.23 Output Formatting.....                                                  | 134       |
| 8.24 Comment Words.....                                                      | 135       |
| 8.25 Environmental Query / SOURCE / REFILL / EVALUATE.....                   | 135       |
| 8.26 Tools Word Set.....                                                     | 140       |
| 8.27 ABORT / QUIT Hand-Built Headers.....                                    | 142       |

|                                                          |            |
|----------------------------------------------------------|------------|
| 8.28 Forth Dictionary (ROM Base Dictionary Headers)..... | 143        |
| <b>9. Alphabetical Index.....</b>                        | <b>160</b> |

# 1. Introduction

## 1.1 Overview

ClaudeForth is a 16-bit ANS Forth implementation targeting the Motorola 6809 microprocessor. It was designed and built incrementally, decision by decision, across a single extended conversation: the inner-interpreter model was chosen first, then the cell size, character set, memory map, and every word in the Core and Core Extension word sets, followed by double-precision arithmetic, string handling, exception handling, and the Tools word set. This document describes the resulting architecture, the reasoning behind its major design choices, and provides a complete glossary of every word implemented.

## 1.2 Inner-Interpreter Model: Subroutine Threading

ClaudeForth uses subroutine-threaded code (STC) rather than indirect or direct threading. A colon definition compiles as a flat sequence of 6809 JSR instructions terminated by a compiled RTS. This means the 6809's own JSR/RTS mechanism is the inner interpreter — there is no separate NEXT dispatch loop for straight-line execution. This was chosen because the 6809's hardware stack (register S) is naturally suited to serving as both the CPU's call stack and Forth's return stack simultaneously, and because STC is generally the fastest threading model available on this processor, at the cost of slightly larger compiled code than indirect threading would produce (each call is 3 bytes — a JSR opcode plus a 16-bit address — rather than a 2-byte threaded pointer).

## 1.3 Register Assignment

U is the data (parameter) stack pointer, manipulated with PSHU/PULU. S is the return stack pointer, using the CPU's native hardware stack directly, so JSR/RTS work as the inner interpreter with no extra bookkeeping. X and Y are general-purpose working registers, used throughout for scanning, indexing, and address computation. D (the combined A:B accumulator) holds top-of-stack values in flight and is the primary vehicle for 16-bit arithmetic, matching the cell size described below.

## 1.4 Cell Size and Character Set

One cell is 16 bits (two bytes), matching the 6809's native D, X, Y, U, and S registers exactly. This gives a single-cell signed range of -32768 to 32767, unsigned 0 to 65535, and means a cell can always hold a full address across the 6809's entire 64K address space. Because the 6809 has no alignment restriction, ALIGN and ALIGNED are true no-ops. Characters are single-byte 7-bit ASCII (values 0-127), so CHARS and CHAR+ reduce to a no-op and a plain +1 respectively, while CELLS requires an actual multiply-by-two.

## 1.5 The Three-Region Memory Split

The single most distinctive architectural decision in ClaudeForth is that dictionary headers, compiled code, and user data are tracked as three independent, separately growing regions, each with its own allocation pointer (DPHERE, CODEHERE, and VARHERE respectively) rather than the single unified dictionary most Forth systems use. This departs from the ANS assumption of one HERE; in this system HERE reports CODEHERE specifically, and a parallel word family (V, / VC, / VALLOT / VHERE) exists to manage the variable region directly. The practical consequence is a genuine distinction between immutable data (CONSTANT values, compiled literals, and string literals, all living in code space) and mutable data (VARIABLE, VALUE, and BUFFER: contents, living in variable space) that most single-region Forths do not enforce architecturally.

The intent behind this split is to provide for a ROMable basic Forth system: BASECODE and BASEDICT (Section 2) hold the entire Core and Core Extension implementation as fixed, burned-in ROM content, needing no RAM at all to boot to a working interactive prompt. The RAM regions — APPCODE, APPDICT, and APPVARS — exist specifically to support interactive user application development on top of that fixed base: new words are compiled, tested, and revised freely in RAM without touching the ROM image underneath them. Once an application is complete, the same compiled RAM content (APPCODE and APPDICT together) is intended to be burned into an expanded ROM alongside the base system, turning a finished, debugged application into a second fixed, non-volatile layer rather than something that must be re-compiled from source into RAM on every power-up.

A feature under consideration, not yet designed or implemented, is the ability to strip the dictionary — discarding header (name, LINK, flags) information for some or all ROM or application words once development is finished, while leaving their code intact and callable. A stripped system would trade FIND-ability and interactive extensibility (WORDS, ', and ordinary text interpretation of a word by name) for a smaller ROM footprint and a correspondingly larger RAM variable area, since the dictionary and variable regions currently share the same fixed page-zero and BASEDICT budgets described in Section 2.

## 1.6 Exception Handling

CATCH and THROW are implemented as direct manipulation of the return stack (register S) rather than a linked list of handler records: CATCH saves the current data-stack pointer and the previous HANDLER value as a small frame on S, and a non-local THROW unwinds directly to that frame with a single TFR instruction, discarding every intervening call level in one step. QUIT wraps INTERPRET in a CATCH and, on a caught error, rolls back all three region pointers (and LATEST) to their state at the start of the failing input line, reclaiming space that an uncaught error would otherwise abandon.

## 1.7 What This Document Covers

Section 2 gives the full memory map. Section 3 is the glossary of every word implemented, organized by topic, each with its stack-effect notation and any side effects. Section 4 covers five practical development topics: serial I/O configuration and handshaking, a brief introduction to Forth syntax and stack operations, worked examples of the Tools word set, dictionary management with MARKER, and why the Search-Order word set is not included. Section 5 is a checklist of every item still open, resolved, or deliberately deferred during the build. Section 6 covers build instructions: which files to assemble and the assembler this source targets. Section 8 is the complete assembler source listing, including the ROM base dictionary (Section 8.28). Section 9 is an alphabetical index of every word with a page reference back into the glossary.

## 2. Memory Map

64K address space, addresses shown high to low. Three independent allocation pointers (DPHERE, CODEHERE, VARHERE) track the dictionary, code, and mutable-variable regions respectively; none of the three boundaries below is enforced by a runtime bounds check.

| Address Range | Size/bytes | Region              | Contents                                                                                                                                                     |
|---------------|------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$FFF0-\$FFFF | 16         | VECTORS             | Hardware vectors                                                                                                                                             |
| \$FFA9-\$FFEF | 71         | INITCODE            | COLDSTRT / WARM entry code                                                                                                                                   |
| \$DEEA-\$FF59 | 8304       | BASECODE            | ROM primitives and compiled system words — real size, confirmed by a real assembler run (BASECODESIZE); exactly contiguous with BASEDICT below, zero gap     |
| \$D6FF-\$DEE9 | 2027       | BASEDICT            | ROM dictionary headers — real size, confirmed by a real assembler run (BASEDICTSIZE)                                                                         |
| \$C000-\$COFF | 256        | INOUT               | Input/output device block, sits directly below USROMSTRT; 6850 ACIA registers live at INOUT+8, not the block base                                            |
| \$BD00-\$BFFF | 768        | RSTACK              | Return stack (S), grows downward — exactly contiguous with DSTACK below, no gap                                                                              |
| \$B900-\$BCFF | 1024       | DSTACK              | Data stack (U), grows downward — occupied bottom matches CODETOP exactly                                                                                     |
| \$7000-\$B8FF | 18688      | APPCODE + Transient | Application code (CODEHERE). Also hosts a floating Transient subregion between CODEHERE and CODEHERE+84, used by WORD/HOLD/PAD — see Transient Region, below |
| \$2000-\$6FFF | 20480      | APPDICT             | Application dictionary (DPHERE)                                                                                                                              |
| \$021B-\$1FFF | 7653       | APPVARS             | Mutable variable space (VARHERE) — APPVARSEND self-adjusts to APPDICT-1                                                                                      |
| \$01DA-\$021A | 65         | (unclaimed)         | Freed by the WORD/PAD redesign (formerly the fixed-size SIBUF/WORDBUF buffers, since retired); not reclaimed for any other purpose                           |
| \$018A-\$01D9 | 80         | TIBBUF              | Terminal input buffer (80 bytes)                                                                                                                             |
| \$014A-\$0189 | 64         | OUTBUF              | Output ring buffer (64 bytes)                                                                                                                                |
| \$010A-\$0149 | 64         | INBUF               | Input ring buffer (64 bytes)                                                                                                                                 |
| \$0106-\$0109 | 4          | SERBUF idx          | Serial ring buffer head/tail indices                                                                                                                         |
| \$0100-\$0105 | 6          | MVSCRATCH           | Shared scratch: MVCNT/MVDST/MVSR, aliased by FILLCNT/HSLEN, FILLADDR/HSADDR, MRESULT/FILLCHR                                                                 |
| \$0000-\$00FF | 256        | GLOBALS             | Shared system globals — Direct Page = \$00                                                                                                                   |

### ROM Size Required

VECTORS, INITCODE, BASECODE, and BASEDICT are the four regions that must actually live in ROM; everything else in the table is RAM or the ACIA hardware itself, memory-mapped rather than stored.

USROMSTRT (\$C100) and USROMEND (\$FFEF, one address before VECTORS' start) define the outer boundary these four must fall within, and all four are genuinely contained within it. A real assembler run (see forth6809.lst.txt) confirms the boundaries between them precisely: BASEDICT's real end lands exactly at BASECODE's start (BASEDICTEND = BASECODE = \$DEEA) — zero gap, zero overlap. BASECODE's real end (\$FF5A) sits 79 bytes below INITCODE's start (\$FFA9); rather than an unaccounted gap, this is explicitly padded with \$FF bytes by the assembler (see BASEND's FILL directive) specifically to stop it from being silently absorbed into the .bin output. INITCODE's real end lands exactly at VECTORS' start (INITEND = VECTORS = \$FFF0) — zero gap, zero overlap.

Using the real sizes confirmed by that assembler run (BASECODESIZE = 8304 bytes, BASEDICTSIZE = 2027 bytes — both measured directly, not estimated), VECTORS + INITCODE + BASECODE + BASEDICT comes to 10418 bytes (16 + 71 + 8304 + 2027, about 10.2 KB), which is why the ROM part is a 16K×8 EPROM (a 27128 or equivalent) rather than an 8K×8 part — comfortable headroom for real usage.

The INOUT block shadows 256 bytes in the ROM block: a 16K×8 EPROM occupying the natural \$C000–\$FFFF footprint that size implies would include \$C000–\$C0FF, but INOUT claims that same 256-byte range for the ACIA and other memory-mapped I/O. This is why USROMSTRT starts at \$C100, not \$C000 — the bottom 256 bytes of the chip's raw address footprint are never available for ROM content at all, regardless of how much of the chip is otherwise unused.

ROM unused byte count: a 16K×8 EPROM (16384 bytes) minus INOUT's 256-byte shadow minus the 10418 bytes of real VECTORS + INITCODE + BASECODE + BASEDICT content leaves 5710 bytes genuinely unused — about 35% of the chip's total capacity still free. Unlike the earlier estimate this replaces, every figure here is a real assembler-measured value, not a manual count.

### Transient Region

A floating scratch region within APPCODE, positioned relative to CODEHERE rather than at a fixed address — its actual location moves as compilation proceeds, since every new colon definition or interactively parsed word advances CODEHERE. It serves two purposes: WORD's own text-scanning buffer for the word currently being parsed, and the pictured numeric output buffer built by <# #S HOLD HOLDS SIGN #>. Reused fresh on every WORD call and every <# ... #> conversion, since neither needs to persist past its own use.

Occupies PADOFFSET (84 bytes, PADMINSIZE) between CODEHERE and PAD: PAD = CODEHERE + 84 (see PADW). That 84-byte gap is shared by two independent, opposite-growing buffers — WORD's own buffer growing upward from CODEHERE toward PAD, and HOLD's buffer growing downward from PAD toward CODEHERE — sized so the two can never collide even in the worst case. PAD itself then extends a further PADMINSIZE (84) bytes upward from its own address, per ANS's own /PAD requirement (3.3.3.6) — using the same constant for both roles was a deliberate design choice, not a coincidence.

| Subregion              | Size                                                                               | Address<br>(relative to<br>CODEHERE)                                                                                                 | Description                                                                                                                                                                                                                                                               |
|------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WORD parsing<br>buffer | up to 47 bytes (1<br>count byte + up<br>to<br>WORDMAXCHAR<br>S = 46<br>characters) | CODEHERE, or<br>CODEHERE+3 for<br>S"/."/ABORT"<br>(which reserve 3<br>bytes ahead of<br>CODEHERE first)<br>— extending to<br>at most | WORD's own scratch buffer for the word or<br>string currently being parsed; grows upward<br>from CODEHERE toward PAD, the traditional<br>fig-Forth layout. Capped at WORDMAXCHARS<br>specifically so it can never reach the HOLD<br>buffer below, even in the worst case. |

|                                                |                                                                                                                                 |                                                      |                                                                                                                                                                                                                                                                             |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                |                                                                                                                                 | CODEHERE+49                                          |                                                                                                                                                                                                                                                                             |
| HOLD /<br>pictured<br>numeric<br>output buffer | 34 bytes<br>(HOLDMINSIZE)                                                                                                       | CODEHERE+50<br>to<br>CODEHERE+83 (= PAD-34 to PAD-1) | Built by <# #S HOLD HOLDS SIGN, one character at a time, growing downward from PAD toward CODEHERE; #> finalizes it, and <# restarts the buffer at PAD again next time.                                                                                                     |
| PAD                                            | at least<br>PADMINSIZE (84)<br>bytes, extending<br>upward;<br>effectively<br>unbounded until<br>the next<br>CODEHERE<br>advance | CODEHERE+84<br>onward                                | General-purpose transient scratch area returned by PAD; also used by interpreted-mode S" for its own string storage. ANS (3.3.3.6) requires only 84 characters of guaranteed capacity — the same PADMINSIZE constant used for the CODEHERE-to-PAD offset itself, by design. |

### Dictionary Entry Layout

Every word in BASEDICT (and every application word compiled into APPDICT) shares this same four-field layout, packed with no padding, in this order:

| Code          | Size    | Name               | Comments                                                                              |
|---------------|---------|--------------------|---------------------------------------------------------------------------------------|
| <b>LEN/FL</b> | 1 byte  | Length + Flags     | bit 7 = IMMEDIATE, bit 6 = SMUDGE, bits 4-0 = name length (0-31)                      |
| <b>NAME</b>   | n bytes | Name Field         | n raw ASCII characters, no terminator - n comes from LEN/FL's low bits                |
| <b>LINK</b>   | 2 bytes | Link Field         | address of the previous header in the chain, or 0 if this is the oldest entry         |
| <b>CFA</b>    | 2 bytes | Code Field Address | address FIND returns - the word's own code, or a DODOES trampoline for a CREATED word |

## 3. Glossary

Words are grouped by topic below. Each entry gives the stack-effect notation, a short description, and any side effects beyond the stack transformation shown.

### 3.1 System / Console I/O

#### KEY

( -- char )

Blocking read of one character from the input ring buffer.

*Side effects: Spins with interrupts enabled until a character is available.*

#### KEY?

( -- flag )

Non-blocking test for a waiting input character.

*Side effects: Read-only; never blocks.*

#### EMIT

( char -- )

Queue one character for transmission.

*Side effects: Enables the ACIA transmit interrupt if not already on.*

#### ACCEPT

( addr n -- n' )

Line-edit up to n characters into addr; returns actual count.

*Side effects: Echoes input; handles backspace/delete.*

#### EXPECT

( addr n -- )

Obsolescent line input; same editing as ACCEPT.

*Side effects: Reports its count through SPAN rather than the stack.*

#### QUERY

( -- )

Refill the terminal input buffer and reset >IN.

*Side effects: Sets SRCADDR/SRCLEN/SRCID for the terminal source.*

#### TYPE

( addr len -- )

Output len characters starting at addr.

*Side effects: None.*

#### CR

( -- )

Output a newline (CR then LF).

*Side effects: None.*

#### SPACE

( -- )

Output one space character.

*Side effects: None.*

#### SPACES

( n -- )

Output n spaces.

*Side effects:* No output if  $n \leq 0$ .

## ABORT

( -- ) ( R: ... -- )

Clear the data stack and enter QUIT. Never returns.

*Side effects:* Full return-stack reset; never executes RTS.

## QUIT

( -- ) ( R: ... -- )

The outer interpreter's top-level loop. Never returns.

*Side effects:* Resets the return stack; wraps INTERPRET in CATCH.

# 3.2 Stack Manipulation — Data Stack

## DUP

( n -- n n )

Duplicate the top stack item.

*Side effects:* None.

## DROP

( n -- )

Discard the top stack item.

*Side effects:* None.

## SWAP

( n1 n2 -- n2 n1 )

Exchange the top two items.

*Side effects:* None.

## OVER

( n1 n2 -- n1 n2 n1 )

Copy the second item to the top.

*Side effects:* None.

## ROT

( n1 n2 n3 -- n2 n3 n1 )

Rotate the third item to the top.

*Side effects:* None.

## ?DUP

( n -- n n | 0 )

Duplicate top only if it is nonzero.

*Side effects:* None.

## DEPTH

( -- n )

Push the number of items on the data stack.

*Side effects:* Computed relative to SPO.

## 2DUP

( x1 x2 -- x1 x2 x1 x2 )

Duplicate the top cell pair.

*Side effects:* None.

**2DROP**

```
( x1 x2 -- )
```

Discard the top cell pair.

*Side effects: None.*

**2SWAP**

```
( x1 x2 x3 x4 -- x3 x4 x1 x2 )
```

Exchange the top two cell pairs.

*Side effects: None.*

**2OVER**

```
( x1 x2 x3 x4 -- x1 x2 x3 x4 x1 x2 )
```

Copy the second cell pair to the top.

*Side effects: None.*

**NIP**

```
( x1 x2 -- x2 )
```

Discard the second item.

*Side effects: None.*

**TUCK**

```
( x1 x2 -- x2 x1 x2 )
```

Copy the top item under the second.

*Side effects: None.*

**PICK**

```
( xu ... x0 u -- xu ... x0 xu )
```

Copy the item u cells deep to the top.

*Side effects: No bounds check against actual stack depth.*

**ROLL**

```
( xu ... x0 u -- xu-1 ... x0 xu )
```

Remove the item u cells deep and move it to the top.

*Side effects: Cost scales with u; no bounds check.*

**2ROT**

```
( x1..x6 -- x3 x4 x5 x6 x1 x2 )
```

Rotate the third cell pair to the top.

*Side effects: None.*

### 3.3 Stack Manipulation — Return Stack

Because DO/LOOP (Section 3.8) stores its loop counters on the return stack, >R and R> — and their double-cell counterparts, 2>R and 2R> — must remain balanced within the body of any DO/LOOP they appear inside, not merely within the enclosing definition as a whole. This discipline is necessary to avoid corrupting the loop's own stack counters, and also the calling word's own subroutine return address, since that address sits on the same return stack, just beneath the loop's own counters — an imbalance left active across a LOOP/+LOOP boundary can overwrite either.

**>R**

```
( x -- ) ( R: -- x )
```

Move top of data stack to the return stack.

*Side effects: Must be balanced by R> within the same definition.*

**R>**

```
( -- x ) ( R: x -- )
```

Move top of return stack to the data stack.

*Side effects: None.*

**R@**

```
( -- x ) ( R: x -- x )
```

Copy top of return stack to the data stack.

*Side effects: Non-destructive.*

**2>R**

```
( x1 x2 -- ) ( R: -- x1 x2 )
```

Move top cell pair to the return stack.

*Side effects: Must be balanced by 2R> in the same definition.*

**2R>**

```
( -- x1 x2 ) ( R: x1 x2 -- )
```

Move top cell pair back to the data stack.

*Side effects: None.*

**2R@**

```
( -- x1 x2 ) ( R: x1 x2 -- x1 x2 )
```

Copy top cell pair from the return stack.

*Side effects: Non-destructive.*

## 3.4 Arithmetic — Single Cell

**+**

```
( n1 n2 -- sum )
```

Add.

*Side effects: None.*

**-**

```
( n1 n2 -- diff )
```

Subtract n2 from n1.

*Side effects: None.*

**\***

```
( n1 n2 -- prod )
```

Signed multiply, truncated to one cell.

*Side effects: None.*

**/**

```
( n1 n2 -- quot )
```

Signed symmetric division.

*Side effects: Throws -10 if n2 is zero.*

**MOD**

```
( n1 n2 -- rem )
```

Signed symmetric remainder.

*Side effects: Throws -10 if n2 is zero.*

**/MOD**

```
( n1 n2 -- rem quot )
```

Quotient and remainder together.

*Side effects: Throws -10 if n2 is zero.*

## NEGATE

( n -- -n )

Two's-complement negate.

*Side effects: None.*

## ABS

( n -- |n| )

Absolute value.

*Side effects: None.*

## MIN

( n1 n2 -- min )

Signed minimum.

*Side effects: None.*

## MAX

( n1 n2 -- max )

Signed maximum.

*Side effects: None.*

## 1+

( n -- n+1 )

Add one.

*Side effects: None.*

## 1-

( n -- n-1 )

Subtract one.

*Side effects: None.*

## 2+

( n -- n+2 )

Add two. Not an ANS-standard word.

*Side effects: None.*

## 2\*

( n -- n\*2 )

Arithmetic shift left one bit.

*Side effects: None.*

## 2/

( n -- n/2 )

Arithmetic shift right one bit, sign-preserving.

*Side effects: None.*

## \* /

( n1 n2 n3 -- quot )

Multiply then divide via a double-cell intermediate.

*Side effects: Throws -10 if n3 is zero.*

## \*/MOD

( n1 n2 n3 -- rem quot )

As \*/ but yields remainder and quotient.

*Side effects: Throws -10 if n3 is zero.*

## 3.5 Arithmetic — Mixed & Double Precision

### UM\*

( u1 u2 -- ud )

Unsigned single by single multiply, double result.

*Side effects: None.*

### UM/MOD

( ud u1 -- u2 u3 )

Unsigned double by single divide.

*Side effects: Throws -10 if u1 is zero.*

### M\*

( n1 n2 -- d )

Signed single by single multiply, double result.

*Side effects: None.*

### FM/MOD

( d n1 -- n2 n3 )

Floored double by single division.

*Side effects: Throws -10 if n1 is zero.*

### SM/REM

( d n1 -- n2 n3 )

Symmetric double by single division.

*Side effects: Throws -10 if n1 is zero.*

### D+

( d1 d2 -- d3 )

Double-cell add.

*Side effects: None.*

### D-

( d1 d2 -- d3 )

Double-cell subtract.

*Side effects: None.*

### DNEGATE

( d1 -- d2 )

Double-cell two's-complement negate.

*Side effects: None.*

### DABS

( d1 -- d2 )

Double-cell absolute value.

*Side effects: None.*

### M+

( d1 n -- d2 )

Add a single-cell value into a double.

*Side effects: None.*

### S>D

( n -- d )

Sign-extend a single cell to double.

*Side effects: None.*

## D>S

( d -- n )

Narrow a double to a single cell.

*Side effects: Implementation-defined if d does not fit; the high cell is dropped.*

## DMAX

( d1 d2 -- d3 )

Double-cell signed maximum.

*Side effects: None.*

## DMIN

( d1 d2 -- d3 )

Double-cell signed minimum.

*Side effects: None.*

# 3.6 Logic, Shifts & Address Arithmetic

## AND

( n1 n2 -- n1&n2 )

Bitwise AND.

*Side effects: None.*

## OR

( n1 n2 -- n1|n2 )

Bitwise OR.

*Side effects: None.*

## XOR

( n1 n2 -- n1^n2 )

Bitwise exclusive OR.

*Side effects: None.*

## INVERT

( n -- ~n )

Ones'-complement.

*Side effects: None.*

## LSHIFT

( x1 u -- x2 )

Logical shift left u bits, zero-fill.

*Side effects: Cost scales with u; no early-out for large u.*

## RSHIFT

( x1 u -- x2 )

Logical shift right u bits, zero-fill.

*Side effects: Cost scales with u; no early-out for large u.*

## CELLS

( n -- n\*2 )

Convert a cell count to a byte offset.

*Side effects: None.*

**CELL+**

```
( addr -- addr+2 )
```

Add the size of one cell.

*Side effects: None.*

**CHARS**

```
( n -- n )
```

Convert a character count to a byte offset. No-op on this system.

*Side effects: None.*

**CHAR+**

```
( addr -- addr+1 )
```

Add the size of one character.

*Side effects: None.*

**ALIGN**

```
( -- )
```

Align HERE to a cell boundary. No-op on the 6809.

*Side effects: None.*

**ALIGNED**

```
( addr -- addr )
```

Align a given address. No-op on the 6809.

*Side effects: None.*

## 3.7 Comparison

**=**

```
( n1 n2 -- flag )
```

True if equal.

*Side effects: None.*

**<**

```
( n1 n2 -- flag )
```

True if n1 signed less than n2.

*Side effects: None.*

**>**

```
( n1 n2 -- flag )
```

True if n1 signed greater than n2.

*Side effects: None.*

**0=**

```
( n -- flag )
```

True if n is zero.

*Side effects: None.*

**0<**

```
( n -- flag )
```

True if n is negative.

*Side effects: None.*

**U<**

```
( u1 u2 -- flag )
```

True if u1 unsigned less than u2.

*Side effects: None.*

**<>**

( n1 n2 -- flag )

True if not equal.

*Side effects: None.*

**0<>**

( n -- flag )

True if n is not zero.

*Side effects: None.*

**0>**

( n -- flag )

True if n is greater than zero.

*Side effects: None.*

**U>**

( u1 u2 -- flag )

True if u1 unsigned greater than u2.

*Side effects: None.*

**WITHIN**

( n1 n2 n3 -- flag )

True if  $n2 \leq n1 < n3$ .

*Side effects: Computed via unsigned comparison of offsets, handles wraparound ranges correctly.*

**D=**

( d1 d2 -- flag )

Double-cell equal.

*Side effects: None.*

**D<**

( d1 d2 -- flag )

Double-cell signed less than.

*Side effects: Compares high cells (signed) first, low cells (unsigned) only if equal.*

**DU<**

( d1 d2 -- flag )

Double-cell unsigned less than.

*Side effects: Both tiers compared unsigned.*

## 3.8 Control Flow — Conditionals & Loops

**IF**

Compile: ( -- ) Run: ( flag -- )

Begin a conditional. IMMEDIATE, compile-only.

*Side effects: Compiles ZBRANCH and a forward-patch field.*

**THEN**

( -- )

Resolve an IF or ELSE forward branch. IMMEDIATE, compile-only.

*Side effects: Throws -22 if the control-flow tag does not match.*

**ELSE**

( -- )

Alternate branch of IF. IMMEDIATE, compile-only.

*Side effects:* Throws -22 if the control-flow tag does not match.

## BEGIN

( -- )

Mark a loop target. IMMEDIATE, compile-only.

*Side effects:* None.

## UNTIL

( flag -- )

Branch back to BEGIN while flag is false. IMMEDIATE, compile-only.

*Side effects:* Throws -22 on a tag mismatch.

## AGAIN

( -- )

Unconditional branch back to BEGIN. IMMEDIATE, compile-only.

*Side effects:* Throws -22 on a tag mismatch.

## WHILE

( flag -- )

Conditional exit within a BEGIN loop. IMMEDIATE, compile-only.

*Side effects:* Leaves a second pending forward-patch field.

## REPEAT

( -- )

Close a BEGIN...WHILE loop. IMMEDIATE, compile-only.

*Side effects:* Patches both the back-edge and WHILE's forward exit.

## RECURSE

( -- )

Compile a call to the word currently being defined. IMMEDIATE, compile-only.

*Side effects:* Bypasses FIND's smudge check; reads LATEST's own CFA directly.

## DO

Compile: ( -- ) Run: ( n1|u1 n2|u2 -- ) ( R: -- loop-sys )

Begin a counted loop. IMMEDIATE, compile-only.

*Side effects:* Builds a 4-cell loop frame on the return stack; runs at least once even if limit=index.

## ?DO

Compile: ( -- ) Run: ( n1|u1 n2|u2 -- ) ( R: -- loop-sys )

As DO, but skips the loop entirely if limit=index. IMMEDIATE, compile-only.

*Side effects:* Builds the same loop frame; adds a forward skip branch.

## LOOP

( -- ) ( R: loop-sys -- | loop-sys )

Increment index by 1; loop until index reaches limit. IMMEDIATE, compile-only.

*Side effects:* Discards the loop frame on exit.

## +LOOP

( n -- ) ( R: loop-sys -- | loop-sys )

Increment index by n; exits when the step crosses limit. IMMEDIATE, compile-only.

*Side effects:* Handles positive or negative step; discards the frame on exit.

## I

```
( -- n ) ( R: loop-sys -- loop-sys )
```

Fetch the innermost loop's index.

*Side effects: Non-destructive read of the current loop frame.*

## J

```
( -- n ) ( R: loop-sys1 loop-sys2 -- loop-sys1 loop-sys2 )
```

Fetch the next-outer loop's index.

*Side effects: Only reaches one level of nesting out; no J2 equivalent exists.*

## LEAVE

```
( -- ) ( R: loop-sys -- loop-sys )
```

Force the innermost loop to exit at its next LOOP/+LOOP.

*Side effects: Assumes it is textually inside the loop it affects; unverified across an intervening word call.*

## UNLOOP

```
( -- ) ( R: loop-sys -- )
```

No-op. Previously discarded the innermost loop frame directly; EXIT's own frame-unwinding (via a compile-time count of enclosing DO/?DO frames) already discards it correctly and automatically on every path, including with no enclosing loop at all. Calling UNLOOP immediately before EXIT used to discard the frame twice - once here, once more in EXIT's own unwind - corrupting the return stack. UNLOOP now does nothing, so both "EXIT" alone and "UNLOOP EXIT" are safe.

*Side effects: none. Kept in the dictionary for source compatibility with code written for systems where EXIT does not auto-unwind and UNLOOP is genuinely required first.*

## EXIT

```
( -- )
```

Return from the current definition immediately. IMMEDIATE, compile-only in the sense that it must be used inside a definition.

*Side effects: Counts and discards any open DO/?DO frames at compile time before compiling its own return. This unwinding is automatic and correct on its own - UNLOOP is not required beforehand (see UNLOOP, above, which is a no-op for exactly this reason).*

## CASE

```
( -- )
```

Begin a multi-way selection. IMMEDIATE, compile-only.

*Side effects: None.*

## OF

```
( x n -- | x )
```

Begin one CASE clause's test. IMMEDIATE, compile-only.

*Side effects: Drops the selector only on a matching clause.*

## ENDOF

```
( -- )
```

End one CASE clause. IMMEDIATE, compile-only.

*Side effects: Throws -22 on a tag mismatch.*

## ENDCASE

```
( x -- )
```

Close a CASE structure, discarding the selector.

*Side effects: Throws -22 if the control-flow stack is malformed.*

## 3.9 Defining Words

:

```
( "name" -- )
```

Begin a colon definition.

*Side effects:* Header is smudged until ; snapshot control-flow depth for later checking.

```
;
```

```
( -- )
```

End a colon definition. IMMEDIATE.

*Side effects:* Unsmudges the header; throws -22 if a control structure is left open.

## CREATE

```
( "name" -- )
```

Build a header and a DOES>-ready trampoline.

*Side effects:* PFA defaults to CODEHERE (immutable space).

## DOES>

```
( -- )
```

Attach runtime behavior to a CREATED word. IMMEDIATE, compile-only.

*Side effects:* Patches the trampoline's BEHAVIOR field at run time via a double return.

## VARIABLE

```
( "name" -- )
```

Create a one-cell mutable variable.

*Side effects:* PFA reserved in VARHERE (mutable space); initialized to zero.

## CONSTANT

```
( x "name" -- )
```

Create a word that returns a fixed value.

*Side effects:* Value stored in CODEHERE (immutable space).

## VALUE

```
( x "name" -- )
```

Create a reassignable named value.

*Side effects:* PFA in VARHERE (mutable space); reassigned via TO.

## TO

```
( x "name" -- )
```

Assign a new value to a VALUE. IMMEDIATE.

*Side effects:* Behaves differently interpreting vs. compiling.

## 2VARIABLE

```
( "name" -- )
```

Create a two-cell mutable variable.

*Side effects:* PFA in VARHERE; both cells initialized to zero.

## 2CONSTANT

```
( x1 x2 "name" -- )
```

Create a word returning a fixed double value.

*Side effects:* Stores x1 at the lower address, x2 at the higher, matching 2@/2!.

## BUFFER:

```
( u "name" -- )
```

Create a word returning the address of u reserved, uninitialized bytes.

*Side effects:* PFA in VARHERE.

## DEFER

```
( "name" -- )
```

Create a word whose action can be set later.

*Side effects:* Default action throws -21 until IS/DEFER! is used.

### DEFER@

```
( xt-defer -- xt )
```

Fetch a deferred word's current target.

*Side effects:* None.

### DEFER!

```
( xt xt-defer -- )
```

Set a deferred word's target.

*Side effects:* None.

### IS

```
( xt "name" -- )
```

Set a deferred word's target by name. IMMEDIATE.

*Side effects:* Behaves differently interpreting vs. compiling.

### ACTION-OF

```
( "name" -- xt )
```

Fetch a deferred word's current target by name. IMMEDIATE.

*Side effects:* Behaves differently interpreting vs. compiling.

### MARKER

```
( "name" -- )
```

Create a word that, when executed, forgets itself and everything defined after it.

*Side effects:* Restores all three HERE pointers and LATEST from a snapshot.

## 3.10 Compiling Words

### IMMEDIATE

```
( -- )
```

Mark the most recently defined word as IMMEDIATE.

*Side effects:* Sets a header flag bit.

### STATE

```
( -- addr )
```

Variable: nonzero while compiling.

*Side effects:* None.

```
[
```

```
( -- )
```

Enter interpret state. IMMEDIATE.

*Side effects:* None.

```
]
```

```
( -- )
```

Enter compile state.

*Side effects:* None.

```
,
```

```
( "name" -- xt )
```

Look up a word's execution token.

*Side effects:* Throws -13 if not found. Works identically interpreting or compiling.

**COMPILE,**

( xt -- )

Compile a call to the given execution token.

*Side effects: None.*

**LITERAL**

( x -- )

Compile x as a literal. IMMEDIATE, compile-only.

*Side effects: None.*

**[ ' ]**

( "name" -- )

Compile-time tick: compile a word's xt as a literal. IMMEDIATE, compile-only.

*Side effects: Throws -14 if used while interpreting; -13 if the name is not found.*

**POSTPONE**

( "name" -- )

Defer a word's compiling behavior into the current definition. IMMEDIATE, compile-only.

*Side effects: Throws -14 if used while interpreting; -13 if the name is not found.*

**[COMPILE]**

( "name" -- )

Obsolescent predecessor to POSTPONE: unconditionally compile a direct call to the found word, right now. IMMEDIATE, compile-only.

*Side effects: Throws -14 if used while interpreting; -13 if the name is not found. Unlike POSTPONE, makes no immediate/non-immediate distinction - prefer POSTPONE for a word whose own compilation semantics is not simply "compile a call to me".*

**>BODY**

( xt -- pfa )

Recover a CREATED word's parameter field address.

*Side effects: Assumes the standard trampoline layout; fixed offset arithmetic.*

**EXECUTE**

( xt -- )

Execute the word with the given execution token.

*Side effects: None.*

**SLITERAL**

( addr len -- )

Compile a runtime string as a literal. IMMEDIATE, compile-only.

*Side effects: Throws -14 if used while interpreting.*

**ABORT"**

Compile: ( "msg" -- ) Run: ( flag -- )

Conditionally type a message and abort. IMMEDIATE, compile-only.

*Side effects: On true flag: types the message, throws -2. Shares WORD's own WORDMAXCHARS cap (46 characters) during parsing, and reserves 3 bytes ahead of CODEHERE before calling WORD for the same reason S" does - see S", above.*

## 3.11 Memory

**@**

( addr -- x )

Fetch a cell.

*Side effects: None.*

**!**  
 ( x addr -- )  
 Store a cell.  
*Side effects: None.*

**C@**  
 ( addr -- char )  
 Fetch a byte, zero-extended.  
*Side effects: None.*

**C!**  
 ( char addr -- )  
 Store the low byte only.  
*Side effects: None.*

**+!**  
 ( n addr -- )  
 Add n into the cell at addr.  
*Side effects: None.*

**2@**  
 ( addr -- x1 x2 )  
 Fetch a double cell.  
*Side effects: Lower address is x1, higher address is x2.*

**2!**  
 ( x1 x2 addr -- )  
 Store a double cell.  
*Side effects: x1 stored at the lower address, x2 at the higher.*

**,**  
 ( n -- )  
 Append one cell to the immutable/code region (CODEHERE).  
*Side effects: Advances CODEHERE.*

**C,**  
 ( char -- )  
 Append one byte to the immutable/code region.  
*Side effects: Advances CODEHERE.*

**ALLLOT**  
 ( n -- )  
 Reserve (or release, if negative) bytes in the immutable/code region.  
*Side effects: No bounds check against adjoining regions.*

**HERE**  
 ( -- addr )  
 Current allocation point of the immutable/code region.  
*Side effects: Reports CODEHERE.*

**V,**  
 ( n -- )  
 Append one cell to the mutable/variable region (VARHERE).  
*Side effects: Advances VARHERE.*

**VC,**

( char -- )

Append one byte to the mutable/variable region.

*Side effects:* Advances VARHERE.

**VALLLOT**

( n -- )

Reserve (or release) bytes in the mutable/variable region.

*Side effects:* No bounds check against adjoining regions.

**WHERE**

( -- addr )

Current allocation point of the mutable/variable region.

*Side effects:* Reports VARHERE.

**PAD**

( -- addr )

A scratch buffer address, offset above CODEHERE by PADMINISIZE (84 characters, matching the ANS minimum).

Floats relative to CODEHERE, not a separately reserved region - its address changes as CODEHERE advances.

*Side effects:* Also used by interpreted S" (see S", below) for its own string storage, per ANS's allowance for non-standard words to use PAD as long as the use is documented. This means PAD and interpreted S" text now share the same space - using PAD directly right after an interpreted S" call (or vice versa) will overwrite one with the other.

**UNUSED**

( -- u )

Bytes remaining in the region HERE addresses.

*Side effects:* Reports distance to the return/data-stack boundary.

**VUNUSED**

( -- u )

Bytes remaining in the mutable/variable region.

*Side effects:* Reports distance to APPCODE's start; not itself enforced.

**MOVE**

( src dst u -- )

Copy u bytes, overlap-safe.

*Side effects:* Chooses direction automatically based on address order.

**FILL**

( addr u char -- )

Fill u bytes with char.

*Side effects:* None.

**ERASE**

( addr u -- )

Fill u bytes with zero.

*Side effects:* None.

**CMOVE**

( src dst u -- )

Copy u bytes low address to high.

*Side effects:* Not overlap-safe in the high-over-low direction.

**CMOVE>**

( src dst u -- )

Copy u bytes high address to low.

*Side effects: Not overlap-safe in the low-over-high direction.*

## 3.12 Strings & Parsing

### COUNT

( c-addr -- addr len )

Convert a counted string to address/length form.

*Side effects: None.*

### WORD

( char -- c-addr )

Parse a delimited token into a counted string.

*Side effects: WORDMAXCHARS-character cap (46) - well above the ANS minimum of 33 (WORDMINSIZE), and above the old, separately-buffered cap of 31 this system used previously, which was actually slightly below that ANS minimum.*

*Scans directly into the CODEHERE-to-PAD gap rather than a separate fixed buffer, reserving room at the PAD end for the pictured numeric output buffer (HOLD and friends), which shares this same gap growing the opposite direction. Skips leading delimiters. The returned region isn't advanced past by WORD itself - like PAD, it's expected to be overwritten by whatever gets compiled or allocated next, per standard transient-region semantics.*

### CHAR

( "name" -- char )

Parse the next word, leave its first character.

*Side effects: Inherits WORD's WORDMAXCHARS cap (46, irrelevant here since only the first character is kept).*

### [CHAR]

( "name" -- )

Compile-time CHAR: compiles the character as a literal. IMMEDIATE, compile-only.

*Side effects: Throws -14 if used while interpreting.*

### PARSE

( char "ccc<char>" -- addr len )

Parse a delimited token, no length cap.

*Side effects: Does not skip leading delimiters, unlike WORD.*

### PARSE - NAME

( "<spaces>name<space>" -- addr len )

Parse a space-delimited token, no length cap.

*Side effects: Skips leading spaces.*

### S"

Compile: ( "str" -- ) Run: ( -- addr len )

Parse and compile, or stage, a string literal.

*Side effects: Compiling: appends inline to CODEHERE - the definition's own permanent storage, unaffected by anything below. Interpreting: copies into PAD (see PAD, above). Both modes are limited by WORD's own WORDMAXCHARS cap (46 characters) during parsing, before either storage path runs - a longer string is truncated by WORD itself, not by S"'s own storage. The compiled path reserves 3 bytes ahead of CODEHERE before calling WORD, so the runtime trampoline compiled afterward doesn't overwrite the text WORD just staged there. PAD's interpreted-mode address is computed fresh from CODEHERE on every call, but only changes when something is compiled or allocated in between - a second S" call before an earlier string is actually used will still silently overwrite it. For any use needing more than one string to coexist (e.g. REPLACES's two arguments), wrap each in its own colon definition (: name S" text" ;) instead - compiled S" never touches PAD at all, giving each string its own permanent storage.*

### "

( "str" -- )

Compile a string that types itself at run time. IMMEDIATE, compile-only.

*Side effects: No interpretation semantics, per ANS. Shares WORD's own WORDMAXCHARS cap (46 characters) during parsing, and reserves 3 bytes ahead of CODEHERE before calling WORD for the same reason S" does - see S", above.*

## COMPARE

```
( addr1 len1 addr2 len2 -- n )
```

Compare two strings, signed three-way result.

*Side effects: None.*

## SEARCH

```
( addr1 len1 addr2 len2 -- addr3 len3 flag )
```

Find a substring within a string.

*Side effects: Empty needle reports not-found (an implementation choice).*

## -TRAILING

```
( addr len1 -- addr len2 )
```

Trim trailing spaces from a string's reported length.

*Side effects: None.*

## /STRING

```
( addr1 len1 n -- addr2 len2 )
```

Trim n characters from the front of a string.

*Side effects: None.*

## REPLACES

```
( addr1 len1 addr2 len2 -- )
```

Register a name/value substitution pair for SUBSTITUTE - addr1/len1 is the replacement text, addr2/len2 is the substitution name (matched later between % delimiters).

*Side effects: Single-slot only, not a table; a second call overwrites the first. Both strings must come from stable storage, not from S" typed directly at the interpreter prompt - interpreted S" writes into PAD (see PAD and S", above), whose address only changes when something is compiled or allocated in between, so a later S" call before an earlier string is actually used (including the one for SUBSTITUTE's own source string) will still silently overwrite it. Wrap each string in its own colon definition (: name S" text" ;) for stable, independent storage instead - compiled S" never touches PAD.*

## SUBSTITUTE

```
( addr1 len1 addr2 len2 -- addr2 len3 n )
```

Scan addr1/len1 for text between '%' delimiter pairs. A pair enclosing the name registered via REPLACES has the whole %name% span - delimiters included - replaced by the registered substitution text; %% collapses to a single %; an unmatched name, or a trailing % with no closing delimiter, is passed through unchanged. Result is stored at addr2, length len3.

*Side effects: n is the number of substitutions made (0 or positive on success). Throws -1 on destination overflow; single-slot only (see REPLACES).*

## SNAME

```
( xt -- addr len )
```

Reverse dictionary lookup: xt to name. Non-standard.

*Side effects: Returns zero length if not found.*

## UNESCAPE

```
( c-addr1 u1 c-addr2 -- c-addr2 u2 )
```

Replace every '%' character in the source string with two '%' characters; every other character passes through unchanged. Passing a string through UNESCAPE and then SUBSTITUTE returns the original string - this is UNESCAPE's specific purpose, protecting a literal '%' from being read as a SUBSTITUTE delimiter. Has nothing to do with backslash sequences (\n, \t, etc.) despite the name's similarity to that idea in other languages - confirmed against the ANS reference implementation.

*Side effects: u2 can be longer than u1 - each '%' in the source adds one extra character to the output. No destination-buffer-overflow check, matching the ANS reference implementation, which also has none (an ambiguous condition per the standard, not a required error case).*

### 3.13 Numeric Output

**<#**

( -- )

Begin pictured numeric output conversion.

*Side effects: Sets HLD to PAD's address. To get a proper ud1 onto the stack for the # family below, type the number with a trailing '.' (e.g. "123." rather than "123") - the text interpreter converts any number immediately followed by a decimal point to a double-cell number, per ANS's own input convention ([forth-standard.org/standard/usage#usage:numbers](http://forth-standard.org/standard/usage#usage:numbers), 3.4.1.3). "123" alone stays single-cell and is not a valid argument to #.*

**#**

( ud1 -- ud2 )

Convert one digit.

*Side effects: Uses the current BASE. Expects a genuine double-cell ud1 - see <# above for how to get one onto the stack from typed input.*

**#S**

( ud1 -- ud2 )

Convert remaining digits until the value is zero.

*Side effects: Uses the current BASE.*

**#>**

( ud -- addr len )

End pictured conversion, discarding ud.

*Side effects: None.*

**HOLD**

( char -- )

Insert one character into the pictured output buffer.

*Side effects: Buffer grows downward from PAD; no bound enforced.*

**HOLDS**

( addr len -- )

Insert an entire string into the pictured output buffer.

*Side effects: Depends on HOLD's exact decrement-by-one behavior.*

**SIGN**

( n -- )

Insert a minus sign if n is negative.

*Side effects: None.*

**.**

( n -- )

Print signed, trailing space.

*Side effects: None.*

**U.**

( u -- )

Print unsigned, trailing space.

*Side effects: None.*

**.R**

( n width -- )

Print signed, right-justified in a field.

*Side effects: No trailing space; width is a minimum, not a cap.*

## U.R

( u width -- )

Print unsigned, right-justified in a field.

*Side effects: No trailing space; width is a minimum, not a cap.*

## ?

( addr -- )

Fetch and print the cell at addr, signed.

*Side effects: None.*

## D.

( d -- )

Print signed double, trailing space.

*Side effects: None.*

## D.R

( d width -- )

Print signed double, right-justified.

*Side effects: No trailing space.*

## 3.14 Base / Radix Control

### BASE

( -- addr )

Variable holding the current number base.

*Side effects: None.*

### DECIMAL

( -- )

Set BASE to 10.

*Side effects: None.*

### HEX

( -- )

Set BASE to 16.

*Side effects: None.*

### BINARY

( -- )

Set BASE to 2. Not an ANS-standard word.

*Side effects: None.*

## 3.15 Exception Handling

### CATCH

( xt -- 0 | n )

Execute xt, trapping any THROW.

*Side effects: Restores data stack depth and HANDLER on either path.*

### THROW

( n -- )

Raise exception n, or do nothing if n is zero.

*Side effects:* Nonzero: non-local exit to the matching CATCH, or JMP ABORT if none is active.

### 3.16 Comments

(

( "ccc<paren>" -- )

Discard input up to the next close paren. IMMEDIATE.

*Side effects:* Works interpreting or compiling.

\

( "ccc<eol>" -- )

Discard the remainder of the current input line. IMMEDIATE.

*Side effects:* Follows SOURCE, so it operates correctly inside EVALUATE.

### 3.17 Environmental & System Queries

#### ENVIRONMENT?

( c-addr u -- false | i\*x true )

Query implementation-defined characteristics.

*Side effects:* Supports /COUNTED-STRING (255), MAX-N (32767), MAX-U (65535), ADDRESS-UNIT-BITS (8), /HOLD (34), /PAD (84), FLOORED (0 - this system uses symmetric, not floored, division), MAX-CHAR (255), MAX-D and MAX-UD (double-cell; push low then high before the true flag), RETURN-STACK-CELLS (384), and STACK-CELLS (512). WORDLISTS is deliberately unrecognized (returns plain false, not 0 true) since the Search-Order word set isn't implemented.

#### SOURCE

( -- addr len )

The current input source.

*Side effects:* Follows terminal input or an active EVALUATE string.

#### SOURCE - ID

( -- 0 | -1 )

Identify the current input source. 0 = terminal, -1 = string.

*Side effects:* None.

#### REFILL

( -- flag )

Attempt to refill the input source.

*Side effects:* Fails (false) if the current source is a string, per ANS.

#### EVALUATE

( addr len -- )

Interpret a string as Forth source.

*Side effects:* Saves and restores the whole input-source state around the call.

#### TIB

( -- addr )

The fixed terminal input buffer address.

*Side effects:* Always the terminal buffer, unaffected by EVALUATE.

#### #TIB

( -- addr )

Variable: count of valid characters in TIB.

*Side effects:* Always the terminal buffer's count.

**>IN**

( -- addr )

Variable: scan offset into the current input source.

*Side effects: None.*

**SPAN**

( -- addr )

Variable: EXPECT's returned character count.

*Side effects: None.*

**BL**

( -- char )

The blank/space character, 32.

*Side effects: None.*

## 3.18 Tools

**.S**

( -- )

Print the data stack, top to bottom.

*Side effects: Non-destructive.*

**WORDS**

( -- )

List every word in the dictionary.

*Side effects: Walks the full ROM+RAM chain from LATEST.*

**DUMP**

( addr u -- )

Hex and ASCII memory dump, 16 bytes per line.

*Side effects: None.*

## 4. Forth Development Notes

This section is written for the audience who will program the completed system: users developing a Forth application on top of ClaudeForth, not those building or modifying the system itself. It assumes the base system already runs and focuses on what's needed to work productively within it day to day.

This section covers five practical aspects of working with the system that fall outside the word-by-word glossary: how serial communication is configured, a brief introduction to Forth syntax and stack operations, worked examples of the Tools word set, how to reclaim dictionary space during interactive development using MARKER, and why the optional Search-Order word set is not part of this implementation.

### 4.1 Serial I/O and Handshaking

Serial communication parameters — baud-rate divisor, word length, parity, and stop bits — are fixed by the hardware, not selectable from Forth. The 6850 ACIA's control register is written once, at COLDSTRT, to a constant value (CR\_RXON, or CR\_RXTX when the output ring has data queued) encoding a  $\pm 16$  clock, 8 data bits, no parity, 1 stop bit, and receive interrupts enabled. There is no word in this system that changes these settings at run time; the terminal or host at the other end of the link must be configured to match this fixed set of parameters independently.

**Hardware (RTS/CTS) handshaking is implemented; software (XON/XOFF) is not.**

RTS — this system's output telling the remote device when to pause sending — is toggled automatically based on the input ring buffer's fill level. IRQH's receive path calls RTSCHECKHI after storing each incoming byte; once the ring reaches INHIWATER (48 of its 64 bytes), RTS is driven high by writing CR\_RTSHI to the ACIA control register, and RTS stays high until KEY, on the mainline side, drains the ring back down to INLOWATER (16 bytes) and calls RTSCHECKLO to drop it low again. The 32-byte gap between the two marks is deliberate hysteresis, avoiding RTS toggling right at a single threshold.

CTS — the remote device's own flow-control signal telling this system when it may transmit — needs no firmware logic at all: the 6850 ACIA inhibits TDRE in hardware whenever CTS is deasserted, and every transmit path in this system (EMIT, IRQH's TXCHK) already gates on TDRE, so CTS is honored automatically with no code changes required.

One real ACIA constraint shapes this design directly: the 6850's control register ties RTS state and transmit-interrupt-enable to the same two bits, so RTS-high and TX-interrupt-enabled can never be selected simultaneously. While RTS is asserted high, EMIT still queues outgoing bytes into the output ring but deliberately does not enable the transmit interrupt to drain them — RTSCHECKLO re-enables it once RTS drops low, restoring CR\_RXTX if output is still queued at that point. A second, genuine race is worth naming rather than glossing over: RTSCHECKHI runs from inside the interrupt handler, where hardware has already masked further interrupts, but RTSCHECKLO runs from ordinary mainline code (KEY) and could otherwise be interrupted mid-decision by IRQH's own TXOFF path, which also writes the control register — RTSCHECKLO masks IRQ around its own critical section specifically to prevent that.

Software (XON/XOFF) handshaking remains unimplemented: this system still neither transmits XON/XOFF nor recognizes those bytes if a remote device sends them — an incoming \$11 or \$13 is treated as an ordinary character and queued like any other. On a genuine 3-wire serial link with no RTS/CTS wiring at all, that link would still have no flow control of any kind; the hardware path described above only helps when RTS/CTS are physically connected.



## 4.2 Forth Syntax and Stack Operations

Forth has no operators, no operator precedence, and no infix expressions. A line of input is simply a sequence of whitespace-delimited words, read and acted on strictly left to right. Each word is either a number, which is pushed onto the data stack, or a name that is looked up in the dictionary and executed immediately. Arithmetic and every other operation work by taking their arguments off the top of the stack and pushing their result back onto it — this is postfix, or reverse Polish, notation: the operator always comes after its operands, never between or before them.

### A numerical example:

Computing  $(3 + 4) \times 2 - 14$  — is written:

```
3 4 + 2 * .
14    ok
```

Read left to right, tracking the data stack after each word (shown deepest item first, top of stack last):

|   |    |         |                                 |
|---|----|---------|---------------------------------|
| 3 | -> | ( 3 )   | push 3                          |
| 4 | -> | ( 3 4 ) | push 4                          |
| + | -> | ( 7 )   | pop 4 and 3, push their sum     |
| 2 | -> | ( 7 2 ) | push 2                          |
| * | -> | ( 14 )  | pop 2 and 7, push their product |
| . | -> | ( )     | pop and print: 14               |

The stack-effect notation used throughout the Glossary (Section 3) — for example, +’s ( n1 n2 -- sum ) — describes exactly this: the items expected on the stack before a word runs, to the left of "--", and what it leaves in their place afterward, to the right. Reading a word’s stack effect this way is enough to know how to use it correctly without needing to know anything about how it is implemented internally.

This system does not require building up a vocabulary from nothing: a large set of predefined words — arithmetic, stack manipulation, comparison, control flow, string and memory operations, and more — already exists and is documented word by word in the Glossary. At the interactive prompt, executing WORDS lists every one of these preexisting words currently in the dictionary, from the most recently defined word back through every built-in primitive. See Section 4.3 for a worked example of WORDS and the other Tools words.

### 4.3 Tools Word Set: Explanation and Examples

The three Tools words in the Glossary — `.S`, `WORDS`, and `DUMP` — are interactive aids rather than words a program typically calls itself. The examples below show plausible session transcripts. One real characteristic of this system worth knowing before reading them: output from a word is written immediately, with no leading newline of its own — the newline seen before "ok" in a normal session comes from `QUIT`, not from the word that ran. The transcripts below add a line break before each result for readability; on the actual wire, output from `.S`, `WORDS`, or `DUMP` appears immediately after the typed command, on the same line.

#### `.S` — print the data stack, top to bottom, non-destructively

Pushes nothing and removes nothing; it only reads the stack from the current top down to `SP0` and prints each cell with the same signed formatting as `DOT`. This makes it safe to run at any point without disturbing whatever a definition is part-way through building on the stack.

```
1 2 3 .S
3 2 1   ok

\ top of stack (3, pushed last) prints first, deepest
\ item (1, pushed first) prints last
```

#### `WORDS` — list every word in the dictionary

Walks the `LATEST` chain from the most recently defined application word all the way back through every ROM primitive, printing each name separated by a space. With roughly 150 application-level words already in the Glossary plus every ROM-resident primitive behind them, real output is long — the excerpt below shows the shape of it rather than a complete transcript.

```
WORDS
SQUARE FENCE VALUE TO VARIABLE CONSTANT DOES>
CREATE ... DUP DROP SWAP OVER ROT DEPTH ...
+ - * / MOD ... IF THEN ELSE BEGIN ... ok

\ most-recently-defined application words appear first,
\ since the walk follows LATEST backward through LINK;
\ ROM primitives make up the remainder of the list
```

#### `DUMP` — hex and ASCII memory dump, 16 bytes per line

Takes an address and a byte count, and prints that many bytes, 16 per line: each byte as two hex digits, then a matching ASCII column with unprintable bytes shown as a period. Worth noting explicitly since it differs from many `DUMP` implementations: this one does not print a leading address on each line — only the hex bytes and their ASCII rendering — so the caller must track which address a given line corresponds to separately if that matters.

```
HEX
7000 10 DUMP
BD E4 32 8E 00 00 39 00 00 00 00 00 00 00 00 00  .....9.....

\ 16 bytes shown starting at $7000 (the start of APPCODE);
\ actual bytes depend entirely on what has been compiled
\ into code space by the time DUMP is run
```

## 4.4 Dictionary Management with MARKER

ANS Forth provides no general "undefine" word; instead, the standard's mechanism for reclaiming dictionary space is MARKER. Executing MARKER with a name creates an ordinary dictionary entry that, when later executed itself, forgets its own definition and every definition made after it — restoring the dictionary to exactly the state it was in at the moment MARKER was invoked. This is the standard way to experiment interactively without permanently accumulating discarded or superseded definitions.

In this implementation, a word created by MARKER stores a snapshot of all three region pointers — DPHERE (dictionary), CODEHERE (code), and VARHERE (mutable variable space) — plus LATEST, at the moment of its own creation. Executing the marker word (DOMARKER) simply writes those four saved values back, in one step. Nothing needs to be erased or unwound word by word: every allocation word in the system (comma, ALLOT, HEADER, and their VARHERE counterparts) only ever appends at the current pointer, so rolling the pointer back alone is sufficient to make everything defined afterward unreachable and its space available again for reuse.

### Example:

```
MARKER FENCE
: SQUARE DUP * ;
5 SQUARE .           \ prints 25

FENCE                 \ forgets FENCE itself, and SQUARE

5 SQUARE .           \ throws -13, undefined word: SQUARE
```

MARKER FENCE creates FENCE and captures the current DPHERE/CODEHERE/VARHERE/LATEST as its own PFA contents. SQUARE is then compiled normally, advancing all three pointers past that snapshot. Executing FENCE restores the four saved values, which puts LATEST back to whatever it was immediately before FENCE was defined — so SQUARE is no longer reachable by FIND, and FENCE itself is gone too, along with the dictionary, code, and variable space both of them occupied. A second MARKER call after this point would need a new name, since FENCE no longer exists to be reused.

***Caveat:** as noted in the glossary, a MARKER's snapshot only covers the single dictionary chain this system currently maintains. It is not defined against the optional Search-Order word set, which this implementation does not include — see Section 4.5.*

## 4.5 The Search-Order Word Set Is Not Included

ANS Forth's optional Search-Order word set (FORTH-WORDLIST, WORDLIST, GET-CURRENT, SET-CURRENT, GET-ORDER, SET-ORDER, ALSO, ONLY, PREVIOUS, DEFINITIONS, and SEARCH-WORDLIST) is not implemented in this system. Every one of the roughly 150 words in the Glossary is resolved through a single, unified dictionary chain rooted at LATEST — there is only ever one wordlist, and new definitions always link into it. Search-Order's whole purpose is letting a program maintain several independent wordlists (for example, to keep an application's vocabulary separate from the words used only to build it) and to search a chosen, ordered subset of them; without it, every word ever defined is always visible to FIND, in one flat namespace, for the lifetime of the system.

This was a foundational decision, not an oversight discovered late: the single-chain design was fixed from the very first system-initialization code, where LATEST is set to the top of the ROM dictionary at cold start and every application word links onto that same chain afterward. Adding Search-Order now would not be a small addition — it would touch code that already exists throughout the system, not just introduce new words. FIND itself would need restructuring into an outer loop over an active search order, trying each wordlist in turn; HEADER, which every defining word (colon definitions, CREATE, VARIABLE, CONSTANT, VALUE, and the rest) already funnels through, would need to link new definitions into a selectable "current" wordlist rather than unconditionally into LATEST; and a few words that deliberately read LATEST directly today — RECURSE, the unsmudging step in semicolon, and WORDS — would each need the same substitution or they would silently keep operating on the wrong wordlist once more than one existed. MARKER's snapshot would also need to widen to cover every wordlist's head cell, not just LATEST, with the further complication that ANS itself leaves a marker's interaction with wordlists created after it as implementation-defined, not fully specified.

In practice, this means every program written against this system lives in one shared, flat namespace — there is no mechanism to hide a helper word's name from being visible everywhere else, and no way to have two differently-scoped words share the same name. For the scope of a single-user, single-application embedded system, this is a reasonable simplification rather than a defect; it is flagged here specifically so it is not mistaken for an accidental gap in ANS coverage.

## 4.6 Exception Handling

CATCH and THROW (Section 3.15) let a definition signal a problem to its caller without forcing every intermediate word on the call chain to check for it. The example below defines a guarded division word that throws rather than dividing by zero, and a caller that traps that exception with CATCH and reports it with an appropriate message in place of a numeric result.

```

\ SAFE/ throws -10 (the standard ANS code for division
\ by zero) instead of dividing when the divisor is 0
\ GUARDED/ wraps SAFE/'s execution token in CATCH,
\ which traps the THROW and returns its code in place
\ of letting it propagate uncaught; nonzero means SAFE/
\ threw, so GUARDED/ reports a message instead of the
\ numeric result CATCH's own success path would leave

```

```

\ Usage:

```

```

\ 10 5 GUARDED/

```

```

\ 2

```

```

\ 10 0 GUARDED/

```

```

\ can't divide by zero

```

```

: SAFE/ ( n1 n2 -- n1/n2 )

```

```

  DUP 0= IF -10 THROW THEN

```

```

  / ;

```

```

: GUARDED/ ( n1 n2 -- )

```

```

  ['] SAFE/ CATCH

```

```

  IF ." can't divide by zero" CR

```

```

  ELSE . CR THEN ;

```

## 5. Open Items Checklist

Everything below was explicitly flagged during the build as incomplete, unverified, or deliberately deferred. Nothing here is a surprise — each item was named at the point it came up. This is a consolidated list to work from, not a new set of findings. Regenerated to reflect fixes and design decisions made since the original version.

This section is a point-in-time snapshot of the checklist maintained alongside the assembler source, which has since grown considerably beyond what is practical to reproduce here in full. Rather than continuing to expand this section, further additions are tracked in that external checklist directly (OPEN\_ITEMS\_CHECKLIST\_part1.md and OPEN\_ITEMS\_CHECKLIST\_part2.md, distributed alongside the source) — see those files for the current, complete state. This section itself is revised only when an item listed below is genuinely addressed, not to keep pace with ongoing work.

### 5.1 Known bugs — resolved since the original checklist

- ✓ **DUMP's partial-final-line bug** — fixed via DUVALID tracking (25\_tools\_word\_set.asm). The ASCII column no longer reads past the valid bytes on a short final line.
- ✓ **SUBSTITUTE** — completed. It now performs a real bounds-checked copy (prefix / replacement / suffix) via a shared SUBCOPY helper, reusing SEARCHW for the match. Still scoped to a single registered name/value pair, not a full table — see "Open design questions" below; that scope limit is a deliberate simplification, not a bug.
- ✓ **VALUE's PFA** — moved from CODEHERE (immutable space) to VARHERE (mutable space), matching VARIABLE's pattern, so every T0-driven update now writes into the correct region instead of into code space. Not on the original checklist (found and fixed after this list was first generated); logged here for the record.
- ✓ **Every scratch/global cell now has a real RMB-assigned address in page zero** (00\_memory\_map\_and\_globals.asm), applied rather than left as a placeholder. Two real findings came out of doing this: the budget is **253 of 256 bytes used — only 3 bytes of headroom** in the GLOBALS page, and SNEND (documented in the original placeholder list) turned out to be genuinely dead — never read or written anywhere — and was dropped rather than given an address. Any future scratch cell added to this system will need to fit in that remaining 3 bytes or the page-zero fast-addressing property (DP = \$00) stops covering it.
- ✓ **The ROM base dictionary is now real** (27\_forth\_dictionary.asm) — every primitive with actual code got a real header, chained via LINK, CFA pointing straight at its code label. ABORTHDR's LINK field, a placeholder 0 since it was first built, is now resolved to the chain's newest entry. Building this surfaced two real findings, not just mechanical work: the original 1024-byte BASEDICT could not hold the header table (ultimately 1954 bytes needed, once DOES> was added — see below), so it was resized to 2048 bytes, taking the space from BASECODE (now 6080 bytes, was 7104) — a resize based purely on the header-table budget, not on any actual measurement of BASECODE's assembled size, which has never been checked with a real 6809 assembler. Generating this table also surfaced that **DOES> had no corresponding code anywhere in the file** — resolved in a follow-up pass, see below.
- ✓ **DOES> — resolved.** SETDOES (the patching runtime) was added beside DODOES/DOESRT0, and DOES> itself (code label DOESGT, since a literal ">" is not a valid 6809 assembler label) was added right after CREATE. DOESBEH was added to the GLOBALS layout to support it, using 2 of the page's

last 3 free bytes — only 1 byte of headroom now remains in page zero. DOES> also has a real dictionary header now (H\_DOESGT, the chain's newest entry); ABORTHDR's LINK was updated again to point at it.

- ✓ **Four internal loop-label names were reused across unrelated routines: FLOOP, UDLOOP, DRPOS, CMLoop.** Resolved — each pair renamed to a distinct name scoped to its own routine: FIND's FLOOP → FFLOOP; FILLW's FLOOP → FILLLOOP; UDIV16's UDLOOP → UD16LOOP; UDDIGIT's UDLOOP → UDDLOOP; DIVCOMMON's DRPOS → DCRPOS; DDOTR's DRPOS → DDRPOS; CMOVEW's CMLoop → CMVLOOP; COMPAREW's CMLoop → CMPLOOP. Verified zero duplicate labels and zero stray references to any of the old names remain anywhere in the file.
- ✓ **Hardware (RTS) serial handshaking — implemented.** IRQH's receive path and KEY now toggle RTS based on the input ring's fill level (INFILL, RTSCHECKHI, RTSCHECKLO against INHIWATER=48/INLOWATER=16, with hysteresis between them) via a new CR\_RTSHI control byte and RTSSTATE flag — the last free byte in the GLOBALS page, which is now fully packed at 256/256. CTS (the other handshaking direction) needed no firmware logic at all: confirmed against the 6850 datasheet that TDRE is automatically inhibited while CTS is deasserted, so this system's existing TDRE-gated transmit logic already respects it. A real race was identified and closed: mainline code (KEY, via RTSCHECKLO) and the ISR (IRQH's TXOFF) can both decide to write the control register, so RTSCHECKLO masks IRQ around its critical section and TXOFF checks RTSSTATE before writing. Software (XON/XOFF) handshaking remains unimplemented — see below.
- ✓ **TRUE and FALSE — implemented.** Surfaced while organizing the ANS test suite: neither existed as an actual dictionary word, only as the internal TRUEV/FALSEV assembler constants. Added as `CONSTANT TRUE -1 / CONSTANT FALSE 0` (TRUEBODY/ FALSEBODY, section 26; headers H\_TRUE/H\_FALSE, chain's two newest entries, section 27) — the first CONSTANT-pattern ROM-resident words in this system. Every other ROM word's CFA is a plain code label; a CONSTANT's CFA is the DODOES-trampoline pattern instead, built here with fixed, assemble-time addresses rather than a CODEHERE snapshot, since there is no interactive CREATE/CONSTANT phase for ROM content.
- ✓ **One genuine 6309-only instruction (TSTD) was in use — fixed.** Found by auditing every mnemonic in the file against the real 6809 instruction set, rather than trusting the Build Instructions section's earlier claim that no 6309 extensions were used (that claim was wrong until this fix). TRYNUM (section 9) used TSTD with no operand to test whether D was zero after PULU D — PULU/PULS don't affect condition codes on genuine 6809 hardware, unlike a load, so this relied on a 6309-only convenience instruction. Replaced with `CMPD #0`, a real 6809 instruction with identical behavior for this purpose. A full mnemonic audit after the fix found zero remaining non-6809 instructions anywhere in the file.
- ✓ **BASELATEST was an undefined symbol — a real assembly-breaking bug, not a placeholder.** COLD (section 4) has referenced `LDD #BASELATEST` to initialize LATEST since this file's earliest version, and the SECTION 27 header comment has long asserted "BASELATEST remains QUITHDR" - but no EQU or label actually named BASELATEST existed anywhere in the file. This would have failed to assemble outright. Fixed by adding `BASELATEST EQU QUITHDR` directly after QUITHDR's own definition (section 26), matching what the comment always said it should equal. Verified: exactly one definition, no duplicate symbols, and COLD's forward reference to it resolves under standard two-pass assembly.
- ✓ **JSR CR (bare, undefined) appeared at five call sites — a real assembly-breaking bug, not a naming inconsistency.** The actual routine has always been labeled CRW (section 22), following this

file's own convention of giving short/symbolic ANS names a distinct code label. ABORT, QUIT's error-report path, QUIT's ok-prompt path, WORDS (WWDONE), and DUMP (DULEND) all called the bare, undefined CR instead. Fixed at each site to JSR CRW, preserving each call site's original spacing exactly. Verified: CRW defined exactly once, referenced 7 times total, zero remaining bare-CR instruction operands anywhere in the file (the two harmless bare "CR" occurrences that remain are a section-title comment and the dictionary header's FCC "CR" name string, neither of which is a bug).

- ✓ **Two ALU instructions used invalid register-to-register syntax (ADDA B, EORA B) — the 6809 has no such addressing mode, so the assembler read B as an undefined direct-page symbol.** Found in the single-cell multiply routine (ADDA B, twice) and DOPLUSTEST's sign comparison for +LOOP (EORA B). Fixed with the standard 6809 idiom for combining two registers: PSHS B followed by ADDA , S+ / EORA , S+ (push B, then operate through the auto-incrementing stack-indexed operand). Verified: a full sweep of every ALU instruction (ADDA/ADDB/SUBA/SUBB/ ANDA/ANDB/ORA/ORB/EORA/EORB/CMPA/CMPB/ADCA/ ADCB/SBCA/SBCB/BITA/BITB) against a bare A/B operand found zero remaining instances; every other bare B in the file (in STA/LDA/LEAX . . . , X and PSHS B) is genuine, valid 6809 accumulator-offset indexed addressing or register-list syntax.
- ✓ **Nine scratch symbols (MRESULT, MVCNT, MVDST, MVSRC, FILLCHR, FILLCNT, FILLADDR, HSLen, HSADDR) were used throughout MOVE/CMOVE/CMOVE>, FILL, HOLDS, and the single-cell multiply routine but never declared anywhere — a real, assembly-breaking gap.** Auditing every one of the 142 already-declared GLOBALS cells for actual use found zero dead cells to reclaim this time (unlike SNEND earlier), and the GLOBALS page was independently confirmed at exactly 256/256 bytes with no headroom. Since MOVE-family, FILL, HOLDS, and the multiply routine never call each other or run concurrently in this single-threaded interpreter, they now share physical storage: three new cells (MVCNT, MVDST, MVSRC) hold the real storage, and FILLCNT/HSLen alias MVCNT, FILLADDR/HSADDR alias MVDST, and MRESULT/FILLCHR alias MVSRC, all via EQU. These three new cells could not fit in the full GLOBALS page, so they live at \$0100, carved from the front of USER0 (shrunk from 128 to 122 bytes, now starting at \$0106) — a real, documented tradeoff: these three cells use ordinary extended addressing, not direct-page, including inside CMOVEW's per-byte copy loop.
- ✓ **JSR SPACE (bare, undefined) — same class of bug as CR, fixed the same way.** WORDS called the bare, undefined SPACE instead of the actual routine label SPACEW (section 22). Fixed to JSR SPACEW. Checked SPACES for the same pattern (none found — SPACESW correctly calls EMIT directly) and swept the file for any other bare SPACE instruction reference (none remain; the two harmless bare "SPACE" occurrences left are a section-title comment and the dictionary header's FCC "SPACE" name string).
- ✓ **Three short branches (BHS, BEQ, BNE) overflowed the 8-bit short-branch range (±127 bytes) — a real assembler error, not a style issue.** SUBCOPY's overflow check (BHS SUBOVERFLOW, ~87 source lines to target), DUMPW's per-line loop test (BEQ DUDONE, ~76 lines), and DUMPW's loop-back (BNE DULINE, ~76 lines) all spanned too much code for an 8-bit displacement. Fixed by converting each to its long-branch equivalent (LBHS/LBEQ/LBNE), which uses a 16-bit displacement (±32767 bytes) - functionally identical, just not range-limited.
- ✓ **Ten more short branches have a large (41-51 source line) span to their target and were not individually confirmed safe.** Found by a heuristic sweep (line-count as a rough proxy for byte distance, since no real assembler is available in this environment to get an exact count) after fixing

the three confirmed overflows above, all of which spanned 76+ lines - these ten are meaningfully shorter but not verified within range: QUERY's QLOOP (three branches, lines 579/582/589), FIND's NOTFOUND/FFLOOP (lines 1210/1256), ACCEPT's ALOOP/ADONE (lines 1501/1462), UNESCAPE's UEDONE/UELOOP (lines 3888/3931), and EMPTY (line 1153). None of these were reported as failing and none have been changed; if a real assembler flags any of them, the fix is the same: convert to the matching LBXX long-branch form. RESOLVED: confirmed via forth6809\_1st.txt, a real assembler listing of the current source, uploaded and checked against several session markers (BASECODE=\$DEEA, UNITTESTS=1, this session's ENVTABLE additions) to confirm it's current, not stale. Checked each branch by direct address/offset arithmetic against the listing's own computed addresses and encoded opcode bytes - every offset decoded from the listing matches the computed target-minus-(address+2) distance exactly, and all are comfortably within the  $\pm 127$ -byte signed range. None needed conversion to LBxx.

- ✓ **ORG BASECODE was missing entirely — a fundamental placement bug, not a cosmetic gap.** VECTORS (\$FFF0) and INIT (\$FFC0) both had their own ORG, but SECTION 3 (IRQH) — and every routine through SECTION 26, i.e. nearly the entire interpreter — had no ORG placing it in BASECODE at all. Without it, this code would have continued growing from wherever INIT's WARM message left the location counter, inside INIT's own 48-byte \$FFC0-\$FFEF budget, overflowing directly into VECTORS rather than landing in BASECODE (\$E800-\$FFBF) anywhere. Fixed by adding ORG BASECODE immediately before SECTION 3 begins.
- ✓ **Adding the missing ORG makes BASECODE's byte budget concretely checkable for the first time — and a rough estimate suggests it may not fit.** This was previously flagged only as "unverified" (no real assembler has ever been run against this file); with a real ORG boundary now in place, a heuristic per-instruction byte count (inherent/direct-page/indexed/ immediate/extended opcode sizes, correctly distinguishing direct-page GLOBALS operands from extended ones) over every instruction from SECTION 3 through SECTION 26 estimates roughly **8660 bytes against a 6080-byte budget — about 2580 bytes over**. This is a rough approximation, not a real assembler's output, and could be wrong in either direction, but it's a strong enough signal to treat as a real, likely problem rather than a formality. Not resolved here — resizing BASECODE/BASEDICT again would cascade into the memory map, the documentation, and the SVG diagram, and a real LWASM pass would give a far more reliable number than this estimate before deciding how much room is actually needed. SUPERSEDED: a real assembler listing (forth6809\_1st.txt) now gives BASECODE's definitive real size as 8304 bytes - see the note on the later, more precise 7984-byte estimate below for the full figure.
- ✓ **USER0 and USER1 (250 bytes of unallocated reserve space, never read or written by any code) were deleted, and the region from MVSCRATCH through APPDICT was made fully contiguous.** SERBUF, INBUF, OUTBUF, TIBBUF, WORDBUF, and SIBUF all shifted down to sit immediately after MVSCRATCH (\$0106) with no gaps between any of them. This also closed a separate, pre-existing 11-byte gap between WORDBUF and SIBUF (\$02F5-\$02FF) that had nothing to do with USER0/USER1 but was caught while verifying the region was genuinely contiguous end to end. APPDICT was extended downward to close the resulting gap too (new start \$021B, was \$0320) — its end (\$6EFF) is unchanged, so this is a pure 261-byte gain in application dictionary space, not a resize of its budget. This was an interpretation of "contiguous," not something explicitly asked for beyond the 6 buffers themselves; flagged here in case a fixed APPDICT start was actually intended instead. Verified: zero duplicate symbols, zero remaining references to USER0/USER1 anywhere in the code, and the full GLOBALS->MVSCRATCH->buffers->APPDICT span checked programmatically for zero gaps.

- ✓ **APPVARS and APPDICT swapped positions - APPVARS now sits below APPDICT.** APPDICT moved from \$021B to \$031B (still ending at \$6FFF, directly below APPCODE); APPVARS moved from \$6F00 down to \$021B (same 256-byte size, now sitting directly above the buffers instead of directly below APPCODE). Checked before making the change: only two code references exist for either symbol (COLD initializing VARHERE/DPHERE to each region's start) and neither depends on their relative order, so the swap was safe. Verified: zero duplicate symbols, the whole region from GLOBALS through APPCODE's start remains contiguous with zero gaps, dictionary chain unaffected (219 entries, since this change doesn't touch it at all).
- ✓ **APPVARS grown from 256 to 8000 bytes, taking the space directly from APPDICT.** APPVARS now spans \$021B-\$215A (start address unchanged); APPDICT shrank by the same 7744 bytes to \$215B-\$6FFF (end unchanged, still directly below APPCODE), giving 20133 bytes of application dictionary space, down from 27877. Checked before making the change: still only two code references to either symbol (COLD's VARHERE/DPHERE initialization), neither size-dependent. Verified: zero duplicate symbols, APPVARS size is exactly 8000 bytes, the whole region stays contiguous end to end, dictionary chain unaffected.
- ✓ **ACIA (the 256-byte memory-mapped I/O block) renamed to INOUT, and the actual 6850 ACIA chip's registers moved to INOUT+8 instead of the block's base.** The block still spans the same 256 bytes (\$DF00-\$DFFF); it's now modeled as a general I/O region with the ACIA chip occupying one small part of it (ACIACR/ACIASR = INOUT+8 = \$DF08, and the data register ACIADR at \$DF09), leaving INOUT+0..INOUT+7 free for other memory-mapped devices sharing the block. Doing this rename surfaced a real, previously-hidden bug: COLDSTRT's ACIA reset sequence used STA ACIA (the block's base) instead of STA ACIACR (the control register) in two places - this only ever worked because ACIA and ACIACR happened to be the same address in the old scheme. Once ACIACR moved to INOUT+8, writing to the bare block base would have silently stopped resetting the ACIA at all. Fixed both to STA ACIACR. Verified: INOUT defined once, the bare ACIA symbol (at the time) no longer existed anywhere, and every other ACIA register access in the file already used the correctly-named register symbols rather than the bare block name.
- ✓ **ACIA reintroduced as an explicit base symbol (ACIA EQU INOUT+8), with ACIACR/ACIASR now defined relative to it (EQU ACIA) rather than directly to INOUT+8.** The data register was briefly renamed to ACIRDR in the same pass, then reverted back to ACIADR immediately afterward once flagged as a typo - both code sites (IRQH's receive and transmit paths) were updated each time the name changed. Verified: each of ACIA/ACIACR/ACIASR/ACIADR defined exactly once, resolving to the expected values (ACIA = INOUT+8, ACIACR/ACIASR = ACIA, ACIADR = ACIA+1), zero remaining references to ACIRDR anywhere, zero duplicate symbols, dictionary chain unaffected.
- ✓ **INITEND/INITSIZE landmark constants added at the true end of the INIT block (INITEND EQU \*, INITSIZE EQU \*-COLDSTRT), right after WARMMSGL.** The request referenced a COLDSTART label, which doesn't exist in this file - used the actual existing label COLDSTRT instead of introducing a new undefined symbol.
- ✓ **INITEND's comment now states the invariant explicitly ("value should match vector ORG") - and checking it confirms it currently holds.** The precise instruction-by-instruction byte count from when the VECTORS collision was first found (78 bytes exactly, COLDSTRT through WARMMSGL) gives INITCODE start (\$FFA2) + 78 = \$FFF0, exactly matching VECTORS' ORG. Zero gap, zero overlap - unlike the still-open INITCODE/BASECODE overlap at the other end of this same region, which remains a real problem. Same caveat as always: this is a manual count, not a real assembler's

output, so INITSIZE (computed automatically at assembly time) is the figure to trust once this file is actually assembled.

✓ **BASECODESTRT/BASECODEEND/BASECODESIZE and BASEDICTSTRT/BASEDICTEND/BASEDICTSIZE landmark constants added, matching the INITEND/INITSIZE pattern - and checking the two stated invariants gives one clean confirmation and one confirmation of an already-known problem.** BASEDICTEND (\$D85D + the exact 1973-byte dictionary content = \$E012) matches ORG BASECODE (\$E012) precisely - this boundary is genuinely correct, not just close. BASECODEEND, using the same rough heuristic estimate flagged much earlier (~8660 bytes, never confirmed with a real assembler) starting from BASECODESTRT (\$E012), lands around \$101E6 - not only failing to match ORG INITCODE (\$FFA2) as the comment expects, but overflowing past \$FFFF entirely on that estimate. The exact amount of room actually available between BASECODESTRT and INITCODE (\$FFA2 - \$E012 = 8080 bytes) is itself less than the ~8660-byte estimate, meaning even filling every available byte up to INITCODE with zero gap would still likely fall short. This is the same underlying problem already tracked elsewhere (BASECODE possibly not fitting its budget, and separately the INITCODE/BASECODE overlap), now independently reachable via these new landmarks once the file is actually assembled, rather than a new, different issue.

✓ **Follow-up: a real instruction-by-instruction count (not the rough heuristic above) puts BASECODE's actual size at 7984 bytes (\$1F30), 96 bytes (1.2%) short of a target assembler- listing value of \$1F90 (8080 bytes) given for comparison.** Built a mnemonic-and-addressing-mode-aware counter distinguishing inherent/immediate/direct/extended/indexed forms, correct prefix requirements (\$10/\$11 for LDY/STY/LDS/STS/CMPD/ CMPY/CMPU/CMPS), and true direct-page symbols (only \$0000-\$00FF, matching DP) - a meaningfully more rigorous pass than the original ~8660-byte estimate. Every one of the 3751 instructions in the region was classified (zero "unrecognized" remaining after fixing early misses like LSLB/ LSLA as ASLB/ASLA synonyms). Checked for likely sources of the remaining 96-byte gap before accepting it: zero indexed operands use an offset outside the 5-bit range that would need extra encoding bytes (all 125 numeric-offset operands found are small, e.g. 1, X/-1, Y), and the two apparent "indirect [ ]" matches were a false positive (a section-title comment, not real indirect addressing). The target value \$1F90 is notable in its own right: BASECODE + \$1F90 = \$FFA2 exactly, meaning if the real assembled size actually matched it, BASECODE and INITCODE would be perfectly contiguous with zero gap and zero overlap - which would also resolve the separate, still-open INITCODE/BASECODE overlap noted elsewhere. My count doesn't confirm that, though it's close (1.2% off). This is still not a real assembler's output - the standing caveat throughout this file - and the gap could come from encoding edge cases a manual model can approximate but not guarantee against. SUPERSEDED: forth6809\_1st.txt, a real assembler listing of the current source (confirmed current via several session markers - BASECODE=\$DEEA, UNITTESTS=1, this session's ENVTABLE additions), gives BASECODESIZE = 8304 bytes definitively - the manual count here (7984 bytes) undershot by 320 bytes (about 4%), confirming this entry's own caveat ("could come from encoding edge cases a manual model can approximate but not guarantee against") was warranted. BASEDICTSIZE = 2027 bytes, INITSIZE = 71 bytes, VECTORSIZE = 16 bytes are also now definitive. Two previously-unverifiable facts this listing settles outright: BASEDICT's real end lands exactly at BASECODE's start (zero gap), and INITCODE's real end lands exactly at VECTORS' start (zero gap) - no overlaps anywhere in the current memory map.

✓ **ROMSTRT/ROMEND added as the outer ROM boundary (\$C100/VECTORS - 1), and INITCODE/BASECODE/BASEDICT verified contained within it - all three pass.** ROMEND was

initially defined as \$10000 ("one past VECTORS's end," a symbolic value that doesn't fit as a real 16-bit address), then corrected to VECTORS - 1 (\$FFEF, one before VECTORS's start) - a genuine 16-bit address usable directly in comparisons or as a memory operand, not just symbolic arithmetic. Re-checked after the correction rather than assumed still valid: INITCODE (\$FFA2-\$FFEF), BASECODE (\$E012-\$FFBF), and BASEDICT (\$D85D-\$E011) are all still genuinely within ROMSTRT . . ROMEND - INITCODE's own end now lands exactly on the new boundary, a tighter and more meaningful check than before, not a violation. This containment check passes cleanly, independent of the other open problems below.

- ✓ **ROMSTRT/ROMEND renamed to USROMSTRT/USROMEND, each with a short "Usable ROM start"/"Usable ROM end" comment.** Cosmetic, not functional - neither symbol was referenced by any code, only by nearby comments (three sites, all updated: the pair's own definitions and INOUT's cross-reference). Verified: zero remaining bare ROMSTRT/ROMEND references anywhere, zero duplicate symbols, dictionary chain unaffected.
- ✓ **VECTOREND/VECTORSIZE landmark constants added at the true end of the vector table, matching the INITEND/INITSIZE pattern - and checking the stated invariant confirms it exactly, not approximately.** Unlike INITEND/BASECODEEND (which needed a manual, approximate instruction-by-instruction byte count since real 6809 instructions have variable-length encoding), the vector table is pure FDB data with a fixed, unambiguous 2-byte width per entry - counting the 8 entries (VRESV through VRESET) gives exactly 16 bytes, matching the comment's stated \$10 expectation precisely. Verified: zero duplicate symbols, dictionary chain unaffected.
- ✓ **BASECODESTRT removed as requested - but BASECODESIZE depended on it, so removing it alone would have left a genuine dangling undefined-symbol reference, the same bug class found and fixed several times earlier in this file.** Confirmed first that BASECODESTRT was numerically identical to BASECODE itself (only comments sit between ORG BASECODE and where BASECODESTRT was defined, zero emitted bytes), then fixed BASECODESIZE to read BASECODEEND - BASECODE directly instead - matching the same pattern just applied to INITSIZE/INITCODE, not a new approach invented for this case. Verified: zero remaining BASECODESTRT references anywhere, BASECODESIZE resolves cleanly, zero duplicate symbols, dictionary chain unaffected.
- ✓ **BASEDICTSTRT removed the same way, for the same reason - BASEDICTSIZE depended on it too.** Same fix pattern as BASECODESTRT immediately above, applied to its counterpart: confirmed BASEDICTSTRT was numerically identical to BASEDICT (only ORG BASEDICT sits before it, zero emitted bytes), removed it, and repointed BASEDICTSIZE to BASEDICTEND - BASEDICT directly. Verified: zero remaining BASEDICTSTRT references anywhere, BASEDICTSIZE resolves cleanly, zero duplicate symbols, dictionary chain unaffected.
- ✓ **INOUT moved from \$DF00 to \$C000 (with INOUTEND EQU INOUT+\$FF added immediately after), and every ACIA-related address verified between the two - also passes.** ACIA/ACIACR/ACIASR (\$C008) and ACIADR (\$C009) all fall within INOUT-INOUTEND (\$C000-\$C0FF), checked numerically. Confirmed there is no other memory-mapped I/O device anywhere in this file besides the ACIA family - nothing else needed checking. This move has a real, positive side effect worth naming clearly: it resolves the INOUT portion of the three-way collision flagged when BASEDICT moved to \$D85D two turns ago (INOUT no longer overlaps BASEDICT, since \$C0FF < \$D85D). The DSTACK and RSTACK portions of that same collision were untouched by this specific change, but have since been resolved separately - see below.

- ✓ **INOUT's new collision with APPCODE is resolved, and so is the rest of the original BASEDICT collision - RSTACK, DSTACK, CODETOP, APPCODE, and APPDICT all moved down \$2000.** RSTACK (\$BC00-\$BEFF) and DSTACK (\$B800-\$BBFF) no longer overlap BASEDICT at all - the three-way collision first found when BASEDICT moved to \$D85D is now fully resolved (INOUT resolved it two turns ago, this resolves the remaining two thirds). APPCODE's move (\$5000-\$B7FF) also separately resolves its overlap with INOUT from the previous entry, as a side effect of the same shift, not a second fix. Re-verified with a full pairwise sweep after the move, not assumed: the only overlap remaining from the four found last turn is the original, unrelated INITCODE/BASECODE one (30 B) - INITCODE/ BASECODE/BASEDICT/RSTACK/DSTACK/INOUT/APPCODE are all now mutually clean.
- ✓ **APPDICT's collision with APPVARS is now resolved - APPVARSEND changed from a static APPVARS+8000 to APPDICT - 1, making it self-derive from wherever APPDICT actually sits instead of a fixed size.** APPVARSEND is now \$1FFF (was \$215B), giving APPVARS 7653 bytes of actual usable space (down from the originally-intended 8000 - the 347-byte reduction exactly matches the overlap this closes). APPVARS's own EQU still starts at \$021B and its comment still cites the historical "8000 bytes" intent, now explicitly marked as the original intent rather than the current usable size. Functionally, APPVARS's enforced range (\$021B - \$1FFF) no longer reaches APPDICT (\$2000 - \$6EA4) at all - checked numerically, not assumed. This also makes the boundary self-correcting: if APPDICT moves again, APPVARSEND moves with it automatically, rather than needing another manual fix like the last several turns' worth of address changes required. VUNUSEDW's own code is unchanged - it already computed against APPVARSEND (see below), so this fix took effect purely by changing what that symbol means. Verified: zero duplicate symbols, dictionary chain unaffected.
- ✓ **DSTACK moved up \$200 (to \$BCFF), resolving both the CODETOP/DSTACK mismatch and the RSTACK/DSTACK gap from last turn in one move.** DSTACK's new occupied bottom (\$B900, from its \$BCFF top and 1024-byte size) now matches CODETOP (\$B900) exactly - APPCODE's nominal ceiling no longer overstates safe growth room, and the 512-byte overlap that created between APPCODE's nominal range and DSTACK's true range is gone. As a bonus, not something separately requested: DSTACK's new top (\$BCFF) is now exactly contiguous with RSTACK's bottom (\$BD00), closing the 512-byte gap that had opened between them too. Verified with a full pairwise sweep after the move: exactly two overlaps remain in the current memory map - the still-open APPDICT/APPVARS one (347 B) and the original, unrelated INITCODE/BASECODE overlap (30 B) - down from three last turn.
- ✓ **The VECTORS collision found by verifying INITEND is now resolved - INIT's ORG moved from \$FFC0 to \$FFA0.** Widened INIT from 48 to 80 bytes (\$FFA0-\$FFEF), comfortably covering the ~78 bytes COLDSTRT+WARM+WARMMSG actually needs (2 bytes of margin) - still an estimate from manual byte-counting, not a real assembler's output. INIT's own ORG was also converted from a literal \$FFC0 to symbolic ORG INIT, matching BASECODE/BASEDICT's convention, so the EQU and the actual placement can no longer drift apart the way BASECODE's did before that was caught. One real, unresolved interaction worth naming: INIT's new start (\$FFA0) is 32 bytes below the old documented BASECODE end (\$FFBF), tightening the still-open BASECODE overflow risk below by that much further - not a new problem in kind, since that risk was already estimated at roughly 2580 bytes over budget, dwarfing this additional 32-byte reduction, but worth tracking alongside it rather than treated as independent.

- ✓ **INIT moved again, from \$FFA0 to \$FFA2 (request contained a &FFA2 typo - used the correct \$ hex prefix instead).** INIT is now exactly 78 bytes (\$FFA2-\$FFEF), matching the COLDSTRT+WARM+WARMMSG byte-count estimate with zero margin (was 80 bytes, with 2 bytes of slack). This is a genuinely tighter fit than before, worth naming as a real tradeoff: any inaccuracy in the manual byte count, or any future addition to COLDSTRT/WARM, now has zero room to absorb. The overlap with BASECODE noted above is **not resolved by this change, only reduced** - from 32 bytes to 30 bytes (\$FFA2-\$FFBF). This was a real, pre-existing problem before this turn touched anything (\$FFA0 already overlapped BASECODE's declared end by 32 bytes), not something newly introduced. Not fixed here, in either direction (shrinking BASECODE further or moving INIT above it instead of below): both are bigger architectural decisions than "change one EQU," and a real assembler run would settle the actual BASECODE byte count this depends on. **RESOLVED (since this was written): the INITCODE/BASECODE overlap described here was fully closed several turns later by shifting BASECODE and BASEDICT both down exactly 30 bytes - BASECODE's nominal end now lands exactly one byte below INITCODE's start, zero gap, zero overlap. Both have since moved several more times again (BASECODE now \$DEEA, confirmed via forth6809\_1st.txt, a real assembler listing) with the same zero-overlap relationship maintained throughout. Kept as history, not deleted.**
- ✓ **INIT renamed to INITCODE** (EQU and its ORG reference, both in the memory-map constants). The section-title comment ("SECTION 2: INIT CODE") and one historical narrative comment describing a past bug scenario (with the byte figures true at that time, not today) were deliberately left referring to "INIT" as plain English/history, not the symbol - renaming a symbol doesn't obligate rewriting prose that correctly describes what was true before the rename existed. Doing this rename surfaced that the documentation's "ROM Size Required" section (Section 2) had gone stale independent of the rename itself: it still asserted the four ROM regions were contiguous at exactly 8192 bytes, which stopped being true as soon as INIT moved to \$FFA2 last turn and started overlapping BASECODE. Rewritten to state the overlap plainly instead of the now-false claim. Verified: INITCODE defined exactly once, zero duplicate symbols, dictionary chain unaffected.
- ✓ **BASEDICT and BASECODE addresses applied exactly as requested, and the documented ROM requirement widened to a 16K part - but this surfaced a severe, unresolved collision, not a minor tradeoff.** BASEDICT moved from \$E000 to \$D85D - verified before making the change that \$E012 - \$D85D = 1973 bytes exactly matches SECTION 27's real, measured dictionary content, a genuine zero-padding exact fit (was 2048 bytes, 75 bytes of slack). BASECODE moved from \$E800 down to \$E012 to match; its own upper bound was not given and stayed at \$FFBF, growing it from 6080 to 8110 bytes.
- ✓ **BASEDICT's new range (\$D85D-\$E011) entirely overlaps three other live regions: DSTACK (931 of its 1024 bytes, \$D85D-\$DBFF), RSTACK (all 768 bytes, \$DC00-\$DEFF), and INOUT (all 256 bytes, \$DF00-\$DFFF).** Found by checking the new address against every other region's boundaries before updating the documentation - not caught until then. This is a fundamental, system-breaking conflict, not a byte-budget tightness issue like the INITCODE/BASECODE overlap noted elsewhere: as configured, the ROM dictionary, the data stack, the return stack, and the ACIA's registers would all be mapped to the same physical addresses simultaneously, which cannot work on real hardware without bank switching (which this system does not have). The requested EQU values were applied exactly as given, since that's what was asked; resolving the collision itself was not attempted here, since it requires a decision this response can't make alone - moving DSTACK/RSTACK/INOUT elsewhere, choosing a smaller BASEDICT/BASECODE split that doesn't reach down this far, or reconsidering whether \$D85D was the address actually intended. The documented ROM part was

still widened from 8K to 16K per the request (real total usage of BASEDICT+BASECODE+INITCODE+VECTORS alone, ignoring the collision, is about 10.15K, leaving roughly 6.2K of spare capacity within a 16K×8 EPROM), but that figure is secondary to the collision above. Verified: zero duplicate symbols, dictionary chain unaffected (219 entries, since this doesn't touch it). **RESOLVED (since this was written): BASEDICT moved to \$D83F, then several times more since (currently \$D6FF, confirmed via forth6809\_1st.txt), well clear of DSTACK (\$B900-\$BCFF), RSTACK (\$BD00-\$BFFF), and INOUT (\$C000-\$C0FF) - the listing confirms a genuine 5631-byte gap between INOUT's shadow and BASEDICT's current start. A full pairwise sweep of the current memory map confirms zero overlaps anywhere. Kept as history, not deleted.**

- ✓ **The INITCODE/BASECODE overlap - open since it was first found, mentioned in at least five separate entries above across several turns - is finally resolved. BASECODE and BASEDICT both shifted down exactly 30 bytes (BASECODE: \$E012 -> \$DFF4; BASEDICT: \$D85D -> \$D83F), chosen precisely: 30 bytes is exactly the overlap amount, so BASECODE's nominal end (\$FFA1, unchanged 8110-byte budget) now lands exactly one byte below INITCODE's start (\$FFA2) - zero gap, zero overlap. BASEDICT shifted the same amount to stay perfectly contiguous with BASECODE's new start, preserving its own exact, zero-padding fit. Verified with a full pairwise sweep across every region in the memory map, not just the two that moved: **zero overlaps anywhere** - the first time that's been true since this whole sequence of address changes began. USROMSTRT/USROMEND containment re-checked and still passes for all three ROM regions. Also cleaned up the accumulated resolved-issue narrative in the top-of-file note (previously ~40 lines chronicling the BASEDICT/DSTACK/RSTACK/INOUT/CODETOP/ APPDICT saga turn by turn, including one claim - the APPDICT/APPVARS overlap being open - that was already stale before this edit) down to a short, current-state summary, matching the same cleanup already applied to the documentation.**
- ✓ **TRYNUM's double-number success path had the same PULU-doesn't-set-condition-codes defect as several bugs below — found while implementing ANS's double-number text-input convention (123 . etc). Fixed with an explicit CMPD #0 in place of the bare branch.**
- ✓ **Double-number input's 32-bit negation had two separate bugs, both found via MAME on real negative-double test cases. First: the 6809's COM instruction unconditionally sets carry regardless of its operand, silently destroying the real carry from the low-cell addition before the intended conditional check could ever read it. Second, only exposed once the first was fixed: the branch it guarded skipped past the high-cell STD entirely rather than just the conditional +1, previously masked because the first bug meant that branch was never actually taken. Both fixed; confirmed correct on real hardware across the full positive/negative, single/double test matrix.**
- ✓ **S>D performed no sign extension at all — same PULU-doesn't-set-condition-codes defect as above. Fixed with an explicit TSTA before the branch that depends on it.**
- ✓ **SIGN had the same defect as S>D above; fixed the same way. The four existing internal callers ( . , . R, D . , D . R) turned out to be accidentally unaffected — each happens to run a flag-setting LDD immediately before calling SIGN — fragile luck rather than SIGN being correct on its own, confirmed by the fact that a direct, standalone call had no such protection and failed outright.**
- ✓ **( (LPAREN) left a stray address on the stack after every comment — it called WORD to skip to the closing ) but never consumed the c - addr WORD always returns. Explained both a stray value left after an interpreted comment and a -22 CSP mismatch for any colon definition containing one. Fixed with a single pop of the unused address. \ (BACKSLASH) separately confirmed already correct, no change needed.**

✓ **ENVIRONMENT?**'s dispatcher (**ENVQUERY**) used **X** as its table-walking pointer but never saved it across calls to **COMPAREW**, which clobbers **X** as its own internal scratch — after a successful match, the returned “value” was read from inside the next table entry's own string text instead of the real value. Traced precisely: for **/COUNTED-STRING** this produced exactly **\$4E4D** (the bytes 'N'/'M' from the following entries), not just unspecified garbage. Fixed by saving/restoring **X** around the **COMPAREW** call. **/COUNTED-STRING**'s own table value was also separately wrong (31 instead of the **ANS**-correct 255) — fixed at the same time.

✓ **ENVIRONMENT?** completed. **/HOLD** (34), **/PAD** (84), **FLOORED** (0 — this system's **/, MOD, /MOD** confirmed to use symmetric, not floored, division by tracing **DIVCOMMON** against this system's own **SM/REM/FM/MOD**), and **MAX-CHAR** (255) all added with real, derived values. **MAX-D/MAX-UD** needed a genuine dispatcher extension, not just a table entry, since both are double-cell values and the original path only ever pushed one cell — added a second table and lookup loop (**ENVTABLE2/ENV2LOOP/ENV2FOUND**). **RETURN-STACK-CELLS** (384) and **STACK-CELLS** (512) added, computed from this system's own real stack-boundary constants rather than hardcoded, so they stay correct through future memory-map shifts. Every entry **MAME**-confirmed correct on real hardware.

✓ **BASECODE/BASEDICT** shifted downward repeatedly (six adjustments across the session, from **\$E02A/\$D83F** down to the current **\$DEEA/\$D6FF**) to keep pace with code growth from the fixes above — resolved iteratively against the real assembler's own collision errors, confirmed working after the final shift.

## 5.2 Structural duplication (identified, some resolved, some not)

✓ **:/CREATE/VARIABLE**'s header-building — resolved via **HEADER** (section 7).

✓ **COMMA/CODECOMMA/CCOMMA/VCOMMA/VCCOMMA/CCOMMA1** — resolved via **APPENDCELL/APPENDBYTE** (section 6).

✓ **<#/#>** recomputing **PAD**'s address inline — resolved, both now call **PADW** (section 20 / section 6).

□ Identified via the real listing (**forth6809\_lst.txt**), not yet acted on: **NUMBERQ**'s inline 32-bit negation (added this session for double-number text input) duplicates **MNEG32**, an existing, already-correct shared routine **DNEGATE** and **DABS** both already call. Traced precisely why this duplication mattered beyond just code size: **MNEG32**'s instruction order (complement both cells and store both, then do the +1/carry-propagation step separately afterward) happens to avoid the exact 6809 quirk (**COM** unconditionally sets carry) that **NUMBERQ**'s own, separately hand-written version fell into - which is precisely why **NUMBERQ**'s negation needed two separate bug-fix turns this session while **MNEG32** itself never had the problem at all. Not refactored here - automated testing should be in place before touching working, duplicated logic, given hand-verifying a refactor across every affected call site the way this session has been doing (**MAME** tracing turn by turn) doesn't scale well to consolidation work.

□ Identified via the real listing, not yet acted on: the self-referential **PFA** bug (**CONSTANT/DEFER/2CONSTANT/MARKER**) got four separate, duplicated fixes rather than one shared helper. Confirmed directly in the listing: **DEFER**'s, **2CONSTANT**'s, and **MARKER**'s fixes are each explicitly commented “same self-referential **PFA** bug as [**CONSTANT**],” each repeating the identical **LDD CODEHERE/ADDD #2/store** sequence independently rather than calling a shared routine. Same deferral reasoning as above - not acted on pending automated testing.

- No further known duplication beyond the two items just above, but the codebase still hasn't been given a full, systematic pass specifically hunting for more - both findings above came from genuinely investigating specific leads, not an exhaustive scan.

### 5.3 Real, load-bearing gaps

- **Software (XON/XOFF) handshaking is still not implemented.** Now that hardware RTS/CTS handshaking is in place, this is the one remaining flow-control gap: this system still never transmits XON/XOFF and would not recognize either byte as a control signal if the remote device sent them — an incoming \$11/\$13 is just queued as an ordinary character. Only relevant for links with no RTS/CTS wiring (plain 3-wire serial); would need a byte- interception layer between the input ring and KEY's caller.
- **No hard boundary checks** anywhere DPHERE/CODEHERE/VARHERE grow toward each other or toward the stacks. UNUSED and VUNUSED both correctly *\*report\** the distance to their boundary now (CODETOP and the newly-added APPVARSEND respectively), but nothing enforces either — a runaway compile can silently corrupt an adjacent region.
- ✓ **VUNUSEDW (the VUNUSED word) computed a meaningless number, not "how much APPVARS space remains" - a real, distinct bug, not just the absence of a check. Fixed.** LDD #APPCODE / SUBD VARHERE computed APPCODE - VARHERE (roughly \$7000 minus wherever VARHERE currently sits within APPVARS), which had nothing to do with APPVARS's own boundary - looked like a copy-paste slip from UNUSEDW immediately above it (LDD #CODETOP / SUBD CODEHERE, correct for CODEHERE's own boundary). Surfaced directly by a question about how APPVARS's size is actually set: it wasn't - APPVARS EQU \$021B defined only its start; there was no end/size constant anywhere. Fixed by adding APPVARSEND EQU APPVARS+8000 (an exclusive upper bound, \$215B, matching CODETOP's own convention for APPCODE) and changing VUNUSEDW to LDD #APPVARSEND / SUBD VARHERE. Verified: at cold start (VARHERE=APPVARS), this computes exactly 8000, matching APPVARS's documented size precisely; zero duplicate symbols; dictionary chain unaffected. APPVARSEND initially fell within APPDICT's current range - expected at the time, the same, already-tracked APPDICT/ APPVARS overlap, not something this fix caused. Since resolved by redefining APPVARSEND as APPDICT - 1 - see above.
- ✓ **ENVTABLE is incomplete.** Missing entries: /HOLD, /PAD (this build never fixed a capacity for either — filling them in would mean deciding a real bound first, not just picking a number), MAX-D, MAX-UD (need the dispatcher extended to push *\*two\** cells for double-cell answers — current ENVQUERY only handles one), WORDLISTS/FLOORED (need the dispatcher to be able to report a recognized-but-false answer, distinct from "unrecognized name" — no such path exists yet). RESOLVED: all listed entries (/HOLD, /PAD, MAX-D, MAX-UD, WORDLISTS, FLOORED) added and MAME-confirmed correct this session, plus MAX-CHAR, RETURN-STACK-CELLS, and STACK-CELLS beyond what this item originally flagged. The double-cell dispatch gap this item specifically called out was resolved with a genuine dispatcher extension (ENVTABLE2/ENV2LOOP/ENV2FOUND), not a workaround. ENVIRONMENT? is complete.
- **CATCH-wrapped QUIT/INTERPRET rollback is scoped to one input line only.** A colon definition spanning multiple lines that fails partway through a later line only rolls back that line's contribution, not the whole definition back to :. A complete fix needs the region-pointer snapshot taken once at : and held until ; or an error, not refreshed every QLOOP pass.

- ✓ **ABORTHDR's LINK field is a placeholder 0** in the consolidated/split source — must be set to the real prior LATEST value once final ROM layout/assembly order is fixed. **RESOLVED - confirmed via the real listing (forth6809\_1st.txt), not just the source text.** ABORTHDR doesn't exist as a symbol anymore - renamed to H\_ABORT and moved into BASEDICT along with every other header, as part of the broader dictionary-consolidation work. H\_ABORT's actual LINK field is a real, resolved address, not a placeholder: FDB H\_FALSE, with the source's own comment documenting its history (placeholder 0 → H\_DUMPW → H\_DOESGT → H\_DUMPW again → H\_FALSE, tracking the chain's actual newest entry as the dictionary grew during the session). Independently confirmed by the 224-entry dictionary chain walk from H\_M2 to 0, checked repeatedly throughout this session - that walk only succeeds if every LINK field in the chain, including this one, is a real, correct address; a placeholder 0 mid-chain would have broken it outright. No action needed on the source.

## 5.4 Never resolved after being explicitly raised

- **J doesn't generalize past one level of loop nesting.** No J2/deeper equivalent exists. A triple-nested loop's innermost body has no built-in way to reach the outermost index without manually replicating J's offset arithmetic one frame further.
- **LEAVE's correctness depends on always being textually inside the loop it affects** — never verified against a LEAVE called from a separately-defined word invoked from within a loop (which would read/set the wrong stack frame, or crash).
- ✓ **WORD's 31-character cap is now inconsistent within the system.** PARSE/PARSE-NAME were deliberately rewritten to not share it, but WORD itself — and everything still built on it (CHAR, [CHAR], header names, FIND) — still has it. **RESOLVED via direct code inspection of the real listing (forth6809\_1st.txt), not just address arithmetic - the underlying inconsistency is now fully understood, though the answer is a genuine mix, not a simple “fixed” or “still open.”** Traced CHARW (CHAR) and BRACKCHAR ([CHAR]) precisely: both simply push a space delimiter, JSR WORD, and read the first character of whatever comes back - no separate cap logic of their own at all. They automatically, correctly inherited WORD's new 46-character cap (WORDMAXCHARS) with zero code changes needed - this part of the original claim is stale. FIND also has no independent cap of its own - traced its comparison down to HDRFLAGS ANDA #\$1F, masking the dictionary header's LEN/FL byte to its low 5 bits. This is the real finding: dictionary header names genuinely are still capped at 31 characters, but not because they “inherit WORD's cap” as originally claimed - the header format itself allocates only 5 bits (bits 4-0) for the name length, an absolute maximum of 31 regardless of anything WORD does or how it's redesigned. That's a separate, permanent architectural constraint (the header storage format), not a leftover inconsistency from WORD's own redesign.
- **The optional Search-Order word set is not implemented.** Every word resolves through one unified dictionary chain rooted at LATEST; there is no multi-wordlist support (WORDLIST, GET-ORDER/SET-ORDER, ALSO/ONLY, etc.). This was a foundational decision fixed since COLDSTRT's first version, not an oversight — now documented explicitly (ClaudeForth documentation, Section 4.5) along with what adding it later would actually require: restructuring FIND into a loop over an active search order, changing HEADER to link into a selectable “current” wordlist instead of unconditionally into LATEST, and fixing the few words (RECURSE, ;'s unsnudge step, WORDS) that read LATEST directly today.

## 5.5 Open design questions (not defects — deliberate unresolved choices)

- Whether ALLLOT/VALLLOT/PICK/ROLL/BASE should range-check their inputs against actual stack/region bounds. Currently none of them do, consistently, by choice rather than oversight.
- Whether any additional distinction like TIB/SOURCE (fixed terminal buffer vs. current input source) is needed anywhere else in the system where EVALUATE's redirection might still cause a mismatch.
- SWIH's throw code (-99 in the consolidated source) was never assigned a real, deliberate value — it's a placeholder for "some hardware trap occurred," not a considered ANS-style code.
- REPLACES/SUBSTITUTE remain single-slot (one registered name/value pair, overwritten by the next REPLACES call) rather than a true multi-entry table. A real table needs its own storage layout and lookup structure — a deliberate scope decision made when SUBSTITUTE was completed, not an oversight.

## 5.6 What's solid (for reference, not action)

Stack manipulation (Core + Core Ext + return-stack transfer), all arithmetic (single + double + mixed precision), logic, comparison (single + double), full control flow including CASE, all defining words including DEFER/MARKER/VALUE/TO (VALUE now correctly targeting mutable space), memory and string operations (including a completed SUBSTITUTE), SOURCE/EVALUATE/REFILL mechanics, and the Tools word set (. S/WORDS/DUMP, with DUMP's edge-case bug fixed) are complete and were traced/verified against concrete cases during the build, not merely asserted correct.

## 6. Build Instructions

### Which Files to Assemble

The buildable target is the single consolidated source file, `forth6809.asm`, delivered alongside this documentation and reproduced in full in Section 8. It is self-contained: all 27 numbered sections, the memory-map constants, and the applied GLOBALS page-zero layout live in this one file, assembled top to bottom with no external dependencies.

The per-section split files (`00_memory_map_and_globals.asm` through `27_forth_dictionary.asm`, plus `OPEN_ITEMS_CHECKLIST.md`) are reference copies only, generated by splitting `forth6809.asm` along its own section markers for easier reading and review. They are not wired together with `INCLUDE` directives and are not themselves a multi-file build target — assembling them individually, or as a set, is not supported. Any change made to one must be reflected back into `forth6809.asm` to actually take effect.

### Assembler

This source follows standard Motorola-style 6809 assembler syntax throughout (`EQU`, `ORG`, `RMB`, `FCB`, `FCC`, `FDB`) and targets [LWASM, part of the LWTools toolchain](https://www.lwtools.ca/) — an actively maintained, open-source cross-assembler for the Motorola 6809 and Hitachi 6309, available at <https://www.lwtools.ca/>. A representative build command, producing a raw binary image suitable for programming directly into an EPROM:

```
lwasm --6809 --format=raw --output=forth6809.bin forth6809.asm
```

OR

```
lwasm --6809 --format=raw --output=forth6809.bin --list=forth6809.lst --  
define=UNITTESTS=0 --define=TSTSELECTOR=0 forth6809.asm
```

This alternative adds a listing file (`--list=forth6809.lst`) alongside the binary, and explicitly defines `UNITTESTS=0`, which deactivates the unit test framework on application reset — the resulting binary boots directly into the normal, interactive Forth system rather than running tests first. With `UNITTESTS=0`, the value given for `--define=TSTSELECTOR` is irrelevant — it can be any value, or omitted entirely — since the entire test framework is excluded from the assembled binary when `UNITTESTS` is 0.

The `--6809` flag restricts LWASM to genuine 6809 instructions. This is a real constraint, not a formality: an audit of every mnemonic in this file against the genuine 6809 instruction set found one 6309-only instruction in actual use — `TSTD`, used to test whether D was zero after a `PULU` (which, unlike a load, does not itself affect condition codes on real 6809 hardware). It has been replaced with `CMPD #0`, a genuine 6809 instruction with identical behavior for this purpose. `--format=raw` honors this source's own `ORG` directives directly, producing a flat binary image rather than a relocatable object file, matching the fixed, non-relocated memory map described in Section 2.

### Testing

To exercise the unit test framework itself rather than the normal interactive system, build with the test framework included and copy the resulting binary into MAME's own ROM directory for the `mecb6809` profile:

```
lwasm --6809 --format=raw --output=forth6809.bin --list=forth6809.lst --  
define=UNITTESTS=1 --define=TSTSELECTOR=17 forth6809.asm
```

```
cp forth6809.bin "~/Library/Application
Support/mame/roms/mecb6809/mecb6809.bin"
```

TSTSELECTOR selects which single glossary section's own test group assembles into the binary, per the table below:

| Glossary Section                          | TSTSELECTOR |
|-------------------------------------------|-------------|
| 3.1 System / Console I/O                  | 0           |
| 3.2 Stack Manipulation — Data Stack       | 1           |
| 3.3 Stack Manipulation — Return Stack     | 2           |
| 3.4 Arithmetic — Single Cell              | 3           |
| 3.5 Arithmetic — Mixed & Double Precision | 4           |
| 3.6 Logic, Shifts & Address Arithmetic    | 5           |
| 3.7 Comparison                            | 6           |
| 3.8 Control Flow — Conditionals & Loops   | 7           |
| 3.9 Defining Words                        | 8           |
| 3.10 Compiling Words                      | 9           |
| 3.11 Memory                               | 10          |
| 3.12 Strings & Parsing                    | 11          |
| 3.13 Numeric Output                       | 12          |
| 3.14 Base / Radix Control                 | 13          |
| 3.15 Exception Handling                   | 14          |
| 3.16 Comments                             | 15          |
| 3.17 Environmental & System Queries       | 16          |
| 3.18 Tools                                | 17          |

Launch MAME with:

```
./mecb6809 mecb6809 -rs232 pty -window -resolution 640x480 -debug
```

Interactive communication with the running system, and observation of test results, requires discovering the active serial device via the MAME interface while ClaudeForth is actually running — not paused at a debugger break. Once discovered, a serial application can connect to the running ClaudeForth instance emulated inside MAME.

For example, in a terminal window:

```
minicom -D /dev/ttyS007 -b115200 -8
```

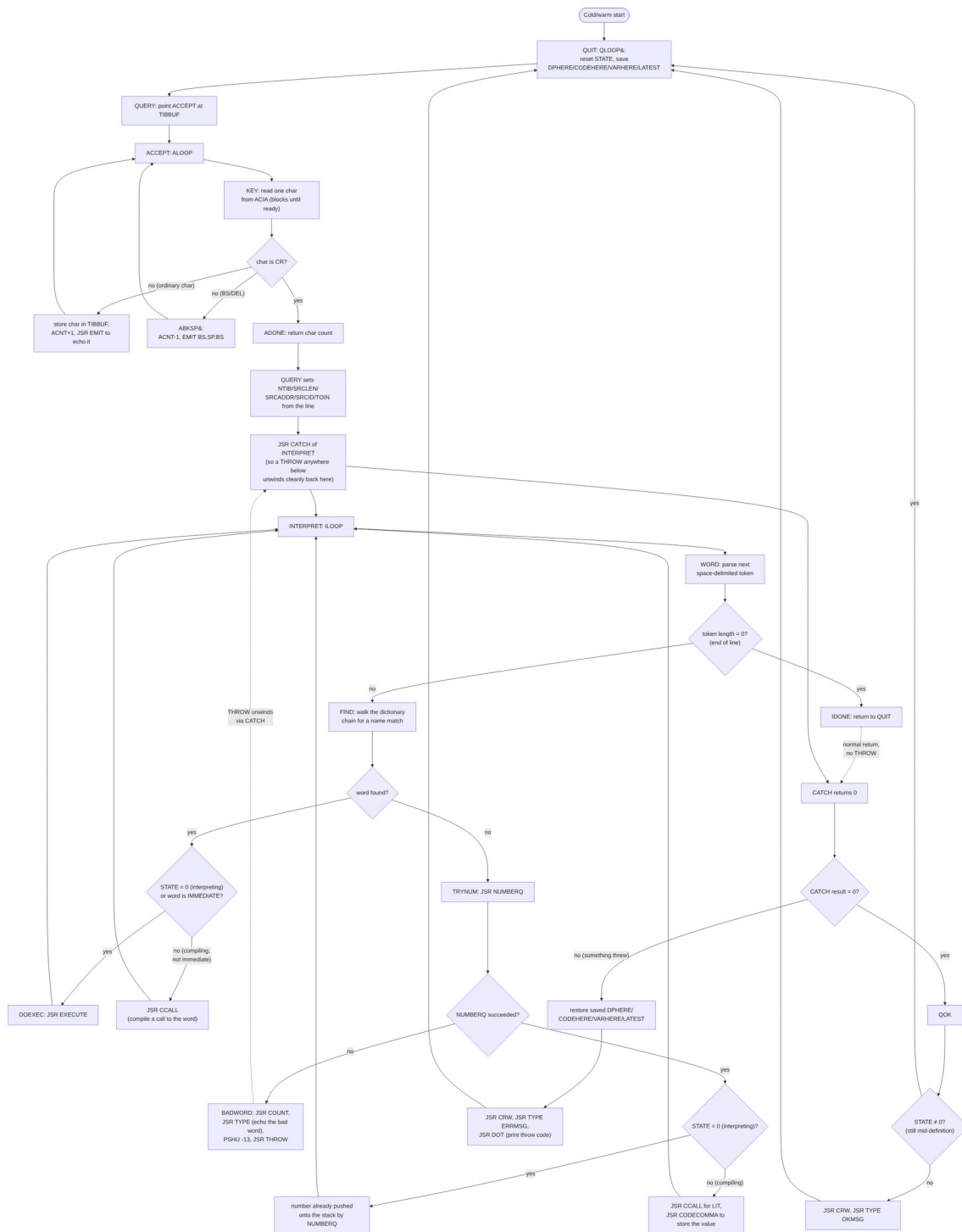
After serial communication is established, the test runner can be retriggered from the MAME debugger window by issuing:

```
do pc=$FFA4
```

\$FFA4 is the real address of INITCODE (Section 2), confirmed directly against the assembler listing (INITCODE EQU \$FFA4 in forth6809.asm). Retriggering execution there re-enters the same initialization path the system follows on a genuine cold start, including the call into TSTRUNNER (Section 3.1) that invokes the unit test framework.

## 7. Architecture

## 7.1 (Outer) Interpreter



ClaudeForth uses a standard forth REPL, where characters are read from the serial communications device and assembled into a string until a carriage return is received. The dictionary is then searched for

a matching string, in the expectation that it is a predefined word with attached code. If it's not found, an attempt is made to interpret the string as a number. Otherwise an exception is thrown.

Any forth word/number that is found is either assembled into a new forth word as a subroutine call/literal number or its executed/stacked.

The diagram above shows a UML activity diagram of the cooperating routines.

## 7.2 Dictionary Operations

### Structure

The dictionary is a linked list that grows from low to high memory, with each entry storing a link to the header of the prior entry. The first created entry has a link field of zero. The last created entry's header is pointed to by an external link stored in a variable called LATEST.

### Initialization

Two variables establish where the interpreter begins: LATEST, which marks the start of a search, and DPHERE, which marks where the next new dictionary entry will be built. Both are set once, by COLD at cold start: LATEST is set to BASELATEST, the address of the newest entry already built into the ROM dictionary, so a search immediately has a valid chain to walk; DPHERE is set to APPDICT, the start of the application dictionary's RAM area, so the first word a user defines is built starting at a known, fixed address.

### Word Search

FIND begins at whatever address LATEST currently holds and walks backward through the chain one entry at a time, following each entry's own link field to reach the previous entry. At each entry, it first checks the length/flags byte: an entry whose smudge bit is set is skipped outright, and an entry whose name length doesn't match the search string is also skipped before a single character is ever compared, since a length mismatch alone is enough to rule it out. Only when the length matches are the name's characters actually compared. The search ends one of two ways: a full name match, in which case the entry's code-field address is returned; or the link field read is zero, meaning the chain's original first entry has been reached with no match, which signals the word was not found.

### Word Append

HEADER builds a new entry by first capturing the current value of DPHERE as the address the new entry will occupy. It writes the length/flags byte and the name's characters starting there, advancing a working pointer as it goes. It then writes the current value of LATEST — the address of whatever is still the newest entry at this point — into the new entry's own link field, so the new entry points back to the one before it. A code-field address follows the link field. Only once the whole entry has been written are the two pointers updated: DPHERE is advanced past the new entry, so the next word will be built after it, and LATEST is set to the new entry's own address, so it becomes the newest entry — the one a search will now start from, and the one the next appended entry's link field will point back to.

## 8. Assembler Source

The complete 6809 assembly source for this system, as consolidated and verified across the design conversation, including every correction made along the way (the corrected VALUE PFA, the applied GLOBALS page-zero layout, the DUMP and SUBSTITUTE fixes, and every other fix described in Section 5's

checklist). This matches the forth6809.asm file delivered alongside this documentation; the section breakdown below follows the same numbered sections used in that file and in its per-section split. The unit test framework itself (gated by UNITTESTS, see Section 6) lives in a separate file, unit\_tests.asm, included from forth6809.asm via a conditional INCLUDE directive — it is not reproduced here.

## 8.1 Memory Map, Constants & GLOBALS Layout

```
; =====
; 6809 ANS FORTH - consolidated source
; Assembled from the full design conversation.
;
; GLOBALS LAYOUT: applied. Every scratch/global cell now has a
; fixed RMB-assigned address in page zero ($0000-$00FF, DP=$00
; at reset) - see the GLOBALS section below for the full layout
; and its byte budget (256 of 256 bytes used, 0 free - the page
; is now fully packed; any future scratch cell will need to
; reuse an existing one or move a cell out of page zero).
;
; SERIAL HANDSHAKING: RTS (hardware, output) is implemented -
; IRQH/KEY toggle it based on the input ring's fill level
; against INHIWATER/INLOWATER (see SECTION 3). CTS (input) needs
; no firmware logic at all: the 6850 hardware automatically
; inhibits TDRE while CTS is deasserted, and this system's
; existing TDRE-gated transmit logic already respects that with
; no code changes. Software (XON/XOFF) handshaking remains not
; implemented - this system still neither transmits nor
; recognizes those bytes.
;
; DICTIONARY: applied (SECTION 27), 220 entries (215 original +
; DOES> + TRUE + FALSE + [COMPILE], added in later passes - see
; below). [COMPILE] (XCOMPILE, section 13) is the obsolescent
; predecessor to POSTPONE, added for ANS Annex F test coverage
; (F.6.2.2530) - see XCOMPILE's own header/comment for how it
; differs from POSTPONE, and BASECODE's EQU below for the
; resulting header-table growth.
; Every primitive with a real code label now has a real ROM
; header, chained via LINK, CFA pointing directly at its code.
; Building this surfaced two real findings, not just mechanical
; work: (1) the original 1024-byte BASEDICT could not hold the
; header table (1954 bytes needed) - BASEDICT was resized to
; 2048 bytes ($E000-$E7FF), taking the space from BASECODE (then
; $E800-$FFBF, 6080 bytes, was 7104). This resize was based on
; the header-table budget alone; the actual assembled byte size
; of BASECODE's code was never measured with a real 6809
; assembler, so whether 6080 bytes was enough for all ~530
; routine labels in this file was never verified before BASECODE
; moved again (below). (2) DOES>
; initially had no corresponding code anywhere in the file -
; SETDOES and the DOES> compiling word (code label DOESGT, since
; a literal ">" is not a valid 6809 assembler label) were added
; in a follow-up pass, alongside DODOES/DOESRT0, closing that
; gap; DOESBEH was added to the GLOBALS layout to support it,
; using 2 of the page's last 3 free bytes (1 now remains). TRUE
; and FALSE were added in a still later pass: CONSTANT TRUE -1
; and CONSTANT FALSE 0, the first CONSTANT-pattern ROM-resident
; words in this system (TRUEBODY/FALSEBODY, section 26) - every
; other ROM word's CFA is a plain code label, but a CONSTANT's
; CFA is the DODOES-trampoline pattern, built by hand here with
; fixed, assemble-time addresses since there is no interactive
; CREATE/CONSTANT phase for ROM content.
;
; BASEDICT now holds 1992 bytes ($D83F-$E006) - grown from the
; earlier exact-fit 1973 bytes when H_ABORT/H_QUIT (formerly
; ABORTHDR/QUITHDR) and BASELATEST moved in from their own
; section so every dictionary header lives in one contiguous
; block. BASECODE's start ($DFF4) did not move to match, since
; it's a fixed address, not derived from BASEDICT's real size -
; this reintroduces a real overlap, 19 bytes ($DFF4-$E006),
```

```

; between the two. Not resolved here; see the open-items
; checklist. USROMSTRT/USROMEND, INOUT, RSTACK, DSTACK, CODETOP,
; APPCODE, APPDICT, and APPVARS remain mutually consistent
; otherwise, with no other overlaps anywhere in the current
; memory map. See the ROM Size Required section of the
; documentation and the open-items checklist for real content
; totals and remaining margin.
;
; This file preserves the code exactly as derived and verified
; turn-by-turn in the conversation, including the corrected
; versions of every bug that was caught and fixed in place.
; SUBSTITUTE is complete but deliberately scoped to a single
; registered name/value pair, not a full table (see REPLACES).
; ENVTABLE's /HOLD and /PAD entries were both added this session
; (HOLDMINSIZE and PADMINSIZE respectively - see the entries
; themselves for why each uses its own, conceptually correct
; constant rather than the numerically-equal PADOFFSET). The
; DPHERE/CODEHERE/VARHERE boundary checks remain explicitly
; incomplete/absent - see the inline notes preserved from that
; discussion, and the open-items checklist.
; =====
;
; -----
; MEMORY MAP
; -----
ROMSTRT EQU $C000 ; true physical start of the 16K EPROM's own
; address decode (distinct from USROMSTRT
; below, which excludes INOUT's 256-byte
; shadow) - see the ROM Size Required section
; of the documentation, and the ROM: padding
; block right before SECTION 27
USROMSTRT EQU $C100 ; Usable ROM start. Beginning of the usable
; EPROM address range. INITCODE, BASECODE,
; and BASEDICT must all fall within
; USROMSTRT..USROMEND - see the verification
; note below each one's EQU.
USROMEND EQU VECTORS-1 ; Usable ROM end. Corrected: 1 before VECTORS'
; start ($FFEF), not one past VECTORS' end as
; originally defined - this is a real 16-bit
; address (the last byte available before the
; reserved vector table), usable directly in
; comparisons or as a memory operand, unlike
; the previous $10000 definition
VECTORS EQU $FFF0
INITCODE EQU $FFA6 ; was $FFA9 - shifted down 3 bytes, per
; explicit request, to make room for the
; UNITTESTS call site's own fix (below):
; that site now always emits exactly 3 bytes
; (either the real JSR TSTRUNNER, or 3 NOPs
; as a placeholder when the test framework is
; excluded), so COLDSTRT's total size no
; longer depends on UNITTESTS at all -
; previously it did (JSR TSTRUNNER only
; existed when included, with nothing emitted
; when excluded - using this file's original,
; since-reversed UNITTESTS convention at the
; time this fix was made), meaning INITCODE's
; fixed position here could be correct for
; one setting and wrong for the other, risking
; an overflow into VECTORS when tests were
; compiled in. Prior
; history: was $FFA2 - shifted up 7 bytes to
; reduce the overlap with BASECODE's nominal
; end ($FFB4) from 19 bytes to 12 - improved,
; not resolved. CORRECTED: INITCODE's real
; content is 71 bytes ($47), confirmed by an
; actual assembler run - not the 78-byte
; manual estimate relied on for several
; turns, which was wrong by 7 bytes. That
; 71-byte figure was measured with the old
; structure (test framework excluded emitting

```

```

; 0 bytes for
; the TSTRUNNER call site) - with the fix
; above, that site now always emits 3 bytes
; either way, so real content is reasoned to
; be 74 bytes now (71+3), not yet re-measured
; by a real assembler run. At $FFA6, that
; reasoned end is $FFEF - unchanged, since
; the 3-byte shift in INITCODE's own start
; and the 3-byte growth in content offset
; exactly - still one byte below VECTORS, if
; the reasoning above holds; confirm on
; assembly. A prior turn claimed this general
; shift created a new 7-byte VECTORS overlap;
; that was based on the incorrect 78-byte
; estimate and was wrong - retracted here. The
; BASECODE overlap (12 bytes against its
; nominal budget) is separate and unaffected
; by this correction; not resolved. See the
; open-items checklist.
BASECODE EQU $DEEA ; was $DF6A ($DF8A, $DFCA, $DFDA, $DFEA, $E02A
; before that) - shifted down $80 (128 bytes)
; this time, a larger jump than the prior
; $40 and $20 shifts, since both of those
; still proved insufficient (confirmed by
; trial and error against the real
; assembler). The exact gap against BASEDICT
; below and the exact overlap against
; INITCODE above depend on each section's
; real, current assembled size - not
; recomputed here without a real assembler
; run; confirm on assembly/MAME rather than
; trust a static estimate. See the open-items
; checklist.
BASEDICT EQU $D6FF ; was $D77F ($D79F, $D7DF, $D7EF, $D7FF, $D83F
; before that) - shifted down $80 (128 bytes)
; this time, same reason as BASECODE above.
; Not recomputed against real, current
; assembled sizes here - confirm on
; assembly/MAME. See the open-items checklist.
INOUT EQU $C000 ; was $DF00 - moved so INOUT (256 B) sits
; directly below USROMSTRT ($C100), contiguous,
; no gap. This also resolves the INOUT portion
; of the collision flagged when BASEDICT moved
; to $D85D: INOUT no longer overlaps BASEDICT
; ($D85D-$E011), since $C0FF < $D85D. The
; DSTACK and RSTACK portions of that same
; collision were NOT touched by this specific
; change, but were resolved separately when
; those two regions moved (see below). This
; move ALSO overlapped APPCODE at the time
; ($7000-$D7FF then) - since resolved too, when
; APPCODE moved down $2000 (see below).
INOUTEND EQU INOUT+$FF
RSTACK EQU $BFFF ; was $BEFF - occupied range is $BD00-$BFFF
; (768 bytes, unchanged size). RESOLVED: once a
; 512-byte gap sat between this and DSTACK
; below; DSTACK moving up $200 closed it - now
; exactly contiguous, no gap
DSTACK EQU $BCFF ; was $BAFF - moved up $200. RESOLVED (both):
; occupied range is now $B900-$BCFF, which
; exactly matches CODETOP ($B900) as its true
; bottom - the 512-byte mismatch is gone - and
; is now exactly contiguous with RSTACK's
; bottom ($BD00), closing that gap too
CODETOP EQU $B900 ; was $B800 - code space ceiling (data stack
; begins here). RESOLVED: this once no longer
; matched DSTACK's true occupied bottom (was
; $B700, a 512-byte mismatch) - now that DSTACK
; moved up $200 to $B900, CODETOP matches it
; exactly again

```

```

; ANS transient-region minimums (forth-standard.org/standard/usage,
; section 3.3.3.6 "Other transient regions"), verified against this
; system's actual layout and used to replace the bare "84" that used
; to sit directly inside PADW with a named, documented constant:
;   PADMINSIZE - PAD's own scratch region, size in characters. WORD
;               originally used a separate, fixed WORDBUF that
;               happened to meet its own 33-character minimum
;               exactly (WORDBUF's span up to where SIBUF used to
;               begin was precisely 33 bytes, confirmed by address
;               subtraction) - but WORD has since been redesigned
;               (see below, and WORD itself in the Header/Compiling
;               section) to use this same CODEHERE-to-PAD gap
;               instead, matching the traditional fig-Forth layout,
;               so WORDBUF itself is retired along with SIBUF.
;   HOLDMINSIZE - pictured numeric output buffer, size in characters,
;               (2 * n) + 2 where n = bits per cell (16 here) = 34.
;               LTNUM ("<#") anchors HLD to PADW's own return value
;               and HOLD decrements before storing, so this buffer
;               grows downward from PAD into the CODEHERE-to-PAD
;               gap, not into PAD's own region above it - confirmed
;               by tracing LTNUM/HOLD. The gap must be at least
;               HOLDMINSIZE wide for this to fit.
;   PADOFFSET (below) satisfies both: it equals PADMINSIZE, which is
;   itself well above HOLDMINSIZE (84 vs 34, 50 bytes of margin) - kept
;   as a single, larger-than-either-strict-minimum offset rather than
;   two separately-tuned numbers, matching "larger than minimum is
;   satisfactory" for a small system.
PADMINSIZE EQU 84
WORDMINSIZE EQU 33
HOLDMINSIZE EQU 34
PADOFFSET EQU PADMINSIZE ; CODEHERE-to-PAD gap; also governs how
; much of PAD's own upward-growing region
; (now used by both the user and, per this
; change, interpreted-mode S" text) is
; guaranteed available before something
; else might claim it
WORDMAXCHARS EQU PADOFFSET-HOLDMINSIZE-1-3 ; max characters WORD can
; scan when using the CODEHERE-to-PAD gap
; (see WORD, below) - reserves HOLDMINSIZE
; bytes at the PAD end for the pictured
; numeric output buffer (which grows
; downward from PAD into this same gap),
; using the remaining space at the
; CODEHERE end, less 1 byte for WORD's own
; leading count byte, less a further 3
; bytes for the worst case: S"/./ABORT"
; all reserve 3 bytes ahead of CODEHERE
; before calling WORD (see SQUOTE/
; DOTQUOTE/AQSTOK), so their text lands 3
; bytes further into this same gap than
; plain WORD-parsing (e.g. a dictionary
; name for FIND) does. Sized for that
; worst case uniformly, rather than giving
; different callers different effective
; limits - simpler and safer than trying
; to track which caller needs which cap.
; 84-34-1-3 = 46.

APPCODE EQU $7000 ; was $5000 - back to its original address.
; RESOLVED: this once overlapped DSTACK's true
; range by 512 bytes ($B700-$B8FF), a direct
; consequence of the CODETOP/DSTACK mismatch -
; now that DSTACK moved and CODETOP matches it
; again, APPCODE's nominal range (up to
; CODETOP-1) no longer reaches into DSTACK
APPDICT EQU $2000 ; was $015B - moved up, size unchanged (20133
; bytes, now $2000-$6EA4). REDUCED BUT NOT
; RESOLVED: still overlaps APPVARS below by 347
; bytes ($2000-$215A), though SIBUF/WORDBUF/
; TIBBUF/OUTBUF (swallowed by the previous
; APPDICT address) are now clear. See the

```

```

; open-items checklist
APPVARS EQU $021B ; grown from 256 to 8000 bytes (end now $215A,
; was $031A), taking the space directly from
; APPDICT above it; start address unchanged.
; That 8000-byte figure describes the original
; intent, not the current actual usable size -
; see APPVARSEND below, which now tracks
; APPDICT's real position instead
APPVARSEND EQU APPDICT-1 ; was APPVARS+8000 ($215B) - now derives
; directly from wherever APPDICT actually
; starts, currently $1FFF (7653 bytes usable,
; down from the static 8000). Self-correcting:
; this can no longer go stale if APPDICT moves
; again, unlike the previous fixed-size
; definition. VUNUSEDW (below) is unchanged -
; it already computed against APPVARSEND
; SIBUF EQU $01FB retired - was interpreted-mode S"'s fixed, dedicated
; 32-byte scratch buffer; S" now writes into PAD instead (see SQINTERP,
; String Words section), which is both larger and correctly per-call
; rather than a single buffer shared by every S" call in a session.
; The $01FB-$021A range it used to occupy is left unclaimed rather
; than reassigned - APPVARS below still starts at its own, unchanged
; address ($021B) - to avoid any risk to the rest of this carefully-
; verified memory map for the sake of reclaiming 32 bytes on a system
; with headroom to spare; reclaiming it would need APPVARS to move,
; a separate, larger change not undertaken here.
; WORDBUF EQU $01DA also retired, for the same reason as SIBUF above -
; WORD now uses the CODEHERE-to-PAD gap directly (see WORD in the
; Header/Compiling section, and WORDMAXCHARS above) rather than a
; fixed, separately-allocated buffer capped at 31 characters. The
; $01DA-$01FA range it used to occupy is left unclaimed, same
; reasoning as SIBUF's retirement above.
TIBBUF EQU $018A ; was $0284
TIBBUFL EQU 80
SERBUF EQU $0106 ; was $0200 - USER0/USER1 removed entirely (see
; below); the 4 buffers (SERBUF's 4-byte index
; block, INBUF, OUTBUF, TIBBUF) still sit
; contiguously right after MVSCRATCH, with no
; gap - WORDBUF and SIBUF, which used to sit
; right after TIBBUF and WORDBUF respectively
; (closing what was once an 11-byte gap,
; $02F5-$02FF, in an earlier address scheme),
; are now both retired (see above) rather than
; part of this contiguous run

INHEAD EQU SERBUF
INTAIL EQU SERBUF+1
OUTHEAD EQU SERBUF+2
OUTTAIL EQU SERBUF+3
INBUFSZ EQU 64
OUTBUFSZ EQU 64
INBUF EQU SERBUF+4
OUTBUF EQU SERBUF+4+INBUFSZ
GLOBALS EQU $0000

SP0 EQU DSTACK+1
RP0 EQU RSTACK+1

```

```

; -----
; SERIALPOLL - conditional-assembly switch for serial I/O.
; 1 (default): KEY/KEYQ/EMIT poll ACIASR directly, no interrupts,
; no ring buffers, no RTS/CTS handshaking - IRQH is a harmless RTI
; stub, matching the other unused vectors (NMIH/FIRQH/SWI2H/
; SWI3H). 0: the original interrupt-driven implementation, with
; INBUF/OUTBUF ring buffers serviced by IRQH and RTS-based flow
; control via INFILL/RTSCHECKHI/RTSCHECKLO. Uses LWASM's IFEQ/
; ELSE/ENDC (a numeric-expression test, not IFDEF/IFNDEF, since
; this is a value to compare, not a symbol's mere presence).
; -----
SERIALPOLL EQU 1

```

```

; ACIA (6850) constants - the chip sits at INOUT+8, not at the
; base of the I/O block, leaving INOUT+0..INOUT+7 free for other
; memory-mapped devices sharing this 256-byte region
; -----
ACIA      EQU    INOUT+8
ACIACR    EQU    ACIA
ACIASR    EQU    ACIA
ACIADR    EQU    ACIA+1
SR_RDRF   EQU    $01
SR_TDRE   EQU    $02
SR_IRQ    EQU    $80
CR_RESET  EQU    $03
CR_RXON   EQU    $95
CR_RXTX   EQU    $B5
CR_POLL   EQU    $15      ; bit7=0 (RX interrupt disabled), bits6-5=00 (RTS
                           ; low, TX interrupt disabled) - CR_RXON ($95) with
                           ; only the RX-interrupt-enable bit cleared. Used
                           ; only when SERIALPOLL=1; RTS stays permanently
                           ; low (asserted), since polling mode has no ring
                           ; buffer to overflow and so needs no flow control
CR_RTSHI  EQU    $D5      ; bits6-5=10: RTS high, TX int disabled, RX int enabled -
                           ; derived from CR_RXON ($95) with bits6-5 changed from
                           ; 00 to 10; the ACIA has no combination offering RTS
                           ; high AND TX interrupt enabled simultaneously (bits6-5
                           ; only has 00/01/10/11, and only 01 enables TX interrupt,
                           ; which always ties RTS low) - EMIT/IRQH's TXCHK must
                           ; respect this and defer transmission while RTS is high

INHIWATER EQU    48      ; input ring fill level (of 64) at/above which RTS is
                           ; asserted high, telling the remote device to pause
INLOWATER EQU    16      ; fill level at/below which RTS is reasserted low;
                           ; deliberately well below INHIWATER (hysteresis) so
                           ; RTS doesn't chatter right at a single threshold

; -----
; Flag / opcode constants
; -----
TRUEV     EQU    $FFFF
FALSEV    EQU    $0000
OPJSR     EQU    $BD
OPRTS     EQU    $39      ; used by section 3.8's control-flow test
                           ; harness to terminate each compiled test
                           ; snippet, compiled via CCOMMA
RTSOPC    EQU    $39

; -----
; Control-flow compile-time tags
; -----
TAGFWD     EQU    1
TAGBACK    EQU    2
TAGDO      EQU    3
TAGCASE    EQU    4
TAGOF      EQU    5
TAGENDOF   EQU    6

; =====
; GLOBALS - real layout, applied. Every scratch/state cell
; referenced across the whole build now has a fixed address
; in page zero (DP = $00 at reset, set in COLDSTRT), laid out
; in RMB order below. Total: 256 of 256 bytes used - the page
; is fully packed. (DOESBEH and RTSSTATE were added in later
; passes, after this comment was first written with 253/256;
; they used up the last 3 bytes of headroom entirely.)
;
; SNEND, which appeared in the original placeholder list
; (documented under S"/./SLITERAL/ABORT"'s string runtime),
; was dropped here: a full pass over every reference confirmed
; it is never actually read or written anywhere - SCNT/SPTR
; alone carry that runtime. CSAVE was removed earlier (see the
; COMMA/CODECOMMA history) and was never part of this layout.
; =====

```

```

      ORG    $0000      ; GLOBALS page - Direct Page (DP) set to $00 at reset
STATE     RMB    2      ; offset $00
BASE      RMB    2      ; offset $02
LATEST    RMB    2      ; offset $04
DPHERE    RMB    2      ; offset $06
CODEHERE  RMB    2      ; offset $08
VARHERE   RMB    2      ; offset $0A
HANDLER    RMB    2      ; offset $0C
THROWN    RMB    2      ; offset $0E
TOIN      RMB    2      ; offset $10
NTIB      RMB    2      ; offset $12
DELIM     RMB    1      ; offset $14
WSTART    RMB    2      ; offset $15
SLEN      RMB    1      ; offset $17
SNAMEP    RMB    2      ; offset $18
FNDPTR    RMB    2      ; offset $1A
HDRPTR    RMB    2      ; offset $1C
HDRFLAGS  RMB    1      ; offset $1E
CADDR     RMB    2      ; offset $1F
CNTREM    RMB    1      ; offset $21
NUMNEG    RMB    1      ; offset $22
NADDR     RMB    2      ; offset $23
NCNT      RMB    2      ; offset $25
MULBASE   RMB    1      ; offset $27
CARRY     RMB    1      ; offset $28
MSCR      RMB    2      ; offset $29
MSCR2     RMB    2      ; offset $2B
MSCR3     RMB    2      ; offset $2D
MSCR4     RMB    2      ; offset $2F
HLD       RMB    2      ; offset $31
DEPTHTMP  RMB    2      ; offset $33
AMAX      RMB    2      ; offset $35
ABUFP     RMB    2      ; offset $37
ACNT      RMB    2      ; offset $39
ACH       RMB    1      ; offset $3B
EMITCH    RMB    1      ; offset $3C
NEWHDR    RMB    2      ; offset $3D
NAMEP     RMB    2      ; offset $3F
NAMELEN   RMB    1      ; offset $41
PTARGET   RMB    2      ; offset $42
PFIELD    RMB    2      ; offset $44
NEWFLD    RMB    2      ; offset $46
CSP       RMB    2      ; offset $48
EXITCNT   RMB    2      ; offset $4A
EXITPTR   RMB    2      ; offset $4C
HDRSMUDGE RMB    1      ; offset $4E
SCNT      RMB    2      ; offset $4F
SPTR      RMB    2      ; offset $51
SAVEN     RMB    2      ; offset $53
DRWIDTH   RMB    2      ; offset $55
DRLEN     RMB    2      ; offset $57
DRADDR    RMB    2      ; offset $59
DRPAD     RMB    2      ; offset $5B
PRODHI    RMB    2      ; offset $5D
PRODLO    RMB    2      ; offset $5F
PSIGN     RMB    1      ; offset $61
DIVNUM    RMB    2      ; offset $62
DIVDEN    RMB    2      ; offset $64
DIVREM    RMB    2      ; offset $66
DIVCNT    RMB    1      ; offset $68
DNSIGN    RMB    1      ; offset $69
DVSIGN    RMB    1      ; offset $6A
DVOWNSIGN RMB    1      ; offset $6B
MAHI      RMB    1      ; offset $6C
MALO      RMB    1      ; offset $6D
MBHI      RMB    1      ; offset $6E
MBLO      RMB    1      ; offset $6F
MSIGN     RMB    1      ; offset $70
REM       RMB    2      ; offset $71
DCNT      RMB    1      ; offset $73
UDHI      RMB    2      ; offset $74

```

(shared: -, \*, comparisons, WITHIN...)

|             |     |   |                                 |
|-------------|-----|---|---------------------------------|
| UDLO        | RMB | 2 | ; offset \$76                   |
| PRSIGN      | RMB | 1 | ; offset \$78                   |
| R2A         | RMB | 2 | ; offset \$79                   |
| R2B         | RMB | 2 | ; offset \$7B                   |
| RDST        | RMB | 2 | ; offset \$7D                   |
| RVAL        | RMB | 2 | ; offset \$7F                   |
| TR1         | RMB | 2 | ; offset \$81                   |
| TR2         | RMB | 2 | ; offset \$83                   |
| SHCNT       | RMB | 1 | ; offset \$85                   |
| SHCNT2      | RMB | 2 | ; offset \$86                   |
| TYPECNT     | RMB | 2 | ; offset \$88                   |
| TYPEADDR    | RMB | 2 | ; offset \$8A                   |
| PDELIM      | RMB | 1 | ; offset \$8C                   |
| PSTART      | RMB | 2 | ; offset \$8D                   |
| PLEN        | RMB | 2 | ; offset \$8F                   |
| CMPA1       | RMB | 2 | ; offset \$91                   |
| CMPL1       | RMB | 2 | ; offset \$93                   |
| CMPA2       | RMB | 2 | ; offset \$95                   |
| CMPL2       | RMB | 2 | ; offset \$97                   |
| CMPMIN      | RMB | 2 | ; offset \$99                   |
| SRCH1       | RMB | 2 | ; offset \$9B                   |
| SRCH1L      | RMB | 2 | ; offset \$9D                   |
| SRCH2       | RMB | 2 | ; offset \$9F                   |
| SRCH2L      | RMB | 2 | ; offset \$A1                   |
| SRCHPOS     | RMB | 2 | ; offset \$A3                   |
| SRCHI       | RMB | 2 | ; offset \$A5                   |
| UEADDR      | RMB | 2 | ; offset \$A7                   |
| UESRCLEN    | RMB | 2 | ; offset \$A9                   |
| UEDST       | RMB | 2 | ; offset \$AB                   |
| UEOUTLEN    | RMB | 2 | ; offset \$AD                   |
| SNXT        | RMB | 2 | ; offset \$AF                   |
| SNTARGET    | RMB | 2 | ; offset \$B1                   |
| REPLNAME    | RMB | 2 | ; offset \$B3                   |
| REPLNLEN    | RMB | 2 | ; offset \$B5                   |
| REPLVAL     | RMB | 2 | ; offset \$B7                   |
| REPLVLEN    | RMB | 2 | ; offset \$B9                   |
| SUBDESTCAP  | RMB | 2 | ; offset \$BB                   |
| SUBDESTADR  | RMB | 2 | ; offset \$BD                   |
| SUBSRCADR   | RMB | 2 | ; offset \$BF                   |
| SUBSRCLEN   | RMB | 2 | ; offset \$C1                   |
| SUBOUTLEN   | RMB | 2 | ; offset \$C3                   |
| SUBWPTR     | RMB | 2 | ; offset \$C5                   |
| SUBCOPYCNT  | RMB | 2 | ; offset \$C7                   |
| SUBCOPYSRC  | RMB | 2 | ; offset \$C9                   |
| MKDP        | RMB | 2 | ; offset \$CB                   |
| MKCODE      | RMB | 2 | ; offset \$CD                   |
| MKVAR       | RMB | 2 | ; offset \$CF                   |
| MKLATEST    | RMB | 2 | ; offset \$D1                   |
| EVSAVEA     | RMB | 2 | ; offset \$D3                   |
| EVSAVEL     | RMB | 2 | ; offset \$D5                   |
| EVSAVEI     | RMB | 2 | ; offset \$D7                   |
| EVSAVET     | RMB | 2 | ; offset \$D9                   |
| SRCADDR     | RMB | 2 | ; offset \$DB                   |
| SRCLN       | RMB | 2 | ; offset \$DD                   |
| SRCID       | RMB | 2 | ; offset \$DF                   |
| SPAN        | RMB | 2 | ; offset \$E1                   |
| DSPTMP      | RMB | 2 | ; offset \$E3                   |
| WWALK       | RMB | 2 | ; offset \$E5                   |
| DUMPADDR    | RMB | 2 | ; offset \$E7                   |
| DUMPCNT     | RMB | 2 | ; offset \$E9                   |
| DUMPCOL     | RMB | 1 | ; offset \$EB                   |
| HEXBUF      | RMB | 2 | ; offset \$EC                   |
| DUVALID     | RMB | 1 | ; offset \$EE                   |
| ENVLEN      | RMB | 2 | ; offset \$EF                   |
| ENVADDR     | RMB | 2 | ; offset \$F1                   |
| QSAVEDP     | RMB | 2 | ; offset \$F3                   |
| QSAVECODE   | RMB | 2 | ; offset \$F5                   |
| QSAVEVAR    | RMB | 2 | ; offset \$F7                   |
| QSAVELATEST | RMB | 2 | ; offset \$F9                   |
| QTHROWCODE  | RMB | 2 | ; offset \$FB                   |
| DOESBEH     | RMB | 2 | ; offset \$FD - SETDOES scratch |

```
RTSSTATE   RMB   1   ; offset $FF - 0 = RTS low (normal), nonzero = RTS
                ; high (paused) - see CR_RTSHI. GLOBALS page is now
                ; fully packed: 256 of 256 bytes used, 0 free.
```

```
GLOBALS_USED EQU 256 ; total bytes used, of 256 available - fully packed
```

```
; -----
; MVSCRATCH - three cells shared, one at a time, by routine
; families that never call each other or run concurrently in
; this single-threaded interpreter: MOVE/CMOVE/CMOVE>, FILL, and
; HOLDS (plus the single-cell multiply routine). Sharing avoids
; needing 3x the physical storage for what is provably the same
; scratch need at different times; the tradeoff is that these
; three cells use ordinary extended addressing (3-byte LDD/STD),
; not direct-page (2-byte), since page zero has no room left.
;
; MVCNT      - MOVE/CMOVE's remaining-byte count
; HSLLEN     EQU MVCNT      - HOLDS's remaining-char count
; MVDST      - MOVE/CMOVE's destination address
; HSADDR     EQU MVDST      - HOLDS's source address
; MVSRC      - MOVE/CMOVE's source address
; MRESULT    EQU MVSRC      - single-cell multiply's 16-bit result
; FILLCHR    EQU MVSRC      - FILL's fill character (1 byte, uses
;                           MVSRC's first byte only)
; FILLCNT/FILLADDR (formerly aliasing MVCNT/MVDST) were removed once
; genuinely unused - FILLW now keeps its count and address directly
; in Y/X rather than round-tripping through memory each iteration.
; -----
```

```
                ORG   $0100
MVCNT          RMB   2
MVDST          RMB   2
MVSRC          RMB   2
HSLLEN        EQU   MVCNT
HSADDR        EQU   MVDST
MRESULT        EQU   MVSRC
FILLCHR        EQU   MVSRC
```

```
; Provide padding, to ensure the correct ROM & .bin file size (and
; opcode offsets) for the MAME emulation and flash memory burn.
                ORG   USROMSTRT
```

```
                IFNDEF UNITTESTS
UNITTESTS SET 0 ; Fallback default value if -D wasn't passed.
                ; Flag meaning reversed from this file's original
                ; UNITTESTS convention (0=included,1=excluded under
                ; IFEQ) to support overriding via lwasm's -D command
                ; line option: default (0, no -D given) now means
                ; EXCLUDED - production builds get plain FILL
                ; padding here with no test-framework code, by
                ; default. Pass -DUNITTESTS=1 (or any nonzero value)
                ; to INCLUDE the test framework - tested and
                ; confirmed working on real MAME hardware under the
                ; old convention; this reversal only changes how the
                ; choice is made, not what including it does.
                ENDC
```

```
                IFNDEF TSTSELECTOR
TSTSELECTOR SET 2 ; Fallback default value if -D wasn't passed.
                ENDC
```

```
                IFNE  UNITTESTS ; >>>>>>>>>
```

```
; -----
; The unit test framework itself (explanatory comment plus all
; test code) lives in unit_tests.asm, a separate file included
; here only when UNITTESTS is nonzero - see unit_tests.asm for
; the full framework description, or OPEN_ITEMS_CHECKLIST_
; part1.md/part2.md for its own development history.
; -----
                INCLUDE unit_tests.asm
```

```

        ENDC ; <<<<<<<<<

ROM:
        FILL $FF,BASEDICT-ROM

```

## 8.2 Hardware Vector Table

```

; =====
; SECTION 1: HARDWARE VECTOR TABLE
; =====
        ORG     VECTORS          ; VECTORS is $FFF0
VRESV   FDB     $0000
VSWI3   FDB     SWI3H
VSWI2   FDB     SWI2H
VFIRQ   FDB     FIRQH
VIRQ    FDB     IRQH
VSWI    FDB     SWIH
VNMI     FDB     WARM           ; NMI -> warm restart
VRESET  FDB     COLDSTRT

VECTOREND EQU    *           ; Verify vectors size, value should match $10.
VECTORSIZE EQU    VECTOREND-VECTORS

; =====
; END OF CONSOLIDATED SOURCE
; =====

```

## 8.3 Init Code (COLDSTRT / WARM)

```

; =====
; SECTION 2: INIT CODE (COLDSTRT / WARM)
; =====
        ORG     INITCODE        ; INITCODE is $FFA9 (was $FFA2, before that $FFA0, before that
literal $FFC0)
COLDSTRT:
        ORCC    #$50
        LDS     #RSTACK+1
        LDU     #DSTACK+1
        CLRA
        TFR     A,DP

        LDX     #GLOBALS
        LDB     #0
CLRGLOB: CLR     ,X+
        DECB
        BNE     CLRGLOB

        LDX     #SERBUF
        CLR     ,X+
        CLR     ,X

        JSR     INITSERIAL

        IFNE    UNITTESTS ; >>>>>>>>
        JSR     TSTRUNNER
        ELSE    ; <<<<<>>>>
        NOP          ; BUG FIX (see the historical note above this
        NOP          ; call site's own comment, describing the
        NOP          ; original bug): this call site used to emit
                        ; 0 bytes when UNITTESTS' flag meaning
                        ; excluded the test framework, meaning
                        ; COLDSTRT's own size varied by 3 bytes
                        ; depending on UNITTESTS - with INITCODE's
                        ; own position fixed regardless, that risked
                        ; the code overflowing into VECTORS whenever
                        ; UNITTESTS was toggled on. Three NOPs here
                        ; are byte-for-byte the same size as the
                        ; JSR TSTRUNNER they replace, so this block
                        ; now always contributes exactly 3 bytes to

```

```

; COLDSTRT either way - COLDSTRT's total size
; no longer depends on UNITTESTS at all.
ENDC ; <<<<<<<<<

JMP COLD

WARM: ORCC #$50
      CLRA
      TFR A, DP
      LDU #SP0
      LDS #RP0
      LDX #WARMMSG
      PSHU X
      LDD #WARMMSGGL
      PSHU D
      JSR TYPE
      ANDCC #SAF
      JMP ABORT

WARMMSG: FCC " warm"
WARMMSGGL EQU *-WARMMSG

INITEND EQU * ; Verify no collision with vectors, value should match vector ORG
INITSIZE EQU INITEND-INITCODE

```

## 8.4 ACIA Interrupt Handler

```

;
; SECTION 3: ACIA INTERRUPT HANDLER
; =====
;
; ORG      BASECODE      ; BASECODE is $E02A. This ORG was missing
;                        ; entirely - every routine from here through
;                        ; SECTION 26 (IRQH, COLD/ABORT/QUIT, and
;                        ; every primitive) would otherwise have
;                        ; continued growing from wherever SECTION 2's
;                        ; WARM message left the location counter,
;                        ; inside INIT's 48-byte $FFC0-$FFEF budget,
;                        ; overflowing directly into VECTORS ($FFF0)
;                        ; instead of landing in BASECODE at all
;
; -----
; INITSERIAL - initializes the ACIA: master reset, then selects
; interrupt-driven or polling operation depending on SERIALPOLL.
; Extracted from COLDSTRT, which now just JSRs here - moved into
; this section so the ACIA's own init code sits next to the rest
; of its interrupt/polling logic rather than inline in COLDSTRT.
; -----
INITSERIAL: LDA    #$03
             STA    ACIACR      ; was "STA ACIA" - only correct by
                                ; coincidence while ACIA and ACIACR were
                                ; the same address; now genuinely distinct
             IFEQ   SERIALPOLL ; >>>>>>>>>
             LDA    #CR_RXON    ; interrupt-driven mode: RX interrupt on
             ELSE   ; <<<<<<>>>>
             LDA    #CR_POLL    ; polling mode: no interrupts, RTS held low
             ENDC   ; <<<<<<<<<
             STA    ACIACR      ; was "STA ACIA" - same fix
             RTS

             IFEQ   SERIALPOLL ; >>>>>>>>>
;
; -----
; INFILL - ( -- A=fill level, 0-63 ) input ring's current fill
; level. INBUFSZ is a power of two, and both indices are always
; kept in 0..INBUFSZ-1, so a plain masked subtraction gives the
; true mod-64 distance even across the wrap point.
; -----
INFILL: LDA    INHEAD
          SUBA  INTAIL
          ANDA  #INBUFSZ-1

```

## RTS

```

; -----
; RTSCHECKHI - called from IRQH's own RX path (interrupts
; already masked by hardware during ISR execution, so no
; explicit masking needed here). If the input ring has reached
; INHIWATER and RTS is not already asserted high, assert it -
; telling the remote device to pause sending. Per the 6850, RTS
; has no automatic tie to reception; this is ordinary firmware
; flow control, not a chip feature.
; -----
RTSCHECKHI: JSR  INFILL
             CMPA #INHIWATER
             BLO  RTSCHIDONE
             TST  RTSSTATE
             BNE  RTSCHIDONE      ; already high - nothing to do
             LDA  #CR_RTSHI
             STA  ACIACR
             LDA  #1
             STA  RTSSTATE
RTSCHIDONE: RTS

; -----
; RTSCHECKLO - called from mainline code (KEY), NOT from the
; ISR, so it must mask IRQ around its critical section: IRQH's
; own TXOFF path also writes ACIACR, and an interrupt landing
; mid-decision here could otherwise race it. If the ring has
; drained to INLOWATER or below and RTS is currently high,
; reassert RTS low - restoring TX-interrupt-enable too if
; output happens to be queued, since the ACIA has no control
; byte combination offering RTS-high with TX-interrupt-enabled
; simultaneously (see CR_RTSHI's comment).
; -----
RTSCHECKLO: JSR  INFILL
             CMPA #INLOWATER
             BHI  RTSCLODONE
             TST  RTSSTATE
             BEQ  RTSCLODONE      ; already low - nothing to do
             ORCC #$10           ; mask IRQ for the critical section
             CLR  RTSSTATE
             LDB  OUTTAIL
             CMPB OUTHEAD
             BEQ  RTSCLONOTX
             LDA  #CR_RXTX
             STA  ACIACR
             BRA  RTSCLOUNMASK
RTSCLONOTX: LDA  #CR_RXON
             STA  ACIACR
RTSCLOUNMASK: ANDCC #$EF
RTSCLODONE: RTS

IRQH:  LDA  ACIASR
       BITA #SR_IRQ
       BEQ  IRQDONE

       BITA #SR_RDRF
       BEQ  TXCHK

       LDB  INHEAD
       LDA  ACIADR
       LDX  #INBUF
       STA  B,X
       INCB
       ANDB #INBUFSZ-1
       CMPB INTAIL
       BEQ  IRQDONE
       STB  INHEAD
       JSR  RTSCHECKHI
       BRA  IRQDONE

TXCHK:  LDB  OUTTAIL

```

```

        CMPB  OUTHEAD
        BEQ   TXOFF
        LDX   #OUTBUF
        LDA   B,X
        STA   ACIADR
        INCB
        ANDB  #OUTBUFSZ-1
        STB   OUTTAIL
        BRA   IRQDONE

TXOFF:   TST   RTSSTATE
        BNE   IRQDONE          ; RTS is asserted high - leave ACIACR alone,
                                ; or this would incorrectly drop it back low

        LDA   #CR_RXON
        STA   ACIACR

IRQDONE: RTI

IRQH:    ELSE ; <<<<<>>>>
        RTI          ; polling mode (SERIALPOLL=1) - ACIA
                    ; interrupts are never enabled (see
                    ; COLDSTRT's CR_POLL init), so this should
                    ; never fire; kept as a safe stub matching
                    ; the other unused vectors below

        ENDC ; <<<<<<<<<<

SWI3H:   RTI
SWI2H:   RTI
FIRQH:   RTI
NMIH:    RTI          ; unused now that NMI -> WARM
SWIH:    LDD   #-99    ; placeholder hardware-trap code; push and
        PSHU  D        ; JMP THROW, per the CATCH/THROW turn
        JMP   THROW

```

## 8.5 COLD / ABORT / QUIT

```

; =====
; SECTION 4: COLD / ABORT / QUIT (with CATCH-wrapped INTERPRET)
; =====
COLD:    LDD   #APPVARS
        STD   VARHERE
        LDD   #APPCODE
        STD   CODEHERE
        LDD   #APPDICT
        STD   DPHERE
        LDD   #BASELATEST
        STD   LATEST

        LDD   #10
        STD   BASE

        LDD   #TIBBUF
        STD   SRCADDR
        LDD   #0
        STD   SRCLEN
        STD   SRCID

        LDX   #SIGNON
        PSHU  X
        LDD   #SIGNONL
        PSHU  D
        JSR   TYPE
        JSR   CRW
        ; falls through into ABORT

ABORT:   LDU   #SP0
        ; falls through into QUIT

QUIT:    LDS   #RP0

```

```

LDD #0 ; BUG FIX: this reset used to live at QLOOP
STD STATE ; below, running on EVERY line - including
; lines in the middle of a colon definition
; that spans more than one line of input.
; COLON (sets STATE=-1) and SEMI (sets
; STATE=0, after checking CSP) are the
; correct, sole places STATE should change
; during normal operation - the old QLOOP
; reset was a third, redundant one that
; silently dropped back to interpret mode
; every time QUERY read a new line,
; regardless of whether ";" had actually
; been reached. A colon definition split
; across multiple lines would have every
; word on every line after the first
; interpreted (and, for anything with a
; stack effect, executed) instead of
; compiled - "TRUE" pushing TRUEV onto the
; data stack instead of being compiled,
; leaving a stray cell CSP would correctly
; catch as a mismatch at ";" (-22). Moved
; here rather than removed outright: QUIT is
; only re-entered on cold boot or an
; uncaught error routing back through ABORT
; (confirmed - ordinary successful lines
; loop back to QLOOP directly, never QUIT),
; so this still correctly forces interpret
; state exactly when ANS's own QUIT
; semantics call for it - just not on every
; single line of an otherwise-uninterrupted
; session. Confirmed via MAME debugger.

QLOOP: JSR QUERY

LDD DPHERE
STD QSAVEDP
LDD CODEHERE
STD QSAVECODE
LDD VARHERE
STD QSAVEVAR
LDD LATEST
STD QSAVELATEST

LDD #INTERPRET
PSHU D
JSR CATCH
PULU D
STD QTHROWCODE
CMPD #0
BEQ QOK

LDD QSAVEDP
STD DPHERE
LDD QSAVECODE
STD CODEHERE
LDD QSAVEVAR
STD VARHERE
LDD QSAVELATEST
STD LATEST
LDU #SP0

JSR CRW
LDX #ERRMSG
PSHU X
LDD #ERRMSGGL
PSHU D
JSR TYPE
LDD QTHROWCODE
PSHU D
JSR DOT
BRA QLOOP

```

```

QOK:      JSR    CRW          ; BUG FIX: was JSR CRW AFTER the STATE check
                                   ; below, so "BNE QLOOP" (still compiling)
                                   ; skipped both the CR echo and the ok
                                   ; message together. The CR reflects a real
                                   ; keystroke - the user genuinely pressed
                                   ; Enter for that line - and should echo
                                   ; regardless of interpret/compile state;
                                   ; only the "ok" message itself should stay
                                   ; conditional (correctly not shown mid-
                                   ; definition). Previously, every line of a
                                   ; multi-line colon definition after the
                                   ; first had its line ending silently
                                   ; dropped, so the echoed source ran
                                   ; together onto one line with no
                                   ; resemblance to what was actually typed.
                                   ; Confirmed via MAME debugger.

        LDD     STATE
        BNE     QLOOP
        LDX     #OKMSG
        PSHU    X
        LDD     #OKMSGL
        PSHU    D
        JSR     TYPE
        BRA     QLOOP

SIGNON:   FCC    "6809 FORTH v1.0"
SIGNONL   EQU    *-SIGNON
OKMSG:    FCC    "  ok"
OKMSGL    EQU    *-OKMSG
ERRMSG:   FCC    "  ERROR "
ERRMSGL   EQU    *-ERRMSG

```

## 8.6 Inner-Interpreter Support

```

; =====
; SECTION 5: INNER-INTERPRETER SUPPORT (LIT, ZBRANCH, BRANCH,
; DODOES, DODEFER, EXECUTE)
; =====
LIT:     PULS    X
        LDD      ,X++
        PSHU     D
        PSHS     X
        RTS

ZBRANCH: PULU     D
        PULS     X
        CMPD     #0
        BNE      ZSKIP
        LDD      ,X
        LEAX     D,X
        PSHS     X
        RTS

ZSKIP:   LEAX     2,X
        PSHS     X
        RTS

BRANCH:  PULS     X
        LDD      ,X
        LEAX     D,X
        PSHS     X
        RTS

DODOES:  PULS     X
        LDY      ,X++
        LDD      ,X
        PSHU     D
        JMP      ,Y

```

DOESRT0: RTS

```

; -----
; SETDOES - compiled via JSR by DOES>'s immediate action.
; Patches LATEST's trampoline BEHAVIOR field, then returns
; two levels up - skipping the rest of the defining word's
; body entirely, straight back to whoever invoked it.
; -----
SETDOES: PULS  X          ; X = addr right after "JSR SETDOES" - new BEHAVIOR
          STX   DOESBEH

          LDX   LATEST
          LDA   ,X
          STA   HDRFLAGS
          LEAX  1,X
          LDB   HDRFLAGS
          ANDB  #$1F
          CLRA

          LEAX  D,X          ; skip name -> LINK field
          LEAX  2,X          ; skip LINK -> CFA field
          LDD   ,X          ; D = CFA (trampoline address)
          ADDD  #3          ; +3 -> BEHAVIOR field (past JSR DODOES)
          TFR   D,X

          LDD   DOESBEH
          STD   ,X          ; patch it

          PULS  X          ; X = the OUTER defining word's own return addr
          JMP   ,X          ; jump there directly - "double RTS"

DODEFER: PULU  X
          LDD   ,X
          TFR   D,X
          JMP   ,X

DOABORTUNDEF: LDD #-21
              PSHU D
              JMP  THROW

DOMARKER: PULU  X
          LDD   ,X
          STD   DPHERE
          LDD   2,X
          STD   CODEHERE
          LDD   4,X
          STD   VARHERE
          LDD   6,X
          STD   LATEST
          RTS

EXECUTE: PULU  X
          JSR   ,X
          RTS

```

## 8.7 Comma Family

```

; =====
; SECTION 6: COMMA FAMILY (factored via APPENDCELL/APPENDBYTE)
; =====
APPENDCELL: PULU D
            LDY   ,X
            STD   ,Y++
            STY   ,X
            RTS

APPENDBYTE: PULU D
            LDY   ,X
            STB   ,Y+
            STY   ,X

```

```

        RTS

COMMA:   LDX   #CODEHERE
        JMP   APPENDCELL

CODECOMMA: LDX   #CODEHERE
        JMP   APPENDCELL

CCOMMA:   LDX   #CODEHERE
        JMP   APPENDBYTE

CCOMMA1:  LDX   #CODEHERE
        JMP   APPENDBYTE

VCOMMA:   LDX   #VARHERE
        JMP   APPENDCELL

VCCOMMA:  LDX   #VARHERE
        JMP   APPENDBYTE

ALLOT:    PULU  D
        LDX   CODEHERE
        LEAX  D,X
        STX   CODEHERE
        RTS

VALLLOT:  PULU  D
        LDX   VARHERE
        LEAX  D,X
        STX   VARHERE
        RTS

HEREW:    LDD   CODEHERE
        PSHU  D
        RTS

VHEREW:   LDD   VARHERE
        PSHU  D
        RTS

PADW:     LDD   CODEHERE
        ADDD  #PADOFFSET
        PSHU  D
        RTS

UNUSEDW:  LDD   #CODETOP
        SUBD  CODEHERE
        PSHU  D
        RTS

VUNUSEDW: LDD   #APPVAREND      ; was #APPCODE - a real bug, not just a
        SUBD  VARHERE          ; missing check: computed a meaningless
        PSHU  D                ; distance to an unrelated region instead
        RTS                    ; of remaining APPVARS space

```

## 8.8 HEADER

```

; =====
; SECTION 7: HEADER (factored from :/CREATE/VARIABLE)
; =====
HEADER:  LDD   #32
        PSHU  D
        JSR   WORD
        PULU  X
        LDA   ,X
        STA   NAMELEN
        LEAX  1,X
        STX   NAMEP

```

```

        PULU D
        STB HDRSMUDGE

        LDD DPHERE
        STD NEWHDR
        LDX DPHERE
        LDA NAMELEN
        TST HDRSMUDGE
        BEQ HDNOSM
        ORA #$40
HDNOSM: STA ,X+
        LDY NAMEP
        LDB NAMELEN
        BEQ HDNONM
HDCPY:  LDA ,Y+
        STA ,X+
        DECB
        BNE HDCPY
HDNONM: LDD LATEST
        STD ,X++
        LDD CODEHERE
        STD ,X++
        STX DPHERE
        LDD NEWHDR
        STD LATEST
        RTS

```

## 8.9 Defining Words

```

; =====
; SECTION 8: DEFINING WORDS
; =====
COLON:  LDD #TRUEV
        PSHU D
        JSR HEADER
        TFR U,D
        STD CSP
        LDD #-1
        STD STATE
        RTS

SEMI:   LDD #RTSOPC
        PSHU D
        JSR CCOMMA1
        TFR U,D
        CMPD CSP
        BEQ SEMIOK
        JSR CFERR
SEMIOK: LDX LATEST
        LDA ,X
        ANDA #$BF
        STA ,X
        LDD #0
        STD STATE
        RTS

CREATE: LDD #0
        PSHU D
        JSR HEADER
        LDD #DODOES
        PSHU D
        JSR CCALL
        LDD #DOESRT0
        PSHU D
        JSR CODECOMMA
        LDD CODEHERE ; BUG FIX: same self-referential PFA bug as
        ADDD #2      ; CONSTANT/DEFER/2CONSTANT/MARKER (CONSTANT's
        PSHU D        ; instance confirmed via MAME debugger) -
        JSR CODECOMMA ; without +2 this pointed at itself instead

```

```

; of the value cell that "," appends next.
; Never on the previously-flagged list (only
; DEFER/2CONSTANT/MARKER were), so it went
; unfixed until confirmed by inspection here.
; Confirmed via MAME: ": ENUM CREATE , DOES>
; @ ;" returned the PFA-pointer field's own
; address instead of the value COMMA stored
; one cell further on - exactly "the address
; before the 16-bit constant" instead of the
; address 1 cell further on, matching a
; user-reported MAME trace precisely.

RTS

; -----
; DOES> ( -- ) IMMEDIATE, compile-only. Compiles a call to
; SETDOES. Code label DOESGT, not "DOES>" - a literal ">" is
; not valid in a 6809 assembler label, same reason ?DUP/2DUP/
; etc. all use mnemonic labels rather than their literal names.
; -----
DOESGT: LDD    #SETDOES
        PSHU   D
        JSR    CCALL
        RTS

VARIABLE: LDD    #0
          PSHU   D
          JSR    HEADER
          LDD    #DODOES
          PSHU   D
          JSR    CCALL
          LDD    #DOESRT0
          PSHU   D
          JSR    CODECOMMA
          LDD    VARHERE
          PSHU   D
          JSR    CODECOMMA
          LDD    #0
          LDX    VARHERE
          STD    ,X++
          STX    VARHERE
          RTS

ATSIGN:  PULU   X
          LDD    ,X
          PSHU   D
          RTS

CONSTANT: LDD    #0
          PSHU   D
          JSR    HEADER
          LDD    #DODOES
          PSHU   D
          JSR    CCALL
          LDD    #ATSIGN
          PSHU   D
          JSR    CODECOMMA
          LDD    CODEHERE ; BUG FIX: was storing CODEHERE's value as-is,
          ADDD    #2      ; but at this point CODEHERE points at THIS
          PSHU   D        ; very cell, about to be written by the next
          JSR    CODECOMMA ; CODECOMMA - self-referential, not pointing
                          ; at the value cell 2 bytes further on where
                          ; the JSR COMMA below actually appends 1234.
                          ; +2 accounts for this FDB field's own width,
                          ; landing correctly on the value that follows.
                          ; Confirmed via MAME debugger: executing the
                          ; constant returned its own compile-time
                          ; CODEHERE address instead of the real value.

          JSR    COMMA
          RTS

DOVALUE: PULU   X

```

```

        LDD    ,X
        PSHU   D
        RTS

VALUEW:  LDD    #0
        PSHU   D
        JSR    HEADER          ; not smudged - immediately findable
        LDD    #DODOES
        PSHU   D
        JSR    CCALL
        LDD    #DOVALUE
        PSHU   D
        JSR    CODECOMMA      ; trampoline itself is still code
        LDD    VARHERE        ; PFA = VARHERE, mutable space - was
        PSHU   D              ; CODEHERE; T0 writes through this PFA,
        JSR    CODECOMMA      ; so it must live in mutable space
        JSR    VCOMMA         ; store x into VARHERE via VCOMMA,
                                ; not COMMA (which targets CODEHERE)
        RTS

TOW:     LDD    #32
        PSHU   D
        JSR    WORD
        JSR    FIND
        PULU   D
        CMPD   #0
        BNE    TOFOUND
        PULU   D
        LDD    #-13
        PSHU   D
        JSR    THROW
TOFOUND: JSR    TOBODY
        LDD    STATE
        BEQ    TOIMMED
        JSR    LITERALW
        LDD    #STOREW
        PSHU   D
        JSR    CCALL
        RTS

TOIMMED: PULU   X
        PULU   D
        STD    ,X
        RTS

TWOVARIABLE: LDD #0
              PSHU D
              JSR HEADER
              LDD #DODOES
              PSHU D
              JSR CCALL
              LDD #DOESRT0
              PSHU D
              JSR CODECOMMA
              LDD VARHERE
              PSHU D
              JSR CODECOMMA
              LDD #0
              LDX VARHERE
              STD ,X++
              STD ,X++
              STX VARHERE
              RTS

TWOCONSTANT: LDD #0
              PSHU D
              JSR HEADER
              LDD #DODOES
              PSHU D
              JSR CCALL
              LDD #DFETCH
              PSHU D

```

```

        JSR CODECOMMA
        LDD CODEHERE ; BUG FIX: same self-referential PFA bug as
        ADDD #2      ; CONSTANT (confirmed via MAME debugger) -
        PSHU D        ; without +2 this pointed at itself instead
        JSR CODECOMMA ; of the two-cell value that COMMA appends
                        ; below.
        PULU D        ; x2, off the top
        STD MSCR
        JSR COMMA     ; x1 -> lower address
        LDD MSCR
        PSHU D
        JSR COMMA     ; x2 -> higher address
        RTS

BUFFERCOLON: PULU D
              STD MSCR2
              LDD #0
              PSHU D
              JSR HEADER
              LDD #DODOES
              PSHU D
              JSR CCALL
              LDD #DOESRT0
              PSHU D
              JSR CODECOMMA
              LDD VARHERE
              PSHU D
              JSR CODECOMMA
              LDD MSCR2
              PSHU D
              JSR VALLLOT
              RTS

DEFERW:  LDD #0
        PSHU D
        JSR HEADER
        LDD #DODOES
        PSHU D
        JSR CCALL
        LDD #DODEFER
        PSHU D
        JSR CODECOMMA
        LDD CODEHERE ; BUG FIX: same self-referential PFA bug as
        ADDD #2      ; CONSTANT (confirmed via MAME debugger) -
        PSHU D        ; without +2 this pointed at itself instead
        JSR CODECOMMA ; of the value cell DOABORTUNDEF lands in
                        ; below. Confirmed by inspection: DODEFER's
                        ; own "LDD ,X" would have read the PFA's own
                        ; address and jumped there, crashing on
                        ; execution of any newly-DEFER'd word.
        LDD #DOABORTUNDEF
        PSHU D
        JSR COMMA
        RTS

DEFERFETCH: JSR TOBODY
            PULU X
            LDD ,X
            PSHU D
            RTS

DEFERSTORE: JSR TOBODY
            PULU X
            PULU D
            STD ,X
            RTS

ISW:       LDD #32
        PSHU D
        JSR WORD
        JSR FIND

```

```

PULU D
CMPD #0
BNE ISFOUND
PULU D
LDD #-13
PSHU D
JSR THROW
ISFOUND: PULU X
LDD STATE
BEQ ISIMMED
PSHU X
JSR LITERALW
LDD #DEFERSTORE
PSHU D
JSR CCALL
RTS
ISIMMED: PSHU X
JSR DEFERSTORE
RTS

ACTIONOF: LDD #32
PSHU D
JSR WORD
JSR FIND
PULU D
CMPD #0
BNE AOFFOUND
PULU D
LDD #-13
PSHU D
JSR THROW
AOFFOUND: PULU X
LDD STATE
BEQ AOIMMED
PSHU X
JSR LITERALW
LDD #DEFERFETCH
PSHU D
JSR CCALL
RTS
AOIMMED: PSHU X
JSR DEFERFETCH
RTS

MARKERW: LDD DPHERE
STD MKDP
LDD CODEHERE
STD MKCODE
LDD VARHERE
STD MKVAR
LDD LATEST
STD MKLATEST
LDD #0
PSHU D
JSR HEADER
LDD #DODOES
PSHU D
JSR CCALL
LDD #DOMARKER
PSHU D
JSR CODECOMMA
LDD CODEHERE ; BUG FIX: same self-referential PFA bug as
ADD #2 ; CONSTANT (confirmed via MAME debugger) -
PSHU D ; without +2 this pointed at itself instead
JSR CODECOMMA ; of the four snapshot cells COMMA appends
; below. Confirmed by inspection: DOMARKER's
; own fixed-offset reads (,X / 2,X / 4,X /
; 6,X) would have read garbage relative to
; the intended MKDP/MKCODE/MKVAR/MKLATEST -
; the most severe of the three, since using
; a MARKER-created word would restore

```

```

; corrupted dictionary-state pointers.
LDD MKDP
PSHU D
JSR COMMA
LDD MKCODE
PSHU D
JSR COMMA
LDD MKVAR
PSHU D
JSR COMMA
LDD MKLATEST
PSHU D
JSR COMMA
RTS

```

## 8.10 Outer Interpreter

```

; =====
; SECTION 9: OUTER INTERPRETER (INTERPRET / WORD / FIND / NUMBER?)
; =====
INTERPRET:
ILOOP:  LDD  #32          ; BUG FIX: WORD expects a delimiter char
        PSHU D          ; pushed by its caller (PULU D/STB DELIM)
                        ; - nothing pushed one here before, so
                        ; WORD pulled from an empty U stack,
                        ; landing on live RSTACK content (SP0 is
                        ; RSTACK's own first byte) instead of a
                        ; real delimiter. Confirmed present in
                        ; both SERIALPOLL branches - IRQH never
                        ; touches U, so interrupt-driven mode
                        ; had no incidental workaround either.

        JSR  WORD
        LDX  ,U
        LDA  ,X
        BEQ  IDONE

        JSR  FIND
        PULU D
        TSTB
        LBEQ TRYNUM

        LDA  STATE+1
        BEQ  DOEXEC
        TSTB
        BPL  DOEXEC
        JSR  CCALL
        BRA  ILOOP

DOEXEC:  JSR  EXECUTE
        BRA  ILOOP

TRYNUM:  JSR  NUMBERQ
        PULU D
        CMPD #0          ; was TSTD (6309-only) - PULU doesn't set CC on
                        ; genuine 6809, so compare D against 0 directly

        BEQ  BADWORD

        CMPD #1          ; NEW: distinguishes NUMBERQ's double-number
        BEQ  TNDOUBLE    ; success code (1) from single (-1), added
                        ; to support ANS's double-number input
                        ; convention (forth-standard.org/standard/
                        ; usage#usage:numbers) - "123." parses as
                        ; a double-cell number, "123" as single.

        LDD  STATE
        BEQ  ILOOP
        LDD  #LIT
        PSHU D
        JSR  CCALL

```

```

        JSR    CODECOMMA
        BRA    ILOOP

TNDDOUBLE: ; U currently holds [UDLO(bottom), UDHI(top)], matching
            ; ANS's (ud1) stack order directly.
        LDD    STATE
        BEQ    ILOOP            ; interpreting: already correctly placed,
            ; nothing more to do
        ; compiling: need two literals compiled in the order they
        ; must execute at runtime - low first (lands deep), high
        ; second (lands on top). UDHI is currently on top of U, so
        ; pop it aside first, freeing UDLO to be compiled first.
        PULU   D
        STD    MSCR4            ; stash UDHI - confirmed safe scratch,
            ; untouched by CCALL/CODECOMMA
        LDD    #LIT
        PSHU   D
        JSR    CCALL
        JSR    CODECOMMA        ; compiles UDLO (still on U underneath)
        LDD    #LIT
        PSHU   D
        JSR    CCALL
        LDD    MSCR4
        PSHU   D
        JSR    CODECOMMA        ; compiles UDHI
        BRA    ILOOP

BADWORD: JSR    COUNT
        JSR    TYPE
        LDD    #-13
        PSHU   D
        JSR    THROW

IDONE:   PULU   X                ; BUG FIX: WORD always pushes a c-addr (WORDBUF),
            ; even via its EMPTY branch - this path used to
            ; branch straight here via a bare peek (LDX ,U,
            ; never popped), stranding that address on U.
            ; JSR FIND (the other path) consumes it via its
            ; own PULU X; this does the same here, matching
            ; that same convention rather than inventing a
            ; different one. Confirmed via MAME debugger:
            ; a spurious WORDBUF address was found sitting
            ; on top of otherwise-correct stack contents.

        RTS

WORD:    PULU   D
        STB    DELIM
        LDD    TOIN
        LDX    SRCADDR
        LEAX   D,X
        LDD    SRCLEN
        SUBD   TOIN
        TFR    D,Y

SKIPLP:  CMPY   #0
        BEQ    EMPTY
        LDA    ,X
        CMPA   DELIM
        BNE    STARTW
        LEAX   1,X
        LEAY   -1,Y
        BRA    SKIPLP

STARTW:  STX     WSTART
        LDB     #0

SCANLP:  CMPY   #0
        BEQ    ENDW
        LDA    ,X
        CMPA   DELIM
        BEQ    CONSUME

```

```

CMPB  #WORDMAXCHARS ; REDESIGN: was "CMPB #31", capping WORD
                        ; at 31 characters regardless of what it was
                        ; parsing - a plain word, a defined name, or
                        ; the text of a compiled/interpreted S"
                        ; string, all fed through the same scan.
                        ; That cap came from WORDBUF's own fixed
                        ; 33-byte allocation (1 count byte + 32 data
                        ; bytes, with 1 byte of that never actually
                        ; used by this check), entirely unrelated to
                        ; CODEHERE or PAD. Traced via MAME: a longer
                        ; S" string was truncated during WORD's own
                        ; scan, before either S"'s interpreted-mode
                        ; (PAD) or compiled-mode (CODEHERE) storage
                        ; path ever got a chance to matter - the
                        ; PAD-based S" redesign a few turns ago
                        ; didn't help here because this is an
                        ; earlier stage entirely. Now uses the
                        ; CODEHERE-to-PAD gap directly, matching the
                        ; traditional fig-Forth layout (WORD's own
                        ; buffer at HERE, growing toward PAD, with
                        ; the pictured numeric output buffer at the
                        ; opposite end growing back toward HERE) -
                        ; WORDMAXCHARS reserves HOLDMINSIZE bytes at
                        ; the PAD end for that buffer, so the two
                        ; don't collide even though ANS itself would
                        ; permit them to (3.3.3.6: "the regions
                        ; returned by WORD and #> may overlap in
                        ; memory"). Confirmed via MAME debugger.

BEQ    ENDW
LEAX   1,X
LEAY   -1,Y
INCB
BRA    SCANLP

CONSUME: LEAX 1,X
          LEAY -1,Y
ENDW:    PSHS B ; BUG FIX: B holds the true character count from
                ; SCANLP's own INCB loop, but B is D's low byte -
                ; TFR X,D below would silently destroy it before
                ; it's stored as the length byte. Save it here,
                ; restore it right before STB ,X+. Affects every
                ; token followed by more input on the same line
                ; (terminated via CONSUME, not by running out of
                ; buffer) - the stored length was TOIN's delta
                ; instead of the true count, one too many (the
                ; consumed delimiter), so the copy loop below
                ; would also copy one byte past the token's real
                ; end. Confirmed via MAME debugger.

TFR    X,D
SUBD   SRCADDR
STD    TOIN
PULS   B

LDX    CODEHERE ; REDESIGN: was "LDX #WORDBUF" - now writes
                ; at CODEHERE directly (see SCANLP above for
                ; the full reasoning). CODEHERE itself is
                ; NOT advanced by this - matches the ANS
                ; transient-region contract, where WORD's
                ; region is expected to be overwritten by
                ; whatever gets compiled/allocated next.

STB    ,X+
LDY    WSTART
COPYLP: TSTB
          BEQ COPYDONE
          LDA ,Y+
          STA ,X+
          DECB
          BRA COPYLP
COPYDONE: LDX CODEHERE ; REDESIGN: was "LDX #WORDBUF", matching
                        ; the copy destination above.

PSHU X

```

```

        RTS

EMPTY:   LDX   CODEHERE   ; REDESIGN: was "LDX #WORDBUF" - same reason.
        CLR   ,X
        PSHU  X
        RTS

FIND:    PULU  X
        LDA   ,X
        STA   SLEN
        LEAX  1,X
        STX   SNAMEP

        LDD   LATEST
        STD   FNDPTR

FFLOOP:  LDD   FNDPTR
        BEQ   NOTFOUND
        STD   HDRPTR
        TFR   D,X
        LDA   ,X
        STA   HDRFLAGS
        BITA  #$40
        BNE   FNEXT
        ANDA  #$1F
        CMPA  SLEN
        BNE   FNEXT
        LEAX  1,X
        LDY   SNAMEP
        LDB   SLEN
        BEQ   FMATCH
CMPLP:   LDA   ,X+
        CMPA  ,Y+
        BNE   FNEXT
        DECB
        BNE   CMPLP

FMATCH:  LDX   HDRPTR
        LEAX  1,X
        LDB   HDRFLAGS
        ANDB  #$1F
        CLRA
        LEAX  D,X
        LEAX  2,X
        LDD   ,X
        PSHU  D
        LDA   HDRFLAGS
        BITA  #$80
        BEQ   FISNORM
        LDD   #1
        BRA   FPUSH
FISNORM: LDD   #-1
FPUSH:   PSHU  D
        RTS

FNEXT:   LDX   HDRPTR
        LEAX  1,X
        LDB   HDRFLAGS
        ANDB  #$1F
        CLRA
        LEAX  D,X
        LDD   ,X
        STD   FNDPTR
        BRA   FFLoop

NOTFOUND: LDX  SNAMEP
        LEAX  -1,X
        PSHU  X
        LDD   #0
        PSHU  D
        RTS

```

```

UDMULADD: STB  CARRY
           LDA  BASE+1
           STA  MULBASE
           LDA  UDLO+1
           LDB  MULBASE
           MUL
           ADDB CARRY
           BCC  UM0
           INCA
UM0:       STB  UDLO+1
           STA  CARRY
           LDA  UDLO
           LDB  MULBASE
           MUL
           ADDB CARRY
           BCC  UM1
           INCA
UM1:       STB  UDLO
           STA  CARRY
           LDA  UDHI+1
           LDB  MULBASE
           MUL
           ADDB CARRY
           BCC  UM2
           INCA
UM2:       STB  UDHI+1
           STA  CARRY
           LDA  UDHI
           LDB  MULBASE
           MUL
           ADDB CARRY
           BCC  UM3
           INCA
UM3:       STB  UDHI
           RTS

NUMLOOP:  LDD  NCNT
           BEQ  NLDONE
           LDX  NADDR
           LDA  ,X
           CMPA #'0'
           BLO  NLDONE
           CMPA #'9'
           BHI  NLALPHA
           SUBA #'0'
           BRA  NLGOT
NLALPHA:  ANDA #$DF
           CMPA #'A'
           BLO  NLDONE
           CMPA #'Z'
           BHI  NLDONE
           SUBA #'A' - 10
NLGOT:    CMPA BASE+1
           BHS  NLDONE
           TFR  A,B
           JSR  UDMULADD
           LDX  NADDR
           LEAX 1,X
           STX  NADDR
           LDD  NCNT
           SUBD #1
           STD  NCNT
           BRA  NUMLOOP
NLDONE:   RTS

TONUMBER: PULU D
           STD  NCNT
           PULU D
           STD  NADDR
           PULU D

```

```

    STD  UDHI
    PULU D
    STD  UDLO
    JSR  NUMLOOP
    LDD  UDLO
    PSHU D
    LDD  UDHI
    PSHU D
    LDX  NADDR
    PSHU X
    LDD  NCNT
    PSHU D
    RTS

NUMBERQ: PULU  X
          STX   CADDR
          LDA   ,X
          BEQ   NQBAD
          STA   CNTREM
          LEAX  1,X

; NEW: ANS double-number input convention (forth-standard.org/
; standard/usage#usage:numbers, 3.4.1.3) - a number immediately
; followed by a decimal point converts to a double-cell number
; instead of single-cell ("1234" -> 1234; "1234." -> 1234 0).
; Detected here to exclude the trailing '.' from CNTREM before
; NUMLOOP runs, so it isn't mistaken for a digit. The decision
; itself is re-derived independently at the end of this routine
; (from CADDR and the original count, both still unchanged at
; that point - confirmed nothing else in this routine, including
; NUMLOOP/UDMULADD, ever touches CADDR) rather than remembered
; in a new persistent flag here - GLOBALS is fully packed at
; 256/256 bytes, and this avoids needing to carve out a byte
; for it.
    LDB  CNTREM
    DECB
    LDA  B,X
    CMPA #'. '
    BNE  NQNOSIGN
    DEC  CNTREM
    BEQ  NQBAD          ; lone "." with nothing before it

NQNOSIGN: CLR  NUMNEG
          LDA  ,X
          CMPA #'-'
          BNE  NQNOSIGN2
          COM  NUMNEG
          LEAX 1,X
          DEC  CNTREM
          BEQ  NQBAD

NQNOSIGN2: STX  NADDR
          CLRA
          LDB  CNTREM
          STD  NCNT
          LDD  #0
          STD  UDHI
          STD  UDLO

          JSR  NUMLOOP

          LDD  NCNT
          BNE  NQBAD

; BUG FIX (as part of this same change): was "LDD UDLO / TST
; NUMNEG / BEQ NQPOS / COMA / COMB / ADDD #1" - negated only
; the low 16 bits. Harmless for single-cell numbers (the high
; cell was never returned before), but wrong once returning a
; genuine 32-bit value for the double case - a negative double
; needs the full 32-bit two's-complement negation, with any
; carry out of the low-cell increment propagated into the

```

```

; high cell.
TST  NUMNEG
BEQ  NQPOS32
LDD  UDLO
COMA
COMB
ADDD  #1
STD  UDLO
PSHS  CC          ; BUG FIX: save the TRUE carry from the
                  ; addition above, before it gets destroyed.
                  ; COM (used on UDHI just below) unconditionally
                  ; SETS carry=1 on the 6809, regardless of its
                  ; operand - a documented quirk, but one that
                  ; silently overwrote the real carry here
                  ; before the BCC check could ever read it,
                  ; making the high-cell increment run
                  ; UNCONDITIONALLY instead of only when the
                  ; low-cell addition actually overflowed.
                  ; Confirmed via MAME: "-123." returned high
                  ; cell 0 instead of -1, and "-123456."
                  ; returned -1 instead of -2 - both exactly
                  ; matching "always increment" rather than
                  ; "increment only on genuine carry".

LDD  UDHI
COMA
COMB
PULS  CC          ; restore the TRUE carry
BCC  NQSTOREHI    ; BUG FIX: this used to be "BCC NQPOS32",
                  ; branching all the way PAST "STD UDHI"
                  ; when there's no carry - the complemented
                  ; value was computed in D but never
                  ; actually stored back to UDHI in that
                  ; case, leaving it at its stale, pre-
                  ; negation value. Masked by the carry bug
                  ; above in practice (that bug meant carry
                  ; was always seen as set, so this branch
                  ; was never actually taken, and STD UDHI
                  ; always ran) - fixing the carry check
                  ; alone would have made this second,
                  ; previously-latent bug live. Now branches
                  ; only past the "+1", never past the
                  ; store itself - the store always runs,
                  ; either way.

ADDD  #1
NQSTOREHI: STD  UDHI

NQPOS32: ; re-derive from CADDR whether the ORIGINAL last character
; was '.' - determines which return convention to use
LDX  CADDR
LDB  ,X
DECB
LEAX  1,X
LDA  B,X
CMPA  #'.'
BNE  NQSINGLE

; DOUBLE success: push low, high, then a distinct success
; code (1, not -1) so TRYNUM can tell single and double
; apart without needing a separate flag of its own either
LDD  UDLO
PSHU  D
LDD  UDHI
PSHU  D
LDD  #1
PSHU  D
RTS

NQSINGLE: ; SINGLE success - unchanged from the original convention
LDD  UDLO
PSHU  D
LDD  #-1

```

```

        PSHU D
        RTS

NQBAD:  LDX  CADDR
        PSHU X
        LDD  #0
        PSHU D
        RTS

```

## 8.11 QUERY / ACCEPT / EXPECT / KEY / KEY? / EMIT

```

; =====
; SECTION 10: QUERY / ACCEPT / EXPECT / KEY / KEY? / EMIT
; =====
; IFEQ SERIALPOLL ; >>>>>>>>>
KEY:    LDA  INHEAD
        CMPA INTAIL
        BEQ  KEY
        LDX  #INBUF
        LDB  INTAIL
        LDA  B,X
        INCB
        ANDB #INBUFSZ-1
        STB  INTAIL
        PSHS A                ; stash the char on the return stack across
                                ; the call - JSR/RTS is self-balancing, so
                                ; this needs no dedicated scratch global

        JSR  RTSCHECKLO
        PULS A
        TFR  A,B
        CLRA
        PSHU D
        RTS

KEYQ:   LDA  INHEAD
        CMPA INTAIL
        BNE  KQTRUE
        LDD  #FALSEV
        PSHU D
        RTS

KQTRUE: LDD  #TRUEV
        PSHU D
        RTS

EMIT:   PULU D
        STB  EMITCH
EMITWT: LDB  OUTHEAD
        INCB
        ANDB #OUTBUFSZ-1
        CMPB OUTTAIL
        BEQ  EMITWT
        LDX  #OUTBUF
        LDB  OUTHEAD
        LDA  EMITCH
        STA  B,X
        INCB
        ANDB #OUTBUFSZ-1
        STB  OUTHEAD
        TST  RTSSTATE
        BNE  EMITNORTS        ; RTS is asserted high - leave ACIACR alone;
                                ; output stays queued until RTS drops low,
                                ; at which point RTSCHECKLO re-enables TX
                                ; interrupt itself if OUTBUF still has data

        LDA  #CR_RXTX
        STA  ACIACR
EMITNORTS: RTS

        ELSE ; <<<<<>>>>>
; -----

```

[illegible]

```

ABKSP:  LDD  ACNT
        BEQ  ALOOP
        SUBD #1
        STD  ACNT
        LDD  #8
        PSHU D
        JSR  EMIT
        LDD  #32
        PSHU D
        JSR  EMIT
        LDD  #8
        PSHU D
        JSR  EMIT
        BRA  ALOOP

ADONE:  LDD  ACNT
        PSHU D
        RTS

EXPECTW: JSR  ACCEPT
        PULU D
        STD  SPAN
        RTS

QUERY:  LDX  #TIBBUF
        PSHU X
        LDD  #TIBBUFL
        PSHU D
        JSR  ACCEPT
        PULU D
        STD  NTIB
        STD  SRCLEN
        LDD  #TIBBUF
        STD  SRCADDR
        LDD  #0
        STD  SRCID
        STD  TOIN
        RTS

```

## 8.12 CATCH / THROW

```

; =====
; SECTION 11: COLON / SEMICOLON support already in section 8
; (COLON/SEMI) - CATCH/THROW, CFERR
; =====
CFERR:  LDD  #-22
        PSHU D
        JSR  THROW
        RTS

CATCH:  PULU X
        LDD  HANDLER
        PSHS D
        PSHS U
        TFR  S,D
        STD  HANDLER

        JSR  ,X

        LEAS 2,S
        PULS D
        STD  HANDLER
        LDD  #0
        PSHU D
        RTS

THROW:  PULU D
        CMPD #0
        BEQ  THDONE

```

```

        STD    THROWN
        LDX    HANDLER
        BEQ    THUNCAU

        TFR    X,S
        PULS   D
        TFR    D,U
        PULS   D
        STD    HANDLER

        LDD    THROWN
        PSHU   D
        RTS
THDONE:  RTS

THUNCAU: LDD    THROWN
        PSHU   D
        JMP    ABORT

```

## 8.13 Control Flow

```

; =====
; SECTION 12: CONTROL FLOW (IF/THEN/ELSE, BEGIN family,
; DO/LOOP/+LOOP/I/J/LEAVE/UNLOOP/?DO, EXIT, CASE family)
; =====
PATCH:  PULU   D
        STD    PFIELD
        PULU   D
        STD    PTARGET
        LDD    PTARGET
        SUBD   PFIELD
        LDX    PFIELD
        STD    ,X
        RTS

IF:      LDD    #ZBRANCH
        PSHU   D
        JSR    CCALL
        LDD    #0
        PSHU   D
        JSR    CODECOMMA
        LDD    CODEHERE
        SUBD   #2
        PSHU   D
        LDD    #TAGFWD
        PSHU   D
        RTS

THEN:    PULU   D
        CMPD   #TAGFWD
        BEQ    THOK
        JSR    CFERR

THOK:    PULU   X
        LDD    CODEHERE
        PSHU   D
        PSHU   X
        JSR    PATCH
        RTS

ELSE:    PULU   D
        CMPD   #TAGFWD
        BEQ    ELOK
        JSR    CFERR

ELOK:    PULU   D          ; BUG FIX: same class as LOOP/E0F0K - was PULU X
        STD    MSCR       ; then used via PSHU X after CCALL/CODECOMMA
                          ; below, both of which clobber X internally.
                          ; Saved to scratch (not parked on U, since
                          ; CODEHERE gets pushed onto U further down,

```

```

; between the save and the retrieve).
LDD #BRANCH
PSHU D
JSR CCALL
LDD #0
PSHU D
JSR CODECOMMA
LDD CODEHERE
SUBD #2
STD NEWFLD
LDD CODEHERE
PSHU D
LDD MSCR
PSHU D
JSR PATCH
LDD NEWFLD
PSHU D
LDD #TAGFWD
PSHU D
RTS

BEGIN: LDD CODEHERE
        PSHU D
        LDD #TAGBACK
        PSHU D
        RTS

UNTIL: PULU D
        CMPD #TAGBACK
        BEQ UNOK
        JSR CFERR

UNOK: PULU D ; BUG FIX: same class as LOOP - was PULU X then
        PSHU D ; TFR X,D after CCALL/CODECOMMA, both of which
                ; clobber X internally. Parked on U instead.

        LDD #ZBRANCH
        PSHU D
        JSR CCALL
        LDD #0
        PSHU D
        JSR CODECOMMA
        LDD CODEHERE
        SUBD #2
        STD PFIELD
        PULU D
        PSHU D
        LDD PFIELD
        PSHU D
        JSR PATCH
        RTS

AGAIN: PULU D
        CMPD #TAGBACK
        BEQ AGOK
        JSR CFERR

AGOK: PULU D ; BUG FIX: same class as LOOP - was PULU X then
        PSHU D ; TFR X,D after CCALL/CODECOMMA, both of which
                ; clobber X internally. Parked on U instead.

        LDD #BRANCH
        PSHU D
        JSR CCALL
        LDD #0
        PSHU D
        JSR CODECOMMA
        LDD CODEHERE
        SUBD #2
        STD PFIELD
        PULU D
        PSHU D
        LDD PFIELD
        PSHU D
        JSR PATCH

```

```

      RTS

WHILE:  LDD  #ZBRANCH
        PSHU D
        JSR  CCALL
        LDD  #0
        PSHU D
        JSR  CODECOMMA
        LDD  CODEHERE
        SUBD #2
        PSHU D
        LDD  #TAGFWD
        PSHU D
      RTS

REPEAT: PULU D
        CMPD #TAGFWD
        BEQ  RPOK1
        JSR  CFERR
RPOK1:  PULU X
        STX  NEWFLD
        PULU D
        CMPD #TAGBACK
        BEQ  RPOK2
        JSR  CFERR
RPOK2:  PULU D      ; BUG FIX: same class as LOOP - was PULU X then
        PSHU D      ; TFR X,D after CCALL/CODECOMMA, both of which
                    ; clobber X internally. Parked on U instead.

        LDD  #BRANCH
        PSHU D
        JSR  CCALL
        LDD  #0
        PSHU D
        JSR  CODECOMMA
        LDD  CODEHERE
        SUBD #2
        STD  PFIELD
        PULU D
        PSHU D
        LDD  PFIELD
        PSHU D
        JSR  PATCH
        LDD  CODEHERE
        PSHU D
        LDD  NEWFLD
        PSHU D
        JSR  PATCH
      RTS

RECURSE: LDX  LATEST
          LDA  ,X
          STA  HDRFLAGS
          LEAX 1,X
          LDB  HDRFLAGS
          ANDB #$1F
          CLRA
          LEAX D,X
          LEAX 2,X
          LDD  ,X
          PSHU D
          JSR  CCALL
          RTS

DO:      LDD  #DOSETUP
          PSHU D
          JSR  CCALL
          LDD  CODEHERE
          PSHU D
          LDD  #TAGDO
          PSHU D
          RTS

```

```

DOSETUP: PULU D
          STD MSCR
          PULU D
          STD MSCR2
          PULS X
          LDD #0
          PSHS D
          LDD MSCR2
          PSHS D
          LDD MSCR
          PSHS D
          PSHS X
          RTS

IWORD:   LDD 2,S      ; BUG FIX: PULS X/PSHS X used to bracket this
          PSHU D      ; read, shifting S by 2 first - so "2,S" landed
          RTS         ; on the limit (what's really at 4,S unshifted)
                  ; instead of the index. The pair served no
                  ; purpose (nothing needed offset 0 for anything
                  ; here) - removed, so 2,S directly and
                  ; correctly targets the index DOSETUP pushed
                  ; there. Confirmed via MAME debugger: "I"
                  ; returned the loop limit on every iteration.

JWORD:   LDD 8,S      ; CORRECTION: an earlier turn changed this to
          PSHU D      ; 10,S, which was itself wrong - it assumed
          RTS         ; DOSETUP's own JSR-pushed return address
                  ; persists on S once each DOSETUP finishes.
                  ; It doesn't: DOSETUP's own RTS pops and
                  ; consumes it to jump into the loop body, so
                  ; it never actually sits on S for J to count
                  ; past. Re-traced by counting every real
                  ; push/pop across a nested D0: after the
                  ; inner DOSETUP finishes, S is [inner-index@0]
                  ; [inner-limit@2][inner-leave@4][outer-
                  ; index@6][outer-limit@8][outer-leave@10] -
                  ; J's own JSR pushes one more cell on top,
                  ; landing outer-index at 8, not 10. Offset 10
                  ; read the outer loop's limit instead - a
                  ; fixed value every iteration, exactly
                  ; matching the observed bug (J stuck at the
                  ; limit, never progressing). Confirmed via a
                  ; real MULT-TABLE test on MAME.

LEAVE:   LDD #TRUEV   ; BUG FIX (earlier): was PULS X first, shifting
          STD 6,S     ; S by 2 before this write, landing 2 bytes
          RTS         ; past the LEAVE flag DOSETUP actually pushed.
                  ; Fixed by deferring PULS X to after the
                  ; write - but that left a PULS X/PSHS X pair
                  ; that had become a genuine no-op (nothing
                  ; between them touches S or X, and this
                  ; routine never uses X's value for anything,
                  ; unlike D0TEST's own deferred PULS X, which
                  ; feeds LDD ,X/LEAX D,X afterward). Removed,
                  ; per the parallel 68000 port's observation,
                  ; confirmed by inspection.

LOOP:    PULU D
          CMPD #TAGDO
          BEQ LOOPOK
          JSR CFERR

LOOPOK:  PULU D      ; BUG FIX: was PULU X, then TFR X,D to retrieve
          PSHU D     ; it after CCALL/CODECOMMA below - but both of
                  ; those clobber X internally (CCALL's own LDX
                  ; CODEHERE, CODECOMMA's LDX #CODEHERE via
                  ; APPENDCELL), so X no longer held the target
                  ; by the time TFR X,D ran - it held the address
                  ; of the CODEHERE variable itself, corrupting
                  ; the branch displacement PATCH computed.
                  ; Parking the target on U instead (untouched by

```

```

; CCALL/CODECOMMA, which only use D/X/A) keeps
; it safe across both calls; retrieved below via
; PULU D in place of the old TFR X,D. Confirmed
; via MAME debugger: the compiled displacement
; came out as +8, executing DOTEST's return to
; a near-zero, invalid address.

LDD    #DOTEST
PSHU   D
JSR    CCALL
LDD    #0
PSHU   D
JSR    CODECOMMA
LDD    CODEHERE
SUBD   #2
STD    PFIELD
PULU   D
PSHU   D
LDD    PFIELD
PSHU   D
JSR    PATCH

LDD    ,U
CMPD   #TAGFWD
BNE    LOOPDONE
PULU   D
PULU   X
LDD    CODEHERE
PSHU   D
PSHU   X
JSR    PATCH
LOOPDONE: RTS

DOTEST: LDD    6,S      ; BUG FIX: PULS X used to run FIRST, shifting S
        BNE    DTEXIT  ; by 2 before these offset reads - "6,S" ended
        LDD    2,S      ; up reading past the 3 cells DOSETUP pushed,
        ADDD   #1       ; into whatever return address was already on
        STD    2,S      ; S below them (e.g. EXECUTE's own), which is
        CMPD   4,S      ; almost always nonzero - so BNE DTEXIT fired
        BEQ    DTEXIT   ; on the very first pass, exiting immediately.
        PULS   X        ; X (the return address to the FDB <offset>
        LDD    ,X       ; field, needed for the back-branch) is only
        LEAX   D,X      ; actually needed here and in DTEXIT below -
        PSHS   X        ; deferred to each path separately instead of
        RTS           ; popped once up front. Confirmed via MAME
                        ; debugger: DOTEST was tested at $E184, a
                        ; leftover EXECUTE return address, instead of
                        ; the intended $0000 LEAVE flag.

DTEXIT: PULS   X
        LEAX   2,X
        LEAS   6,S
        PSHS   X
        RTS

PLUSLOOP: PULU   D
        CMPD   #TAGDO
        BEQ    PLOOPOK
        JSR    CFERR
PLOOPOK: PULU   D      ; BUG FIX: same class as LOOP above - was PULU X
        PSHU   D      ; then TFR X,D after CCALL/CODECOMMA, both of
                        ; which clobber X internally. Parked on U
                        ; instead, retrieved below via PULU D.

        LDD    #DOPLUSTEST
        PSHU   D
        JSR    CCALL
        LDD    #0
        PSHU   D
        JSR    CODECOMMA
        LDD    CODEHERE
        SUBD   #2
        STD    PFIELD
        PULU   D

```

```

    PSHU D
    LDD PFIELD
    PSHU D
    JSR PATCH

    LDD ,U
    CMPD #TAGFWD
    BNE PLOOPDONE
    PULU D
    PULU X
    LDD CODEHERE
    PSHU D
    PSHU X
    JSR PATCH
PLOOPDONE: RTS

DOPLUSTEST: LDD 6,S      ; BUG FIX: same class as DOTEST - PULS X used to
                  BNE DPTEXTIT ; run first, shifting S by 2 before every one of
    PULU D          ; these offset reads (6,S/2,S/4,S), landing past
    STD MSCR        ; the 3 cells D0SETUP pushed. Deferred PULS X to
    LDD 2,S          ; each path separately below, same fix as DOTEST.
    SUBD 4,S
    STD MSCR2
    LDD 2,S
    ADDD MSCR
    STD 2,S
    SUBD 4,S
    STD MSCR3
    LDA MSCR2
    LDB MSCR3
    PSHS B
    EORA ,S+          ; was "EORA B" - not valid 6809 syntax (no
                      ; register-to-register EORA); push B, then
                      ; operate through ,S+ - the standard 6809
                      ; idiom for adding/combining two registers

    BMI DPTEXTIT
    LDD MSCR3
    BEQ DPTEXTIT
    PULS X
    LDD ,X
    LEAX D,X
    PSHS X
    RTS
DPTEXTIT: PULS X
          LEAX 2,X
          LEAS 6,S
          PSHS X
          RTS

QDO: LDD #QD0SETUP
     PSHU D
     JSR CCALL
     LDD #0
     PSHU D
     JSR CODECOMMA
     LDD CODEHERE
     SUBD #2
     PSHU D
     LDD #TAGFWD
     PSHU D
     LDD CODEHERE
     PSHU D
     LDD #TAGDO
     PSHU D
     RTS

QD0SETUP: PULU D
          STD MSCR
          PULU D
          STD MSCR2
          PULS X

```

```

        LDD  MSCR2
        CMPD MSCR
        BNE  QDBUILD
        LDD  ,X
        LEAX D,X
        PSHS X
        RTS
QDBUILD: LEAX 2,X
        LDD  #0
        PSHS D
        LDD  MSCR2
        PSHS D
        LDD  MSCR
        PSHS D
        PSHS X
        RTS

UNLOOP: RTS          ; DESIGN CHANGE: was PULS X/LEAS 6,S/PSHS X/RTS
                    ; (a real discard of the loop-control frame).
                    ; EXIT's own EXITUNLOOP mechanism already
                    ; discards the frame automatically and
                    ; correctly on every path - traced and
                    ; confirmed: with no enclosing D0 (compiled
                    ; count 0), with one enclosing D0 and no
                    ; prior UNLOOP (discards the full 8-byte
                    ; frame correctly). The one combination that
                    ; broke was UNLOOP immediately before EXIT -
                    ; UNLOOP discarding 6 of the 8 bytes itself,
                    ; then EXITUNLOOP unconditionally discarding
                    ; a full 8 more, overshooting into whatever
                    ; sat below (typically the caller's own
                    ; return address) and branching into random
                    ; memory on return. Since UNLOOP has no
                    ; other legitimate use than immediately
                    ; preceding an exit from the definition, and
                    ; EXIT already handles that correctly on its
                    ; own, UNLOOP is now a true no-op rather than
                    ; a second, conflicting discard mechanism.

EXIT:    LDD  #0
        STD  EXITCNT
        TFR  U,D
        STD  EXITPTR
EXSCAN:  LDD  EXITPTR
        CMPD CSP
        BEQ  EXSCANDONE
        LDX  EXITPTR
        LDD  ,X
        CMPD #TAGDO
        BNE  EXNOTDO
        LDD  EXITCNT
        ADDD #1
        STD  EXITCNT
EXNOTDO: LDD  EXITPTR
        ADDD #4
        STD  EXITPTR
        BRA  EXSCAN
EXSCANDONE:
        LDD  #EXITUNLOOP
        PSRU D
        JSR  CCALL
        LDD  EXITCNT
        PSRU D
        JSR  CODECOMMA
        RTS

EXITUNLOOP: PULS X
        LDD  ,X
        TFR  D,Y
EULoop:  CMPY #0
        BEQ  EUDONE

```

```

        LEAS 8,S
        LEAY -1,Y
        BRA  EULLOOP
EUDONE:  PULS Y
        JMP  ,Y

CASEW:  LDD  #TAGCASE ; BUG FIX: was LDD #0/PSHU D here first, pushing
        PSHU D        ; a value nothing downstream ever consumes -
        RTS           ; OF/ENDOF never read this deep, and ENDCASE's
                       ; scan loop stops the instant it sees TAGCASE,
                       ; never popping past it. Left one cell
                       ; permanently stranded below CSP's expected
                       ; depth, so ";" always saw a mismatch and
                       ; threw -22, even for the simplest CASE...
                       ; ENDCASE with no OF clauses at all. Removed;
                       ; confirmed by inspection that nothing reads
                       ; or depends on it anywhere in OF/ENDOF/
                       ; ENDCASE's logic.

OF:      LDD  #OVER
        PSHU D
        JSR  CCALL
        LDD  #EQUALW
        PSHU D
        JSR  CCALL
        LDD  #ZBRANCH
        PSHU D
        JSR  CCALL
        LDD  #0
        PSHU D
        JSR  CODECOMMA
        LDD  CODEHERE
        SUBD #2
        PSHU D
        LDD  #DROP
        PSHU D
        JSR  CCALL
        LDD  #TAGOF
        PSHU D
        RTS

ENDOF:   PULU D
        CMPD #TAGOF
        BEQ  EOFOK
        JSR  CFERR
EOFOK:   PULU D ; BUG FIX: same class as LOOP - was PULU X then
        STD  MSCR ; used via PSHU X after CCALL/CODECOMMA below,
                  ; both of which clobber X internally. Saved to
                  ; scratch instead (not parked on U, since
                  ; CODEHERE gets pushed onto U further down,
                  ; between the save and the retrieve below -
                  ; parking on U would retrieve that instead).

        LDD  #BRANCH
        PSHU D
        JSR  CCALL
        LDD  #0
        PSHU D
        JSR  CODECOMMA
        LDD  CODEHERE
        SUBD #2
        STD  NEWFLD
        LDD  CODEHERE
        PSHU D
        LDD  MSCR
        PSHU D
        JSR  PATCH
        LDD  NEWFLD
        PSHU D
        LDD  #TAGENDOF
        PSHU D
        RTS

```

```

ENDCASE: LDD    #DROP
          PSHU   D
          JSR    CCALL
ECLLOOP: PULU   D
          CMPD   #TAGCASE
          BEQ    ECDONE
          CMPD   #TAGENDOF
          BEQ    ECPATCH
          JSR    CFERR
ECPATCH: PULU   X
          LDD    CODEHERE
          PSHU   D
          PSHU   X
          JSR    PATCH
          BRA    ECLLOOP
ECDONE:  RTS

```

## 8.14 Compiling Words

```

; =====
; SECTION 13: COMPILING WORDS (IMMEDIATE/[ ]/'/COMPILE,/
; LITERAL/[ ]/POSTPONE/>BODY, SLITERAL, ABORT")
; =====
STATEW:  LDD    #STATE
          PSHU   D
          RTS

IMMEDIATE: LDX   LATEST
           LDA   ,X
           ORA   #$80
           STA   ,X
           RTS

LBRACKET: LDD    #0
           STD   STATE
           RTS

RBRACKET: LDD    #-1
           STD   STATE
           RTS

TICK:     LDD    #32
           PSHU   D
           JSR    WORD
           JSR    FIND
           PULU   D
           CMPD   #0
           BNE    TICKOK
           PULU   D
           LDD    #-13
           PSHU   D
           JSR    THROW
TICKOK:   RTS

COMPILECOMMA: JMP  CCALL

CCALL:    LDX    CODEHERE      ; BUG FIX: was PULU D first, then LDA #OPJSR -
           LDA    #OPJSR      ; but A is D's high byte, so that LDA silently
           STA    ,X+         ; destroyed the top byte of the address PULU D
           PULU   D           ; had just pulled, and the STD below wrote out
           STD    ,X++        ; the corrupted result - a JSR to a garbage
                               ; target address. Deferring PULU D until after
                               ; STA ,X+ means D is never live at the same time
                               ; A gets reused for the opcode, so nothing
                               ; clobbers it. Confirmed via MAME debugger:
                               ; execution jumped to random memory.

           STX    CODEHERE
           RTS

```

```

LITERALW: LDD #LIT
           PSHU D
           JSR CCALL
           JSR CODECOMMA
           RTS

BRACKTICK: LDD STATE
           BNE BTSTOK
           LDD #-14
           PSHU D
           JSR THROW
BTSTOK:    LDD #32
           PSHU D
           JSR WORD
           JSR FIND
           PULU D
           CMPD #0
           BNE BTOK
           PULU D
           LDD #-13
           PSHU D
           JSR THROW
BTOK:      JSR LITERALW
           RTS

POSTPONEW: LDD STATE
           BNE PPSTOK
           LDD #-14
           PSHU D
           JSR THROW
PPSTOK:    LDD #32
           PSHU D
           JSR WORD
           JSR FIND
           PULU D
           CMPD #0
           BNE PPFFOUND
           PULU D
           LDD #-13
           PSHU D
           JSR THROW
PPFFOUND:  CMPD #1
           BEQ PPIMM
           JSR LITERALW
           LDD #COMPILECOMMA
           PSHU D
           JSR CCALL
           RTS
PPIMM:     JSR COMPILECOMMA
           RTS

TOBODY:    PULU D
           ADDD #5
           TFR D,X
           LDD ,X
           PSHU D
           RTS

SLITERALW: LDD STATE
           BNE SLSTOK
           LDD #-14
           PSHU D
           JSR THROW
SLSTOK:    PULU D
           STD SCNT
           PULU D
           STD SPTR
           LDD #DOSTR
           PSHU D
           JSR CCALL

```

```

        LDX CODEHERE
        LDB SCNT+1
        STB ,X+
        LDY SPTR
        LDB SCNT+1
        BEQ SLEND
SLCPY:   LDA ,Y+
        STA ,X+
        DECB
        BNE SLCPY
SLEND:   STX CODEHERE
        RTS

DOABORTQUOTE: PULS X
        LDB ,X
        LEAX 1,X
        STX SPTR
        CLRA
        STD SCNT
        LDX SPTR
        LDB SCNT+1
        LEAX B,X
        PULU D
        CMPD #0
        BNE AQTHROW
        PSHS X
        RTS
AQTHROW: LDD SPTR
        PSHU D
        LDD SCNT
        PSHU D
        JSR TYPE
        PSHS X
        LDD #-2
        PSHU D
        JMP THROW

ABORTQUOTE: LDD STATE
        BNE AQSTOK
        LDD #-14
        PSHU D
        JSR THROW
AQSTOK:   LDD #34
        PSHU D
        LDD CODEHERE ; BUG FIX: same class as SQUOTE/DOTQUOTE -
        ADDD #3       ; reserve 3 bytes ahead of WORD's write so
        STD CODEHERE  ; the trampoline compiled below doesn't
                        ; overwrite the text it's about to stage.
        JSR WORD
        PULU X
        LDA ,X
        STA SCNT
        LEAX 1,X
        STX SPTR
        LDD CODEHERE ; restore
        SUBD #3
        STD CODEHERE
        LDD #DOABORTQUOTE
        PSHU D
        JSR CCALL
        LDX CODEHERE
        LDA SCNT
        STA ,X+
        LDY SPTR
        LDB SCNT
        BEQ AQEND
AQCPY:   LDA ,Y+
        STA ,X+
        DECB
        BNE AQCPY
AQEND:   STX CODEHERE

```

```

                RTS

BLW:    LDD    #32
        PSHU   D
        RTS

TOINW:   LDD    #TOIN
        PSHU   D
        RTS

SPANW:   LDD    #SPAN
        PSHU   D
        RTS

TIBW:    LDD    #TIBBUF
        PSHU   D
        RTS

NTIBW:   LDD    #NTIB
        PSHU   D
        RTS

```

## 8.15 Stack Manipulation

```

; =====
; SECTION 14: STACK MANIPULATION (Core + Core Ext + return stack)
; =====
DUP:     LDD    ,U
        PSHU   D
        RTS

DROP:    LEAU   2,U
        RTS

SWAP:    LDD    ,U
        LDX    2,U
        STX    ,U
        STD    2,U
        RTS

OVER:    LDD    2,U
        PSHU   D
        RTS

ROT:     LDD    ,U
        LDX    2,U
        LDY    4,U
        STY    ,U
        STD    2,U
        STX    4,U
        RTS

QDUP:    LDD    ,U
        CMPD   #0
        BEQ    QDUPDONE
        PSHU   D
QDUPDONE: RTS

DEPTH:   TFR    U,D
        STD    DEPTHTMP
        LDD    #SP0
        SUBD   DEPTHTMP
        LSRA
        RORB
        PSHU   D
        RTS

DDUP:    LDD    2,U
        LDX    ,U

```

```

        PSHU D
        PSHU X
        RTS

DDROP:  LEAU 4,U
        RTS

DSWAP:  LDD ,U
        STD MSCR
        LDD 2,U
        LDX 4,U
        LDY 6,U
        STD 6,U
        STX ,U
        STY 2,U
        LDD MSCR
        STD 4,U
        RTS

DOVER:  LDD 6,U
        LDX 4,U
        PSHU D
        PSHU X
        RTS

NIP:    LDD ,U
        STD 2,U
        LEAU 2,U
        RTS

TUCK:   LDD ,U
        LDX 2,U
        PSHU D
        STX 2,U
        STD 4,U
        RTS

PICK:   PULU D
        LSLB
        ROLA
        LDD D,U
        PSHU D
        RTS

ROLL:   PULU D
        CMPD #0
        BEQ ROLLDONE
        LSLB
        ROLA
        STD RDST
        LEAX D,U
        LDD ,X
        STD RVAL
RLOOP:  LDD RDST
        CMPD #2
        BLT RSTORE
        LEAY D,U
        SUBD #2
        LEAX D,U
        LDD ,X
        STD ,Y
        LDD RDST
        SUBD #2
        STD RDST
        BRA RLOOP
RSTORE: LDD RVAL
        STD ,U
ROLLDONE: RTS

DROT:   LDD 10,U
        STD TR1

```

```

LDD 8,U
STD TR2
LDD 6,U
STD 10,U
LDD 4,U
STD 8,U
LDD 2,U
STD 6,U
LDD 0,U
STD 4,U
LDD TR2
STD ,U
LDD TR1
STD 2,U
RTS

TOR:  PULU D
      PULS X
      PSHS D
      PSHS X
      RTS

FROMR: PULS X
        PULS D
        PSHS X
        PSHU D
        RTS

RFETCH: PULS X
         LDD ,S
         PSHS X
         PSHU D
         RTS

TWOTOR: PULU D
        STD R2A
        PULU D
        STD R2B
        PULS X
        LDD R2B
        PSHS D
        LDD R2A
        PSHS D
        PSHS X
        RTS

TWOFROMR: PULS X
           PULS D
           STD R2A
           PULS D
           STD R2B
           PSHS X
           LDD R2B
           PSHU D
           LDD R2A
           PSHU D
           RTS

TWO RFETCH: PULS X
            LDD ,S
            STD R2A
            LDD 2,S
            STD R2B
            PSHS X
            LDD R2B
            PSHU D
            LDD R2A
            PSHU D
            RTS

```

## 8.16 Arithmetic

```

; =====
; SECTION 15: ARITHMETIC (single + double + mixed precision)
; =====
PLUS:    PULU    D
          ADDD    ,U
          STD     ,U
          RTS

; MSCR is declared once in the GLOBALS layout above - no local
; redeclaration needed here.
MINUS:   PULU    D
          STD     MSCR
          LDD     ,U
          SUBD    MSCR
          STD     ,U
          RTS

NEGATE:   LDD     ,U
          COMA
          COMB
          ADDD    #1
          STD     ,U
          RTS

ABSW:     LDD     ,U
          BPL     ABSDONE
          COMA
          COMB
          ADDD    #1
          STD     ,U
ABSDONE:  RTS

MIN:       PULU    D
          CMPD    ,U
          BLT     MINISN2
          RTS
MINISN2:   STD     ,U
          RTS

MAX:       PULU    D
          CMPD    ,U
          BGT     MAXISN2
          RTS
MAXISN2:   STD     ,U
          RTS

ONEPLUS:   LDD     ,U
          ADDD    #1
          STD     ,U
          RTS

ONEMINUS:  LDD     ,U
          SUBD    #1
          STD     ,U
          RTS

TWOPLUS:   LDD     ,U
          ADDD    #2
          STD     ,U
          RTS

STAR:      PULU    D
          STD     MSCR
          LDD     ,U
          CLR     MSIGN
          BPL     SNOFLIP1
          COM      MSIGN
          COMA
          COMB

```

```

        ADDD #1
SNOFLIP1: STA MAHI
          STB MALO
          LDD MSCR
          BPL SNOFLIP2
          COM MSIGN
          COMA
          COMB
          ADDD #1
SNOFLIP2: STA MBHI
          STB MBLO
          LDA MALO
          LDB MBLO
          MUL
          STD MRESULT
          LDA MAHI
          LDB MBLO
          MUL
          LDA MRESULT
          PSHS B
          ADDA ,S+          ; was "ADDA B" - not valid 6809 syntax
          STA MRESULT
          LDA MALO
          LDB MBHI
          MUL
          LDA MRESULT
          PSHS B
          ADDA ,S+          ; was "ADDA B" - not valid 6809 syntax
          STA MRESULT
          LDD MRESULT
          TST MSIGN
          BEQ SDONE
          COMA
          COMB
          ADDD #1
SDONE:   STD ,U
          RTS

TWOSTAR: LDD ,U
          ASLB
          ROLA
          STD ,U
          RTS

UDIV16:  CLR DIVREM
          CLR DIVREM+1
          LDB #16
          STB DIVCNT
UD16LOOP: ASL DIVNUM+1
          ROL DIVNUM
          ROL DIVREM+1
          ROL DIVREM
          LDD DIVREM
          SUBD DIVDEN
          BLO UDSKIP
          STD DIVREM
          INC DIVNUM+1
UDSKIP:  DEC DIVCNT
          BNE UD16LOOP
          RTS

DIVCOMMON: PULU D
           STD DIVDEN
           CMPD #0
           BNE DCOK
           LDD #-10
           PSHU D
           JSR THROW
DCOK:     PULU D
           STD DIVNUM
           CLR DVSIGN

```

```

        CLR    DNSIGN
        TST    DIVNUM
        BPL    DNPOS
        COM    DNSIGN
        COM    DVSIGN
        LDD    DIVNUM
        COMA
        COMB
        ADDD   #1
        STD    DIVNUM
DNPOS:   TST    DIVDEN
        BPL    DVPOS
        COM    DVSIGN
        LDD    DIVDEN
        COMA
        COMB
        ADDD   #1
        STD    DIVDEN
DVPOS:   JSR    UDIV16
        LDD    DIVNUM
        TST    DVSIGN
        BEQ    DQPOS
        COMA
        COMB
        ADDD   #1
        STD    DIVNUM
DQPOS:   LDD    DIVREM
        TST    DNSIGN
        BEQ    DCRPOS
        COMA
        COMB
        ADDD   #1
        STD    DIVREM
DCRPOS:  RTS

SLASH:   JSR    DIVCOMMON
        LDD    DIVNUM
        PSHU   D
        RTS

MODW:    JSR    DIVCOMMON
        LDD    DIVREM
        PSHU   D
        RTS

SLASHMOD: JSR    DIVCOMMON
        LDD    DIVREM
        PSHU   D
        LDD    DIVNUM
        PSHU   D
        RTS

TWOSLASH: LDD    ,U
        ASRA
        RORB
        STD    ,U
        RTS

UMUL32:  LDA    MALO
        LDB    MBLO
        MUL
        STD    PRODLO
        CLR    PRODHI
        CLR    PRODHI+1
        LDA    MAHI
        LDB    MBLO
        MUL
        ADDB   PRODLO
        STB    PRODLO
        ADCA   #0
        ADDA   PRODHI+1

```

```

        STA  PRODHI+1
        BCC  UM32A
        INC  PRODHI
UM32A:  LDA  MALO
        LDB  MBHI
        MUL
        ADDB PRODLO
        STB  PRODLO
        ADCA #0
        ADDA PRODHI+1
        STA  PRODHI+1
        BCC  UM32B
        INC  PRODHI
UM32B:  LDA  MAHI
        LDB  MBHI
        MUL
        ADDD PRODHI
        STD  PRODHI
        RTS

UDIV32: CLR  DIVREM
        CLR  DIVREM+1
        LDB  #32
        STB  DIVCNT
UD32LP: ASL  PRODLO+1
        ROL  PRODLO
        ROL  PRODHI+1
        ROL  PRODHI
        ROL  DIVREM+1
        ROL  DIVREM
        LDD  DIVREM
        SUBD DIVDEN
        BLO  UD32SKIP
        STD  DIVREM
        INC  PRODLO+1
UD32SKIP: DEC  DIVCNT
        BNE  UD32LP
        RTS

MNEG32: LDD  PRODLO
        COMA
        COMB
        STD  PRODLO
        LDD  PRODHI
        COMA
        COMB
        STD  PRODHI
        LDD  PRODLO
        ADDD #1
        STD  PRODLO
        BCC  MN32DONE
        LDD  PRODHI
        ADDD #1
        STD  PRODHI
MN32DONE: RTS

STARSLASHCOMMON:
        PULU D
        STD  DIVDEN
        CMPD #0
        BNE  SSOK
        LDD  #-10
        PSHU D
        JSR  THROW
SSOK:   CLR  PSIGN
        TST  DIVDEN
        BPL  SSN3POS
        COM  PSIGN
        LDD  DIVDEN
        COMA
        COMB

```

```

        ADDD #1
        STD  DIVDEN
SSN3POS: LDA #0
        STA  PRSIGN
        PULU D
        STD  MSCR
        TST  MSCR
        BPL  SSN2POS
        COM  PSIGN
        COM  PRSIGN
        LDD  MSCR
        COMA
        COMB
        ADDD #1
        STD  MSCR
SSN2POS: LDA MSCR
        STA  MBHI
        LDA  MSCR+1
        STA  MBLO
        PULU D
        STD  MSCR
        TST  MSCR
        BPL  SSN1POS
        COM  PSIGN
        COM  PRSIGN
        LDD  MSCR
        COMA
        COMB
        ADDD #1
        STD  MSCR
SSN1POS: LDA MSCR
        STA  MAHI
        LDA  MSCR+1
        STA  MALO
        JSR  UMUL32
        JSR  UDIV32
        LDD  PRODLO
        TST  PSIGN
        BEQ  SSQPOS
        COMA
        COMB
        ADDD #1
        STD  PRODLO
SSQPOS: LDD  DIVREM
        TST  PRSIGN
        BEQ  SSRPOS
        COMA
        COMB
        ADDD #1
        STD  DIVREM
SSRPOS: RTS

STARSLASH: JSR  STARSLASHCOMMON
        LDD  PRODLO
        PSHU D
        RTS

STARSLASHMOD: JSR  STARSLASHCOMMON
        LDD  DIVREM
        PSHU D
        LDD  PRODLO
        PSHU D
        RTS

UMSTAR: PULU D
        STD  MSCR
        PULU D
        STA  MAHI
        STB  MALO
        LDD  MSCR
        STA  MBHI

```

```

        STB    MBL0
        JSR    UMUL32
        LDD    PRODLO
        PSHU   D
        LDD    PRODHI
        PSHU   D
        RTS

UMSLASHMOD: PULU D
            STD  DIVDEN
            CMPD #0
            BNE  UMOK
            LDD  #-10
            PSHU D
            JSR  THROW
UMOK:      PULU D
            STD  PRODHI
            PULU D
            STD  PRODLO
            JSR  UDIV32
            LDD  DIVREM
            PSHU D
            LDD  PRODLO
            PSHU D
            RTS

MSTAR:    PULU D
            STD  MSCR
            PULU D
            CLR  MSIGN
            TSTA
            ; BUG FIX: CLR MSIGN unconditionally sets N=0
            ; (CLR always clears to 0), overwriting D's
            ; own flags from the PULU above - and PULU
            ; doesn't set flags on genuine 6809 anyway
            ; (same class as TRYNUM/STOD/SIGN's own fixes
            ; elsewhere in this file), so BPL was always
            ; testing CLR's result, not n1's actual sign.
            ; BPL always branched, meaning n1 was NEVER
            ; negated even when genuinely negative - found
            ; via MAME: TSTMSTAR's two returned values
            ; came back wrong, which turned out to be
            ; this, not a push-order swap in the test
            ; itself. FMSLASHMOD/SMSLASHREM already do
            ; this correctly (explicit TST PRODHI between
            ; their own CLR calls and the branch) - MSTAR
            ; was simply missing the equivalent re-test.

            BPL  MSN1POS
            COM  MSIGN
            COMA
            COMB
            ADDD #1
MSN1POS:  STA  MAHI
            STB  MAL0
            LDD  MSCR
            BPL  MSN2POS
            COM  MSIGN
            COMA
            COMB
            ADDD #1
MSN2POS:  STA  MBHI
            STB  MBL0
            JSR  UMUL32
            TST  MSIGN
            BEQ  MSDONE
            JSR  MNEG32
MSDONE:   LDD  PRODLO
            PSHU D
            LDD  PRODHI
            PSHU D
            RTS

```

```

SMSLASHREM: PULU D
              STD DIVDEN
              CMPD #0
              BNE SMOK
              LDD #-10
              PSHU D
              JSR THROW
SMOK:        PULU D
              STD PRODHI
              PULU D
              STD PRODLO
              CLR DNSIGN
              CLR DVSIGN
              TST PRODHI
              BPL SMDPOS
              COM DNSIGN
              COM DVSIGN
              JSR MNEG32
SMDPOS:      LDD DIVDEN
              BPL SMDVPOS
              COM DVSIGN
              LDD DIVDEN
              COMA
              COMB
              ADDD #1
              STD DIVDEN
SMDVPOS:     JSR UDIV32
              LDD DIVREM
              TST DNSIGN
              BEQ SMRPOS
              COMA
              COMB
              ADDD #1
SMRPOS:      PSHU D
              LDD PRODLO
              TST DVSIGN
              BEQ SMQPOS
              COMA
              COMB
              ADDD #1
SMQPOS:      PSHU D
              RTS

FMSLASHMOD: PULU D
              STD DIVDEN
              CMPD #0
              BNE FMOK
              LDD #-10
              PSHU D
              JSR THROW
FMOK:        PULU D
              STD PRODHI
              PULU D
              STD PRODLO
              CLR DNSIGN
              CLR DVSIGN
              CLR DVOWNSIGN
              TST PRODHI
              BPL FMDPOS
              COM DNSIGN
              COM DVSIGN
              JSR MNEG32
FMDPOS:      LDD DIVDEN
              BPL FMDVPOS
              COM DVSIGN
              COM DVOWNSIGN
              LDD DIVDEN
              COMA
              COMB
              ADDD #1
              STD DIVDEN

```

```

FMDVPOS:   JSR   UDIV32
            TST   DVSIGN
            BEQ   FMNOFLOOR
            LDD   DIVREM
            BEQ   FMNOFLOOR
            LDD   PRODLO
            ADDD  #1
            STD   PRODLO
            LDD   DIVDEN
            SUBD  DIVREM
            STD   DIVREM
FMNOFLOOR:  LDD   DIVREM
            TST   DVOWNSIGN
            BEQ   FMRPOS
            COMA
            COMB
            ADDD  #1
FMRPOS:    PSHU  D
            LDD   PRODLO
            TST   DVSIGN
            BEQ   FMQPOS
            COMA
            COMB
            ADDD  #1
FMQPOS:    PSHU  D
            RTS

```

```

DPLUS:     PULU  D
            STD   MSCR
            PULU  D
            STD   MSCR2
            PULU  D
            STD   MSCR3
            PULU  D
            ADDD  MSCR2
            STD   MSCR4
            BCC   DPNOCY
            LDD   MSCR3
            ADDD  MSCR
            ADDD  #1
            BRA   DPHIDONE
DPNOCY:    LDD   MSCR3
            ADDD  MSCR
DPHIDONE:  STD   MSCR3
            LDD   MSCR4
            PSHU  D
            LDD   MSCR3
            PSHU  D
            RTS

```

```

DMINUS:    PULU  D
            STD   MSCR
            PULU  D
            STD   MSCR2
            PULU  D
            STD   MSCR3
            PULU  D
            SUBD  MSCR2
            STD   MSCR4
            BCC   DMNOBOR
            LDD   MSCR3
            SUBD  MSCR
            SUBD  #1
            BRA   DMHIDONE
DMNOBOR:   LDD   MSCR3
            SUBD  MSCR
DMHIDONE:  STD   MSCR3
            LDD   MSCR4
            PSHU  D
            LDD   MSCR3
            PSHU  D

```

```

      RTS

DNEGATEW: PULU D
          STD  PRODHI
          PULU D
          STD  PRODLO
          JSR  MNEG32
          LDD  PRODLO
          PSHU D
          LDD  PRODHI
          PSHU D
          RTS

DABSW:   PULU D
          STD  PRODHI
          PULU D
          STD  PRODLO
          TST  PRODHI
          BPL  DABSDONE
          JSR  MNEG32
DABSDONE: LDD  PRODLO
          PSHU D
          LDD  PRODHI
          PSHU D
          RTS

MPLUS:   PULU D
          STD  MSCR2
          BPL  MPPOSN
          LDD  #-1
          BRA  MPSIGNED
MPPOSN:   LDD  #0
MPSIGNED: STD  MSCR
          PULU D
          STD  MSCR3
          PULU D
          ADDD MSCR2
          STD  MSCR4
          BCC  MPNOCY
          LDD  MSCR3
          ADDD MSCR
          ADDD #1
          BRA  MPHIDONE
MPNOCY:   LDD  MSCR3
          ADDD MSCR
MPHIDONE: STD  MSCR3
          LDD  MSCR4
          PSHU D
          LDD  MSCR3
          PSHU D
          RTS

STOD:     PULU D
          PSHU D
          TSTA
          ; BUG FIX: was a bare "BPL SDPOS" right after
          ; PULU - PULU doesn't affect condition codes
          ; on genuine 6809 (same class as TRYNUM's own
          ; PULU-then-compare fix elsewhere in this
          ; file), so BPL was testing whatever flags an
          ; unrelated, earlier instruction happened to
          ; leave set, not the sign of the value just
          ; popped. TSTA explicitly tests D's high byte
          ; (bit 7 of A reflects the full 16-bit value's
          ; sign) immediately before the branch that
          ; depends on it. Confirmed via MAME debugger:
          ; "-123 S>D" returned high cell 0 instead of
          ; -1 - no sign extension was happening at all.

          BPL  SDPOS
          LDD  #-1
          BRA  SDPUSH
SDPOS:    LDD  #0

```

```

SDPUSH:  PSHU  D
          RTS

DTOS:    PULU  D
          RTS

DMAXW:    PULU  D
          STD   MSCR
          PULU  D
          STD   MSCR2
          PULU  D
          STD   MSCR3
          PULU  D
          STD   MSCR4
          LDD   MSCR3
          CMPD  MSCR
          BGT   DMXD1
          BLT   DMXD2
          LDD   MSCR4
          CMPD  MSCR2
          BHS   DMXD1
DMXD2:    LDD   MSCR2
          PSHU  D
          LDD   MSCR
          PSHU  D
          RTS
DMXD1:    LDD   MSCR4
          PSHU  D
          LDD   MSCR3
          PSHU  D
          RTS

DMINW:    PULU  D
          STD   MSCR
          PULU  D
          STD   MSCR2
          PULU  D
          STD   MSCR3
          PULU  D
          STD   MSCR4
          LDD   MSCR3
          CMPD  MSCR
          BLT   DMND1
          BGT   DMND2
          LDD   MSCR4
          CMPD  MSCR2
          BLS   DMND1
DMND2:    LDD   MSCR2
          PSHU  D
          LDD   MSCR
          PSHU  D
          RTS
DMND1:    LDD   MSCR4
          PSHU  D
          LDD   MSCR3
          PSHU  D
          RTS

```

## 8.17 Logic, Shifts & Address Arithmetic

```

; =====
; SECTION 16: LOGIC / SHIFTS / ADDRESS ARITHMETIC
; =====
ANDW:    PULU  D
          ANDA  ,U
          ANDB  1,U
          STD   ,U
          RTS

```

```

ORW:      PULU  D
          ORA   ,U
          ORB   1,U
          STD   ,U
          RTS

XORW:      PULU  D
          EORA   ,U
          EORB   1,U
          STD   ,U
          RTS

INVERT:     LDD   ,U
          COMA
          COMB
          STD   ,U
          RTS

LSHIFT:      PULU  D
          STB   SHCNT
          LDD   ,U
LSLOOP:      LDB   SHCNT
          BEQ   LSDONE
          ASL   1,U
          ROL   ,U
          DEC   SHCNT
          BRA   LSLLOOP
LSDONE:      RTS

RSHIFT:      PULU  D
          STB   SHCNT
RSLLOOP:      LDB   SHCNT
          BEQ   RSDONE
          LSR   ,U
          ROR   1,U
          DEC   SHCNT
          BRA   RSLLOOP
RSDONE:      RTS

CELLSW:      LDD   ,U
          ASLB
          ROLA
          STD   ,U
          RTS

CELLPLUS:    LDD   ,U
          ADDD  #2
          STD   ,U
          RTS

CHARSW:      RTS

CHARPLUS:    LDD   ,U
          ADDD  #1
          STD   ,U
          RTS

ALIGNW:      RTS
ALIGNEDW:    RTS

```

## 8.18 Comparison

```

; =====
; SECTION 17: COMPARISON
; =====
EQUALW:      PULU  D
          CMPD   ,U
          BEQ   EQTRUE
          LDD   #FALSEV

```

```

      STD ,U
      RTS
EQTRUE: LDD #TRUEV
      STD ,U
      RTS

LESSW:  PULU D
      STD MSCR
      LDD ,U
      CMPD MSCR
      BLT LTTRUE
      LDD #FALSEV
      STD ,U
      RTS
LTTRUE: LDD #TRUEV
      STD ,U
      RTS

GREATERW: PULU D
      STD MSCR
      LDD ,U
      CMPD MSCR
      BGT GTTRUE
      LDD #FALSEV
      STD ,U
      RTS
GTTRUE: LDD #TRUEV
      STD ,U
      RTS

ZEROEQ: LDD ,U
      BEQ ZEQTRUE
      LDD #FALSEV
      STD ,U
      RTS
ZEQTRUE: LDD #TRUEV
      STD ,U
      RTS

ZEROLT: LDD ,U
      BMI ZLTTRUE
      LDD #FALSEV
      STD ,U
      RTS
ZLTTRUE: LDD #TRUEV
      STD ,U
      RTS

ULESSW: PULU D
      STD MSCR
      LDD ,U
      CMPD MSCR
      BLO ULTRUE
      LDD #FALSEV
      STD ,U
      RTS
ULTRUE: LDD #TRUEV
      STD ,U
      RTS

NOTEQUAL: JSR EQUALW
      LDD ,U
      COMA
      COMB
      STD ,U
      RTS

ZERONE: JSR ZEROEQ
      LDD ,U
      COMA
      COMB

```

```

        STD    ,U
        RTS

ZEROGT:  LDD    ,U
        BEQ    ZGTFALSE
        BMI    ZGTFALSE
        LDD    #TRUEV
        STD    ,U
        RTS
ZGTFALSE: LDD    #FALSEV
        STD    ,U
        RTS

UGREATER: PULU D
        STD    MSCR
        LDD    ,U
        CMPD   MSCR
        BLO    UGFALSE
        BEQ    UGFALSE
        LDD    #TRUEV
        STD    ,U
        RTS
UGFALSE:  LDD    #FALSEV
        STD    ,U
        RTS

WITHINW: PULU D
        STD    MSCR
        PULU D
        STD    MSCR2
        LDD    ,U
        SUBD   MSCR2
        STD    MSCR3
        LDD    MSCR
        SUBD   MSCR2
        STD    MSCR
        LDD    MSCR3
        CMPD   MSCR
        BLO    WITHTRUE
        LDD    #FALSEV
        STD    ,U
        RTS
WITHTRUE: LDD    #TRUEV
        STD    ,U
        RTS

DEQUAL:  PULU D
        STD    MSCR
        PULU D
        STD    MSCR2
        PULU D
        STD    MSCR3
        PULU D
        CMPD   MSCR2
        BNE    DEQFALSE
        LDD    MSCR3
        CMPD   MSCR
        BNE    DEQFALSE
        LDD    #TRUEV
        PSHU D
        RTS
DEQFALSE: LDD    #FALSEV
        PSHU D
        RTS

DLESSW:  PULU D
        STD    MSCR
        PULU D
        STD    MSCR2
        PULU D
        STD    MSCR3

```

```

        PULU D
        STD  MSCR4
        LDD  MSCR3
        CMPD MSCR
        BLT  DLTRUE
        BGT  DLFALSE
        LDD  MSCR4
        CMPD MSCR2
        BLO  DLTRUE
DLFALSE: LDD  #FALSEV
        PSHU D
        RTS
DLTRUE:  LDD  #TRUEV
        PSHU D
        RTS

DULESSW: PULU D
        STD  MSCR
        PULU D
        STD  MSCR2
        PULU D
        STD  MSCR3
        PULU D
        STD  MSCR4
        LDD  MSCR3
        CMPD MSCR
        BLO  DULTRUE
        BHI  DULFALSE
        LDD  MSCR4
        CMPD MSCR2
        BLO  DULTRUE
DULFALSE: LDD  #FALSEV
        PSHU D
        RTS
DULTRUE:  LDD  #TRUEV
        PSHU D
        RTS

```

## 8.19 Memory

```

; =====
; SECTION 18: MEMORY (fetch/store, block ops)
; =====
STOREW: PULU X
        PULU D
        STD  ,X
        RTS

CFETCH: PULU X
        LDB  ,X
        CLRA
        PSHU D
        RTS

CSTOREW: PULU X
        PULU D
        STB  ,X
        RTS

PLUSSTORE: PULU X
           PULU D
           ADDD ,X
           STD  ,X
           RTS

DFETCH: PULU X           ; BUG FIX: was reading high address first
        LDD  ,X           ; (pushed deep) then low address second
        PSHU D           ; (pushed on top) - the reverse of what
        LDD  2,X         ; DSTORE actually writes (x1 low, x2 high),

```

```

        PSHU D          ; so a 2! 2@ round trip swapped the two
        RTS             ; values. Now reads low first (x1, pushed
                          ; deep) then high second (x2, pushed on
                          ; top), matching DSTORE and correctly
                          ; round-tripping. Flagged by the parallel
                          ; 68000 port, confirmed by inspection.

DSTORE: PULU X
        PULU D
        STD 2,X
        PULU D
        STD ,X
        RTS

CMOVEV: PULU D
        STD MVCNT
        PULU D
        STD MVDST
        PULU D
        STD MVSRC
        LDX MVSRC
        LDY MVDST
CMVLOOP: LDD MVCNT
        BEQ CMDONE
        SUBD #1          ; BUG FIX: was after the byte copy below -
        STD MVCNT        ; LDA ,X+ (needed for the byte itself)
                          ; clobbers A, D's high byte, corrupting the
                          ; count before SUBD used it. Unlike FILL,
                          ; there's no spare 16-bit register to keep
                          ; the count in instead - X and Y are both
                          ; already committed to the source and
                          ; destination addresses - so the fix here is
                          ; reordering: decrement and store while D
                          ; still holds the true count, before the
                          ; copy is free to clobber A. Confirmed via
                          ; MAME debugger: the loop never terminated.

        LDA ,X+
        STA ,Y+
        BRA CMVLOOP
CMDONE: RTS

CMOVEGT: PULU D
        STD MVCNT
        PULU D
        STD MVDST
        PULU D
        STD MVSRC
        LDD MVCNT
        BEQ CGDONE
        LDX MVSRC
        LEAX D,X
        LEAX -1,X
        LDY MVDST
        LEAY D,Y
        LEAY -1,Y
CGLOOP: LDA ,X
        STA ,Y
        LEAX -1,X
        LEAY -1,Y
        LDD MVCNT
        SUBD #1
        STD MVCNT
        BNE CGLOOP
CGDONE: RTS

MOVEV: PULU D
        STD MVCNT
        PULU D
        STD MVDST
        PULU D
        STD MVSRC

```

```

        LDD    MVDST
        CMPD   MVSRC
        BLS    MVLOW
        LDD    MVSRC
        PSHU   D
        LDD    MVDST
        PSHU   D
        LDD    MVCNT
        PSHU   D
        JMP    CMOVEGT
MVLOW:   LDD    MVSRC
        PSHU   D
        LDD    MVDST
        PSHU   D
        LDD    MVCNT
        PSHU   D
        JMP    CMOVEW

FILLW:   PULU   D
        STB    FILLCHR
        PULU   D
        TFR    D,Y          ; BUG FIX: was STD FILLCNT/LDD FILLCNT each
                             ; iteration - and inside the loop, LDA
                             ; FILLCHR (needed for the fill byte) clobbers
                             ; A, which is D's high byte, corrupting the
                             ; count that SUBD #1 then decremented from.
                             ; The high byte took on the fill character's
                             ; value instead of the count's real high
                             ; byte, so the count almost never reached
                             ; zero at the intended point - FILL ran far
                             ; past the requested length. Keeping the
                             ; count in Y instead removes the conflict
                             ; entirely (Y is untouched by loading the
                             ; fill character into A) and removes the
                             ; per-iteration memory round-trip - also
                             ; addresses the redundant scratch usage
                             ; flagged alongside this bug. Confirmed via
                             ; MAME debugger.

        PULU   D
        TFR    D,X          ; address, kept directly in X rather than
                             ; round-tripping through FILLADDR too

FILLLOOP: CMPY   #0
        BEQ    FDONE
        LDA    FILLCHR
        STA    ,X+
        LEAY   -1,Y
        BRA    FILLLOOP
FDONE:   RTS

ERASEW:  LDD    #0
        PSHU   D
        JMP    FILLW

```

## 8.20 String Words

```

; =====
; SECTION 19: STRING WORDS
; =====
DOSTR:   PULS   X
        LDB    ,X
        LEAX   1,X
        PSHU   X
        CLRA
        PSHU   D
        LEAX   B,X
        PSHS   X
        RTS

SQUOTE:  LDD    #34

```

```

PSHU D
LDD CODEHERE ; BUG FIX (caught while verifying the WORD
ADD #3 ; redesign above): reserve 3 bytes ahead of
STD CODEHERE ; where WORD is about to write its parsed
; text. Without this, the compiled path
; below would compile "JSR DOSTR" (3 bytes)
; directly at CODEHERE - the same address
; WORD just used - overwriting the first 2
; characters of the very text being staged,
; before the copy loop even runs. Reserving
; the gap first means the text lands exactly
; where it needs to end up, and the trampoline
; safely goes in front of it instead of on
; top of it.

JSR WORD
PULU X
LDA ,X
STA SCNT
LEAX 1,X
STX SPTR
LDD CODEHERE ; restore - undo the temporary reserve
SUBD #3
STD CODEHERE
LDD STATE
BEQ SQINTERP

LDD #DOSTR
PSHU D
JSR CCALL
LDX CODEHERE
LDA SCNT
STA ,X+
LDY SPTR
LDB SCNT
BEQ SQEND
SQCPY: LDA ,Y+
STA ,X+
DECB
BNE SQCPY
SQEND: STX CODEHERE
RTS

SQINTERP: ; REDESIGN: was a fixed "LDX #SIBUF" here and at SQIEND -
; SIBUF was a dedicated, fixed 32-byte buffer,
; capping every interpreted S" string at 32
; characters and (worse) shared identically
; by every S" call, so a second S" call before
; the first string was actually used (e.g.
; registering two REPLACES strings, or a
; SUBSTITUTE template argument) silently
; overwrote the first - confirmed via MAME
; testing and traced precisely earlier this
; session. Retired per user's own follow-up
; testing/design decision: rather than give
; REPLACES its own dedicated copy-on-register
; storage, wrap each string in its own colon
; definition for stable storage instead - but
; that still leaves interpreted S" itself
; capped at SIBUF's 32 characters for any
; single string. Now computes PAD's current
; address fresh via PADW (dynamic - depends
; on CODEHERE, per ANS's own transient-region
; semantics) and writes there instead, giving
; interpreted S" access to PAD's full
; PADMINISIZE-character region (84, comfortably
; above the old 32-character SIBUF limit) -
; SIBUF itself is now unused and retired (see
; the GLOBALS layout notes above).

JSR PADW
PULU X
STX MSCR4 ; save PAD's own address - X gets advanced

```

```

; by the copy loop below, need the
; original back for the return value

        LDY  SPTR
        LDB  SCNT
        BEQ  SQIEND
SQICPY:  LDA  ,Y+
        STA  ,X+
        DECB
        BNE  SQICPY
SQIEND:  LDX  MSCR4
        PSHU X
        CLRA
        LDB  SCNT
        PSHU D
        RTS

DOTSTR:  PULS  X
        LDB  ,X
        LEAX 1,X
        STX  SPTR
        CLRA
        STD  SCNT
        LEAX B,X
        PSHS X
        LDX  SPTR
        PSHU X
        LDD  SCNT
        PSHU D
        JSR  TYPE
        RTS

DOTQUOTE: LDD  #34
        PSHU D
        LDD  CODEHERE ; BUG FIX: same class as SQUOTE above -
        ADDD #3        ; reserve 3 bytes ahead of WORD's write so
        STD  CODEHERE  ; the trampoline compiled below doesn't
                        ; overwrite the text it's about to stage.

        JSR  WORD
        PULU X
        LDA  ,X
        STA  SCNT
        LEAX 1,X
        STX  SPTR
        LDD  CODEHERE ; restore
        SUBD #3
        STD  CODEHERE
        LDD  #DOTSTR
        PSHU D
        JSR  CCALL
        LDX  CODEHERE
        LDA  SCNT
        STA  ,X+
        LDY  SPTR
        LDB  SCNT
        BEQ  DQEND
DQCPY:  LDA  ,Y+
        STA  ,X+
        DECB
        BNE  DQCPY
DQEND:  STX  CODEHERE
        RTS

TYPE:  PULU  D
        STD  TYPECNT
        PULU  D
        STD  TYPEADDR
TYLOOP: LDD  TYPECNT
        BEQ  TYDONE
        LDX  TYPEADDR
        LDA  ,X+
        STX  TYPEADDR

```

```

        TFR    A,B
        CLRA
        PSHU   D
        JSR    EMIT
        LDD    TYPECNT
        SUBD   #1
        STD    TYPECNT
        BRA    TYLOOP
TYDONE:  RTS

COUNT:  PULU   X
        LDB    ,X
        CLRA
        STD    MSCR
        LEAX   1,X
        PSHU   X
        LDD    MSCR
        PSHU   D
        RTS

CHARW:   LDD    #32
        PSHU   D
        JSR    WORD
        PULU   X
        LDB    1,X
        CLRA
        PSHU   D
        RTS

BRACKCHAR: LDD    STATE
        BNE    BCSTOK
        LDD    #-14
        PSHU   D
        JSR    THROW
BCSTOK:   LDD    #32
        PSHU   D
        JSR    WORD
        PULU   X
        LDB    1,X
        CLRA
        PSHU   D
        JSR    LITERALW
        RTS

PARSEW:   PULU   D
        STB    PDELIM
        LDD    TOIN
        LDX    SRCADDR
        LEAX   D,X
        STX    PSTART
        LDD    SRCLEN
        SUBD   TOIN
        TFR    D,Y
        LDD    #0
        STD    PLEN
PSCAN:    CMPY   #0
        BEQ    PDONE
        LDA    ,X
        CMPA   PDELIM
        BEQ    PFOUND
        LEAX   1,X
        LEAY   -1,Y
        LDD    PLEN
        ADDD   #1
        STD    PLEN
        BRA    PSCAN
PFOUND:   LEAX   1,X
        LEAY   -1,Y
PDONE:    TFR    X,D
        SUBD   SRCADDR
        STD    TOIN

```

```

        LDX  PSTART
        PSHU X
        LDD  PLEN
        PSHU D
        RTS

PARSENAME: LDD  TOIN
           LDX  SRCADDR
           LEAX D,X
           LDD  SRCLEN
           SUBD TOIN
           TFR  D,Y
PNSKIP:    CMPY #0
           BEQ  PNEMPTY
           LDA  ,X
           CMPA #32
           BNE  PNSTART
           LEAX 1,X
           LEAY -1,Y
           BRA  PNSKIP
PNSTART:   STX  PSTART
           LDD  #0
           STD  PLEN
PNSCAN:    CMPY #0
           BEQ  PNDONE
           LDA  ,X
           CMPA #32
           BEQ  PNFOUND
           LEAX 1,X
           LEAY -1,Y
           LDD  PLEN
           ADDD #1
           STD  PLEN
           BRA  PNSCAN
PNFOUND:   LEAX 1,X
           LEAY -1,Y
PNDONE:    TFR  X,D
           SUBD SRCADDR
           STD  TOIN
           LDX  PSTART
           PSHU X
           LDD  PLEN
           PSHU D
           RTS
PNEMPTY:   LDD  SRCLEN
           LDX  SRCADDR
           LEAX D,X
           STD  TOIN
           PSHU X
           LDD  #0
           PSHU D
           RTS

SLASHSTRING: PULU D
            STD  MSCR
            PULU D
            SUBD MSCR
            STD  MSCR2
            PULU D
            ADDD MSCR
            PSHU D
            LDD  MSCR2
            PSHU D
            RTS

DASHTRAILING: LDD  ,U
            STD  PLEN
DTLOOP:      LDD  PLEN
            BEQ  DTDONE
            LDX  2,U
            LEAX D,X

```

```

        LEAX -1,X
        LDA ,X
        CMPA #32
        BNE DTDONE
        LDD PLEN
        SUBD #1
        STD PLEN
        BRA DTLLOOP
DTDONE:  LDD PLEN
        STD ,U
        RTS

COMPAREW: PULU D
        STD CMPL2
        PULU D
        STD CMPA2
        PULU D
        STD CMPL1
        PULU D
        STD CMPA1
        LDD CMPL1
        CMPD CMPL2
        BLS CMMINIS1
        LDD CMPL2
        BRA CMMINSET
CMMINIS1: LDD CMPL1
CMMINSET: STD CMPMIN
        LDY CMPA1
        LDY CMPA2
CMPLOOP: LDD CMPMIN
        BEQ CMTIEBREAK
        LDA ,X+
        CMPA ,Y
        BLO CMLT
        BHI CMGT
        LEAY 1,Y
        LDD CMPMIN
        SUBD #1
        STD CMPMIN
        BRA CMPLOOP
CMTIEBREAK: LDD CMPL1
        CMPD CMPL2
        BLO CMLT
        BHI CMGT
        LDD #0
        PSHU D
        RTS
CMLT:    LDD #-1
        PSHU D
        RTS
CMGT:    LDD #1
        PSHU D
        RTS

SEARCHW: PULU D
        STD SRCH2L
        PULU D
        STD SRCH2
        PULU D
        STD SRCH1L
        PULU D
        STD SRCH1
        LDD SRCH2L
        BEQ SRCHNOTFOUND
        LDD SRCH1L
        SUBD SRCH2L
        BLT SRCHNOTFOUND
        ADDD #1
        STD SRCHPOS
        LDD #0
        STD SRCHI

```

```

SPOSLOOP: LDD  SRCHI
           CMPD SRCHPOS
           BEQ  SRCHNOTFOUND
           LDX  SRCH1
           LDD  SRCHI
           LEAX D,X
           LDY  SRCH2
           LDD  SRCH2L
           STD  MSCR3
SMATCH:   LDD  MSCR3
           BEQ  SFOUND
           LDA  ,X+
           CMPA ,Y+
           BNE  SNOMATCH
           LDD  MSCR3
           SUBD #1
           STD  MSCR3
           BRA  SMATCH
SNOMATCH: LDD  SRCHI
           ADDD #1
           STD  SRCHI
           BRA  SPOSLOOP
SFOUND:   LDX  SRCH1
           LDD  SRCHI
           LEAX D,X
           PSHU X
           LDD  SRCH2L
           PSHU D
           LDD  #TRUEV
           PSHU D
           RTS
SRCHNOTFOUND: LDD SRCH1
              PSHU D
              LDD  SRCH1L
              PSHU D
              LDD  #FALSEV
              PSHU D
              RTS

SNAMEW:   PULU  D
           STD  SNTARGET
           LDD  LATEST
           STD  SNXT
SNLOOP:   LDD  SNXT
           BEQ  SNNOTFOUND
           LDX  SNXT
           LDA  ,X
           STA  HDRFLAGS
           LEAX 1,X
           LDB  HDRFLAGS
           ANDB #$1F
           CLRA
           LEAX D,X
           LEAX 2,X
           LDD  ,X
           CMPD SNTARGET
           BEQ  SNFOUND
           LDX  SNXT
           LEAX 1,X
           LDB  HDRFLAGS
           ANDB #$1F
           CLRA
           LEAX D,X
           LDD  ,X
           STD  SNXT
           BRA  SNLOOP
SNFOUND:  LDX  SNXT
           LEAX 1,X
           PSHU X
           LDX  SNXT
           LDA  ,X

```

```

        ANDA    #$1F
        CLRB
        TFR     A,B
        CLRA
        PSHU    D
        RTS
SNNOTFOUND: LDD    #0
             PSHU    D
             PSHU    D
             RTS

UNESCAPEW: PULU    D        ; BUG FIX: was PULU D/STD UESRCLEN (storing
             STD     UEDST   ; the popped dest-addr into the SOURCE-
                               ; length variable), then LDD ,U (a PEEK, not
                               ; a pop) into BOTH UEADDR and UEDST - so the
                               ; real source length was misread as an
                               ; address, the real source address was never
                               ; popped at all, and the true destination
                               ; address never reached UEDST. Correct
                               ; ANS order is ( c-addr1 u1 c-addr2 -- ):
                               ; c-addr2 (dest) is on top, popped first.
                               ; next: u1 (source length)
        PULU    D
        STD     UESRCLEN
        PULU    D        ; next: c-addr1 (source address)
        STD     UEADDR
        LDD     #0
        STD     UEOUTLEN
        LDX     UEADDR
        LDY     UEDST

        ; ALGORITHM REPLACED (caught by MAME testing after the
        ; argument-order fix above): the entire body below used
        ; to decode backslash escape sequences (\n, \t, \\, \") -
        ; a plausible-looking but entirely wrong reading of what
        ; ANS UNESCAPE does. Confirmed against the actual spec and
        ; its canonical reference implementation (forth-standard.
        ; org/standard/string/UNESCAPE and complang.tuwien.ac.at's
        ; ANS Forth reference text): UNESCAPE has nothing to do
        ; with backslash sequences at all - it replaces every '%'
        ; character with two '%' characters, and nothing else, so
        ; that a literal '%' in text survives an eventual
        ; SUBSTITUTE pass unchanged ("If you pass a string through
        ; UNESCAPE and then SUBSTITUTE, you get the original
        ; string"). Every character that ISN'T '%' passes straight
        ; through, one-for-one. Confirmed via MAME: doubling a
        ; single '%' wasn't happening at all, and the returned
        ; count didn't grow to reflect the added characters -
        ; both are direct, expected consequences of this being the
        ; wrong algorithm entirely, not a smaller bug within it.
UELOOP:    LDD     UESRCLEN
        BEQ     UEDONE
        LDA     ,X+
        CMPA    #'%'
        BNE     UENOTPCT
        LDB     #'%'        ; write the extra '%' first - the character
        STB     ,Y+         ; itself still gets written once more below,
                               ; giving two '%' total for each one in the
                               ; source. Handles multiple '%' characters in
                               ; the same string correctly, since this runs
                               ; independently for each one the loop visits.
UENOTPCT:  STA     ,Y+        ; write the actual character - always, for
                               ; every character, '%' or not
        LDD     UESRCLEN
        SUBD    #1
        STD     UESRCLEN
        BRA     UELOOP
UEDONE:    TFR     Y,D        ; BUG FIX: was "LDD UEOUTLEN / STD ,U" -
        SUBD    UEDST        ; overwriting whatever was left on top of
        STD     UEOUTLEN     ; the stack rather than pushing both
                               ; required return values, and separately,
                               ; UEOUTLEN itself was never correctly

```

```

; tracked by the old backslash-decoding
; loop's own per-character bookkeeping in a
; way that accounted for doubled '%'
; characters. Now computed directly as the
; final write pointer (Y) minus the
; original destination address (UEDST) -
; correct regardless of how many characters
; were doubled, since it's derived from
; where writing actually stopped rather than
; incremented alongside it.
LDD UEDST
PSHU D ; push c-addr2 (destination address)
LDD UEOUTLEN
PSHU D ; push u2 (actual unescaped length)
RTS

; REPLACES/SUBSTITUTE - single-slot simplified version, per
; the explicit scoping-down discussed in the source conversation
REPLACESW: PULU D
STD REPLNLEN
PULU D
STD REPLNAME
PULU D
STD REPLVLEN
PULU D
STD REPLVAL
RTS

SUBCOPY: STD SUBCOPYCNT
STX SUBCOPYSRC
SUBCPLP: LDD SUBCOPYCNT
BEQ SUBCPDONE
LDD SUBOUTLEN
CMPD SUBDESTCAP
LBHS SUBOVERFLOW ; was BHS - out of short-branch range
LDX SUBCOPYSRC
LDA ,X+
STX SUBCOPYSRC
LDY SUBWPTR
STA ,Y+
STY SUBWPTR
LDD SUBOUTLEN
ADDD #1
STD SUBOUTLEN
LDD SUBCOPYCNT
SUBD #1
STD SUBCOPYCNT
BRA SUBCPLP
SUBCPDONE: RTS

SUBSTITUTEW: ; REWRITE: was a plain substring search-and-replace on
; the bare registered name (via SEARCHW), which found
; "girl" and replaced only that span, leaving surrounding
; "%..." delimiters untouched in the output - and never
; returned the substitution count ANS requires as a
; third stack item. Per the ANS spec (forth-standard.org/
; standard/string/SUBSTITUTE), SUBSTITUTE must scan for
; text between '%' (ASCII $25) delimiter pairs
; specifically: "%" collapses to a single '%' (count
; unchanged); a name matching the REPLACES registration
; has the ENTIRE "%name%" span - delimiters included -
; replaced by the substitution text (count incremented);
; a non-matching name is passed through unchanged,
; delimiters and all (count unchanged); an unpaired
; trailing '%' with no closing delimiter passes the
; residue through unchanged. Confirmed via MAME testing
; against a real template. GLOBALS is fully packed
; (256/256), so this reuses MSCR/MSCR2/MSCR3/MSCR4
; (confirmed untouched by SUBCOPY) rather than adding
; dedicated cells: MSCR = current read position, MSCR2 =
; running substitution count, MSCR3/MSCR4 = local scratch

```

```

; per %-pair found. Reuses the existing COMPAREW for name
; matching rather than a new comparison loop.
PULU D
STD SUBDESTCAP
PULU D
STD SUBDESTADR
PULU D
STD SUBSRCLEN
PULU D
STD SUBSRCADR

LDY SUBDESTADR
STY SUBWPTR
LDD #0
STD SUBOUTLEN
STD MSCR          ; read position, starts at 0
STD MSCR2         ; substitution count, starts at 0

SUBSCAN:  LDD MSCR
          CMPD SUBSRCLEN
          LBHS SUBSDONE
          LDX SUBSRCADR
          LEAX D,X
          LDA ,X
          CMPA #$25      ; '%' delimiter
          BEQ SUBPCT
          LDD #1
          JSR SUBCOPY
          LDD MSCR
          ADDD #1
          STD MSCR
          LBRA SUBSCAN

SUBPCT:   LDD MSCR
          ADDD #1
          STD MSCR3      ; scan for the closing '%'

SUBFINDCL: LDD MSCR3
          CMPD SUBSRCLEN
          BLO SUBFC2
          ; no closing delimiter found - residue passed unchanged
          LDD MSCR
          LDX SUBSRCADR
          LEAX D,X
          LDD SUBSRCLEN
          SUBD MSCR
          JSR SUBCOPY
          LDD SUBSRCLEN
          STD MSCR
          LBRA SUBSCAN

SUBFC2:   LDD MSCR3
          LDX SUBSRCADR
          LEAX D,X
          LDA ,X
          CMPA #$25
          BEQ SUBFOUNDCL
          LDD MSCR3
          ADDD #1
          STD MSCR3
          LBRA SUBFINDCL

SUBFOUNDCL: LDD MSCR3
          SUBD MSCR
          SUBD #1
          STD MSCR4      ; enclosed name length
          LBNE SUBHASNAME
          ; namelen 0: "%%" -> single '%' to output, count unchanged
          LDD MSCR
          LDX SUBSRCADR
          LEAX D,X
          LDD #1
          JSR SUBCOPY

```

```

        LDD  MSCR3
        ADDD #1
        STD  MSCR
        LBRA SUBSCAN

SUBHASNAME:  LDD  MSCR
              ADDD #1
              LDX  SUBSRCADR
              LEAX D,X
              PSHU X
              LDD  MSCR4
              PSHU D
              LDD  REPLNAME
              PSHU D
              LDD  REPLNLEN
              PSHU D
              JSR  COMPAREW
              PULU D
              CMPD #0
              BNE  SUBNOMATCH
              ; valid name - replace the whole %name% span
              LDX  REPLVAL
              LDD  REPLVLEN
              JSR  SUBCOPY
              LDD  MSCR2
              ADDD #1
              STD  MSCR2
              LDD  MSCR3
              ADDD #1
              STD  MSCR
              LBRA SUBSCAN

SUBNOMATCH:  ; not a registered name - pass %name% through unchanged
              LDD  MSCR
              LDX  SUBSRCADR
              LEAX D,X
              LDD  MSCR3
              SUBD MSCR
              ADDD #1
              JSR  SUBCOPY
              LDD  MSCR3
              ADDD #1
              STD  MSCR
              LBRA SUBSCAN

SUBSDONE:   LDX  SUBDESTADR
              PSHU X
              LDD  SUBOUTLEN
              PSHU D
              LDD  MSCR2
              PSHU D
              RTS

SUBOVERFLOW: LDD #-1
              PSHU D
              JSR  THROW

```

## 8.21 Numeric Output

```

; =====
; SECTION 20: NUMERIC OUTPUT (pictured + direct)
; =====
LTNUM:   JSR  PADW
          PULU D
          STD  HLD
          RTS

HOLD:    PULU D
          LDX  HLD

```

```

        LEAX    -1,X
        STX     HLD
        STB     ,X
        RTS

HOLDS:   PULU    D
        STD     HSLEN
        PULU    D
        STD     HSADDR
HSLLOOP: LDD     HSLEN
        BEQ     HSDONE
        SUBD    #1
        STD     HSLEN
        LDX     HSADDR
        LDD     HSLEN
        LEAX    D,X
        LDA     ,X
        TFR     A,B
        CLRA
        PSHU    D
        JSR     HOLD
        BRA     HSLLOOP
HSDONE:  RTS

NUMSIGN: PULU    D
        STD     UDHI
        PULU    D
        STD     UDLO
        JSR     UDDIGIT
        LDA     REM
        CMPA    #10
        BLO     NDIGIT
        ADDA    #'A' -10
        BRA     NHOLD
NDIGIT:  ADDA    #'0'
NHOLD:   TFR     A,B
        CLRA
        PSHU    D
        JSR     HOLD
        LDD     UDLO
        PSHU    D
        LDD     UDHI
        PSHU    D
        RTS

UDDIGIT: CLR     REM
        LDB     #32
        STB     DCNT
UDDLLOOP: ASL     UDLO+1
        ROL     UDLO
        ROL     UDHI+1
        ROL     UDHI
        ROL     REM
        LDA     REM
        CMPA    BASE+1
        BLO     UDNEXT
        SUBA    BASE+1
        STA     REM
        INC     UDLO+1
UDNEXT:  DEC     DCNT
        BNE     UDDLLOOP
        RTS

NUMSIGNS: JSR    NUMSIGN
        LDD     UDHI
        BNE     NUMSIGNS
        LDD     UDLO
        BNE     NUMSIGNS
        RTS

SIGN:    PULU    D

```

```

TSTA          ; BUG FIX: was a bare "BPL SIGNDONE" right
               ; after PULU - PULU doesn't affect condition
               ; codes on genuine 6809 (same class as STOD's
               ; own fix elsewhere in this file), so BPL was
               ; testing whatever flags an unrelated, earlier
               ; instruction happened to leave set, not the
               ; sign of the value just popped. TSTA
               ; explicitly tests D's high byte immediately
               ; before the branch that depends on it. The
               ; four existing internal callers (DOT/DOTR/
               ; DDOT/DDOTR, i.e. . / .R / D. / D.R) were
               ; accidentally unaffected - each happens to
               ; run "LDD SAVEN" (a flag-setting load of the
               ; true signed value) immediately before PSHU
               ; D/JSR SIGN, and PSHU doesn't disturb flags -
               ; but this was fragile, caller-side luck, not
               ; SIGN being correct on its own. Confirmed via
               ; MAME: a direct, standalone use of SIGN (not
               ; preceded by an unrelated flag-setting
               ; instruction) never added the minus sign for
               ; either sign of input.

BPL    SIGNDONE
LDD    #'-'
PSHU   D
JSR    HOLD
SIGNDONE: RTS

NUMGT:  PULU   D
        PULU   D
        LDX    HLD
        PSHU   X
        JSR    PADW
        PULU   D
        SUBD   HLD
        PSHU   D
        RTS

DOT:    PULU   D
        STD    SAVEN
        BPL    DABSOK
        COMA
        COMB
        ADDD   #1
DABSOK: PSHU   D
        LDD    #0
        PSHU   D
        JSR    LTNUM
        JSR    NUMSIGNS
        LDD    SAVEN
        PSHU   D
        JSR    SIGN
        JSR    NUMGT
        JSR    TYPE
        LDD    #32
        PSHU   D
        JSR    EMIT
        RTS

UDOT:   LDD    #0          ; SIMPLIFICATION: was PULU D/PSHU D here first -
                           ; a literal no-op (pop the value, immediately
                           ; push it back, nothing in between), leftover
                           ; rather than functional. The value being
                           ; formatted was never actually consumed here;
                           ; this line already pushes 0 on top of it
                           ; unchanged either way. Removed per the
                           ; parallel 68000 port's observation, confirmed
                           ; by inspection.

        PSHU   D
        JSR    LTNUM
        JSR    NUMSIGNS
        JSR    NUMGT

```

```

        JSR  TYPE
        LDD  #32
        PSHU D
        JSR  EMIT
        RTS

DOTR:    PULU D
        STD  DRWIDTH
        PULU D
        STD  SAVEN
        BPL  DRABSOK
        COMA
        COMB
        ADDD #1
DRABSOK: PSHU D
        LDD  #0
        PSHU D
        JSR  LTNUM
        JSR  NUMSIGNS
        LDD  SAVEN
        PSHU D
        JSR  SIGN
        JSR  NUMGT
        PULU D
        STD  DRLEN
        PULU D
        STD  DRADDR
        LDD  DRWIDTH
        SUBD DRLEN
        BLE  DRNOPAD
        STD  DRPAD
DRPADLP: LDD  DRPAD
        BEQ  DRNOPAD
        SUBD #1
        STD  DRPAD
        LDD  #32
        PSHU D
        JSR  EMIT
        BRA  DRPADLP
DRNOPAD: LDX  DRADDR
        PSHU X
        LDD  DRLEN
        PSHU D
        JSR  TYPE
        RTS

UDOTR:   PULU D
        STD  DRWIDTH
        LDD  #0
        ; SIMPLIFICATION: was PULU D/PSHU D here first -
        ; a literal no-op on the value being formatted
        ; (the width above is a real, separate pop -
        ; unaffected). Removed per the parallel 68000
        ; port's observation, confirmed by inspection.

        PSHU D
        JSR  LTNUM
        JSR  NUMSIGNS
        JSR  NUMGT
        PULU D
        STD  DRLEN
        PULU D
        STD  DRADDR
        LDD  DRWIDTH
        SUBD DRLEN
        BLE  UDRNOPAD
        STD  DRPAD
UDRPADLP: LDD  DRPAD
        BEQ  UDRNOPAD
        SUBD #1
        STD  DRPAD
        LDD  #32
        PSHU D

```

```

        JSR  EMIT
        BRA  UDRPADLP
UDRNO PAD: LDX  DRADDR
        PSHU X
        LDD  DRLEN
        PSHU D
        JSR  TYPE
        RTS

QMARK:   PULU X
        LDD  ,X
        PSHU D
        JSR  DOT
        RTS

DDOT:    PULU D
        STD  PRODHI
        PULU D
        STD  PRODLO
        LDD  PRODHI
        STD  SAVEN
        BPL  DDPOS
        JSR  MNEG32
DDPOS:   LDD  PRODLO
        PSHU D
        LDD  PRODHI
        PSHU D
        JSR  LTNUM
        JSR  NUMSIGNS
        LDD  SAVEN
        PSHU D
        JSR  SIGN
        JSR  NUMGT
        JSR  TYPE
        LDD  #32
        PSHU D
        JSR  EMIT
        RTS

DDOTR:   PULU D
        STD  DRWIDTH
        PULU D
        STD  PRODHI
        PULU D
        STD  PRODLO
        LDD  PRODHI
        STD  SAVEN
        BPL  DDRPOS
        JSR  MNEG32
DDRPOS:  LDD  PRODLO
        PSHU D
        LDD  PRODHI
        PSHU D
        JSR  LTNUM
        JSR  NUMSIGNS
        LDD  SAVEN
        PSHU D
        JSR  SIGN
        JSR  NUMGT
        PULU D
        STD  DRLEN
        PULU D
        STD  DRADDR
        LDD  DRWIDTH
        SUBD DRLEN
        BLE  DRDNO PAD
        STD  DRPAD
DRDPADLP: LDD  DRPAD
        BEQ  DRDNO PAD
        SUBD #1
        STD  DRPAD

```

```

        LDD #32
        PSFU D
        JSR EMIT
        BRA DRDPADLP
DRDNOPAD: LDX DRADDR
        PSFU X
        LDD DRLLEN
        PSFU D
        JSR TYPE
        RTS

```

## 8.22 Base / Radix Control

```

; =====
; SECTION 21: BASE / RADIX CONTROL
; =====
BASEW:  LDD #BASE
        PSFU D
        RTS

DECIMAL: LDD #10
        STD BASE
        RTS

HEXW:    LDD #16
        STD BASE
        RTS

BINARYW: LDD #2
        STD BASE
        RTS

```

## 8.23 Output Formatting

```

; =====
; SECTION 22: OUTPUT FORMATTING (CR/SPACE/SPACES)
; =====
CRW:    LDD #13
        PSFU D
        JSR EMIT
        LDD #10
        PSFU D
        JSR EMIT
        RTS

SPACEW: LDD #32
        PSFU D
        JSR EMIT
        RTS

SPACESW: PULU D
        STD SHCNT2
SPL00P:  LDD SHCNT2
        BLE SPDONE
        LDD #32
        PSFU D
        JSR EMIT
        LDD SHCNT2
        SUBD #1
        STD SHCNT2
        BRA SPL00P
SPDONE:  RTS

```

## 8.24 Comment Words

```
; =====
; SECTION 23: COMMENT WORDS
; =====
LPAREN:  LDD    #' )'
         PSHU   D
         JSR    WORD
         PULU   D      ; BUG FIX: WORD always pushes a c-addr at the
                        ; end (the parsed text's address), which this
                        ; word never consumed - "(" only needs WORD's
                        ; side effect of advancing >IN past the closing
                        ; delimiter, not the address itself, but that
                        ; address was left stranded on the stack every
                        ; time. Explains both reported symptoms: a
                        ; stray value visible after a comment in
                        ; interpret mode, and a CSP mismatch (-22) for
                        ; any colon definition containing one, since
                        ; the leftover throws off the compile-time
                        ; stack-depth check between ":" and ";".
                        ; Confirmed via MAME debugger.

         RTS

BACKSLASH: LDD    SRCLEN
          STD     TOIN
          RTS
```

## 8.25 Environmental Query / SOURCE / REFILL / EVALUATE

```
; =====
; SECTION 24: ENVIRONMENTAL QUERY / SOURCE / REFILL / EVALUATE
; =====
SOURCEW: LDD     SRCADDR
         PSHU    D
         LDD     SRCLEN
         PSHU    D
         RTS

SOURCEID: LDD     SRCID
         PSHU    D
         RTS

REFILLW: LDD     SRCID
         BEQ     RFTerm
         LDD     #FALSEV
         PSHU    D
         RTS

RFTerm:  JSR     QUERY
         LDD     #TRUEV
         PSHU    D
         RTS

EVALUATEW: LDD     SRCADDR
          STD     EVSAVEA
          LDD     SRCLEN
          STD     EVSAVEL
          LDD     SRCID
          STD     EVSAVEI
          LDD     TOIN
          STD     EVSAVET
          PULU    D
          STD     SRCLEN
          PULU    D
          STD     SRCADDR
          LDD     #-1
          STD     SRCID
          LDD     #0
          STD     TOIN
          JSR     INTERPRET
```

```

LDD EVSAVEA
STD SRCADDR
LDD EVSAVEL
STD SRCLEN
LDD EVSAVEI
STD SRCID
LDD EVSAVET
STD TOIN
RTS

; ENVIRONMENT? - dispatcher complete; table has entries derived
; without fabricating unfixed capacities. /HOLD and /PAD were both
; added this session, answering HOLDMINSIZE and PADMINISIZE
; respectively (see each entry itself, below, for why each uses its
; own, conceptually correct constant rather than the numerically-
; equal PADOFFSET).
;
; WORDLISTS (16.3.2, Search-Order word set) is deliberately absent
; from ENVTABLE, not an oversight - this system doesn't implement
; the Search-Order word set, so WORDLISTS is genuinely an unknown
; attribute here. Per ENVIRONMENT?'s own spec (6.1.1345): "If the
; system treats the attribute as unknown, the returned flag is
; false" - falling through to ENVNOTFOUND already produces exactly
; that, correctly and completely, with no dispatcher work needed.
; Confirmed via MAME testing.
;
; MAX-D/MAX-UD were resolved this session via a genuine dispatcher
; extension, not just a table entry - both are double-cell values
; per their own ANS data type, which the original single-cell
; ENVFOUND (pushing exactly one value before TRUE) couldn't
; represent. Added a second table (ENVTABLE2) and matching loop
; (ENV2START/ENV2LOOP/ENV2FOUND, below) specifically for double-cell
; entries, rather than reworking the single-cell path to handle both
; shapes - see those labels for the full design.
;
; FLOORED was the third, separate open item flagged previously -
; investigated this session by tracing DIVCOMMON (shared by /, MOD,
; /MOD) against this system's own SM/REM and FM/MOD. Confirmed
; DIVCOMMON matches SM/REM's convention exactly (remainder sign
; restored from the dividend's own original sign, not the quotient's)
; - this system's primary division words use symmetric division, not
; floored. FLOORED now answers 0 (false) - see the entry itself,
; below, for the full trace.
ENVQUERY: PULU D
          STD ENVLEN
          PULU D
          STD ENVADDR
          LDX #ENVTABLE
ENVLOOP:  LDD ,X
          CMPD #0
          BEQ ENV2START ; CHANGED: was BEQ ENVNOTFOUND - now tries
                        ; the double-cell table (below) before
                        ; giving up entirely. See ENV2START/
                        ; ENV2LOOP for MAX-D/MAX-UD, added this
                        ; turn - both are double-cell values per
                        ; their own ANS data type, which this
                        ; single-cell table's ENVFOUND (pushing
                        ; exactly one value before TRUE) can't
                        ; represent correctly.
          PSHS X        ; BUG FIX: COMPAREW uses X as its own
                        ; internal scratch (LDX CMPA1, then LDA
                        ; ,X+ advancing through the comparison),
                        ; clobbering whatever the caller had there.
                        ; ENVQUERY needs X to still point at the
                        ; current table entry after COMPAREW
                        ; returns, both for ENVFOUND's own LDD 4,X
                        ; and for advancing to the next entry -
                        ; without saving it first, a successful
                        ; match left X sitting just past the
                        ; matched string's own text (for
                        ; "/COUNTED-STRING", 15 characters, that

```

```

; lands exactly at the start of the next
; entry's string, "MAX-N") - so ENVFOUND
; read the "value" field from inside the
; NEXT entry's own string text instead of
; the real value. Confirmed via MAME: this
; exact mechanism produces $4E4D precisely
; - the bytes 'N' ($4E) and 'M' ($4D, the
; start of "MAX-U") sitting 4 bytes into
; "MAX-N", read as one cell.

PSHU D
LDD 2,X
PSHU D
LDD ENVADDR
PSHU D
LDD ENVLEN
PSHU D
JSR COMPAREW
PULS X ; restore X before using it again
PULU D
CMPD #0
BEQ ENVFOUND
LEAX 6,X
BRA ENVLOOP
ENVFOUND: LDD 4,X
PSHU D
LDD #TRUEV
PSHU D
RTS

ENV2START: LDX #ENVTABLE2 ; NEW: double-cell query table - kept
; entirely separate from ENVTABLE above
; rather than trying to unify both cases
; into one mechanism, since the two
; genuinely differ in entry size (8 bytes
; here, vs 6 for single-cell) and in how
; many cells a match pushes (two, not
; one) - lower risk than reworking the
; already-tested single-cell path to
; handle both shapes at once.

ENV2LOOP: LDD ,X
CMPD #0
BEQ ENVNOTFOUND
PSHS X ; same COMPAREW X-clobber concern as above
PSHU D
LDD 2,X
PSHU D
LDD ENVADDR
PSHU D
LDD ENVLEN
PSHU D
JSR COMPAREW
PULS X
PULU D
CMPD #0
BEQ ENV2FOUND
LEAX 8,X ; entries are 8 bytes here (ADDR,LEN,LOW,
; HIGH), not 6 (ADDR,LEN,VALUE)
BRA ENV2LOOP
ENV2FOUND: LDD 4,X ; low cell - pushed first/deep, matching
PSHU D ; standard double-cell stack order (3.1.4.1:
; the cell containing the most significant
; part shall be above the cell containing
; the least significant part)
LDD 6,X ; high cell - pushed second/top
PSHU D
LDD #TRUEV
PSHU D
RTS

ENVNOTFOUND: LDD #FALSEV
PSHU D

```

RTS

ENVTABLE2:

```

FDB  EN11,EN11L,$FFFF,$7FFF ; MAX-D: maximum signed
      ; double-cell value, $7FFFFFFF - low
      ; cell $FFFF, high cell $7FFF (the
      ; maximum positive 16-bit signed value in
      ; the high cell, matching how a 32-bit
      ; two's-complement maximum is laid out:
      ; all bits set except the sign bit of the
      ; most significant cell).
FDB  EN12,EN12L,$FFFF,$FFFF ; MAX-UD: maximum unsigned
      ; double-cell value, $FFFFFFFF - both
      ; cells all bits set.
FDB  0
EN11: FCC  "MAX-D"
EN11L EQU  *-EN11
EN12: FCC  "MAX-UD"
EN12L EQU  *-EN12

```

ENVTABLE:

```

FDB  EN1,EN1L,255 ; BUG FIX: was 31 - the maximum size of a
      ; counted string is bounded by its 1-byte
      ; count field (0-255), not 31, which
      ; looks like it was mistakenly copied from
      ; an unrelated constraint (WORD's own,
      ; separate scan cap in an earlier version
      ; of this file). /COUNTED-STRING's own
      ; meaning (forth-standard.org/standard/
      ; usage#usage:env) is specifically the
      ; count byte's own maximum value.
FDB  EN2,EN2L,32767
FDB  EN3,EN3L,65535
FDB  EN6,EN6L,8
FDB  EN7,EN7L,HOLDMINSIZE ; NEW: /HOLD was previously absent
      ; entirely (the top-of-file note calling
      ; out "/HOLD and /PAD... remain explicitly
      ; incomplete/absent" was accurate at the
      ; time). Answers HOLDMINSIZE (34), not the
      ; larger PADOFFSET (84) - HOLDMINSIZE is
      ; specifically the portion of the
      ; CODEHERE-to-PAD gap reserved for the
      ; pictured numeric output buffer, the same
      ; amount WORDMAXCHARS deliberately holds
      ; back from WORD's own use at the other
      ; end of that same shared gap. PADOFFSET
      ; is the total gap width, most of which is
      ; actually earmarked for WORD's own
      ; parsing - reporting it here would
      ; overstate what HOLD can safely use
      ; without risking collision with whatever
      ; WORD is doing in the same space.
FDB  EN8,EN8L,PADMINISIZE ; NEW: /PAD, the last of the two
      ; entries the top-of-file note originally
      ; flagged as absent - now both present.
      ; Answers PADMINISIZE (84) directly, per
      ; ANS's own /PAD meaning (3.3.3.6, "the
      ; size of the scratch area whose address
      ; is returned by PAD") - PAD's own region,
      ; growing upward from PAD itself, distinct
      ; from HOLDMINSIZE above (which answers
      ; for a different region entirely, the
      ; downward-growing pictured-numeric buffer
      ; in the CODEHERE-to-PAD gap). PADOFFSET
      ; happens to equal PADMINISIZE numerically
      ; in this implementation (PADOFFSET EQU
      ; PADMINISIZE), but /PAD is answered with
      ; the conceptually correct constant, not
      ; the coincidentally-equal one.
FDB  EN9,EN9L,0 ; NEW: FLOORED - previously the one item
      ; in this area explicitly left open

```

```

; pending its own investigation (unlike
; WORDLISTS, this is a CORE query, not
; tied to an unimplemented optional word
; set, so "false by omission" wasn't the
; right default). Investigated by tracing
; DIVCOMMON (shared by /, MOD, /MOD)
; against this system's own SM/REM and
; FM/MOD: DIVCOMMON restores the
; remainder's sign from DNSIGN (the
; dividend's own original sign) after
; dividing absolute values - exactly
; SM/REM's own logic, confirmed identical
; by direct comparison. FM/MOD, by
; contrast, has an explicit flooring
; adjustment step (correcting the
; quotient and remainder when the
; quotient would be negative with a
; nonzero remainder) that DIVCOMMON does
; not have. This system's primary
; division words (/, MOD, /MOD) use
; symmetric division, not floored -
; FLOORED is 0 (false), a real, meaningful
; answer (recognized query, value false),
; not the "unrecognized" fallthrough
; WORDLISTS correctly uses.
FDB EN10,EN10L,255 ; NEW: MAX-CHAR - maximum value of any
; character in this implementation's
; character set (3.2.6). This system uses
; 8-bit characters throughout (matching
; ADDRESS-UNIT-BITS=8, already in this
; table) - a single byte's maximum value
; is 255.
FDB EN13,EN13L,(RSTACK-DSTACK)/2 ; NEW: RETURN-STACK-
; CELLS - computed from this system's own
; stack region boundaries rather than
; hardcoded, so this stays correct if the
; memory map shifts again (it has
; repeatedly this session). RSTACK's own
; comment confirms the occupied range is
; $BD00-RSTACK (768 bytes currently) -
; DSTACK+1 is $BD00 exactly (the two
; regions are contiguous, per RSTACK's
; and DSTACK's own comments). Since
; RSTACK-(DSTACK+1)+1 simplifies to
; RSTACK-DSTACK (the +1/-1 cancel), the
; simpler form is used directly: 384
; currently.
FDB EN14,EN14L,(DSTACK-CODETOP+1)/2 ; NEW: STACK-CELLS -
; same reasoning, for the data stack's own
; region (CODETOP to DSTACK, 1024 bytes
; currently, matching CODETOP's own
; comment "code space ceiling (data stack
; begins here)": 512 currently.
FDB 0
EN1: FCC "/COUNTED-STRING"
EN1L EQU *-EN1
EN2: FCC "MAX-N"
EN2L EQU *-EN2
EN3: FCC "MAX-U"
EN3L EQU *-EN3
EN6: FCC "ADDRESS-UNIT-BITS"
EN6L EQU *-EN6
EN7: FCC "/HOLD"
EN7L EQU *-EN7
EN8: FCC "/PAD"
EN8L EQU *-EN8
EN9: FCC "FLOORED"
EN9L EQU *-EN9
EN10: FCC "MAX-CHAR"
EN10L EQU *-EN10
EN13: FCC "RETURN-STACK-CELLS"

```

```

EN13L EQU *-EN13
EN14: FCC "STACK-CELLS"
EN14L EQU *-EN14

```

## 8.26 Tools Word Set

```

; =====
; SECTION 25: TOOLS WORD SET (.S / WORDS / DUMP)
; =====
DOTS:   TFR    U,D
        STD    DSPTMP
DSLLOOP: LDD    DSPTMP
        CMPD   #SP0
        BEQ    DSDONE
        LDX    DSPTMP
        LDD    ,X
        PSHU   D
        JSR    DOT
        LDD    DSPTMP
        ADDD   #2
        STD    DSPTMP
        BRA    DSLLOOP
DSDONE:  RTS

WORDSW:  LDD    LATEST
        STD    WWALK
WWLOOP:  LDD    WWALK
        BEQ    WWDONE
        LDX    WWALK
        LDA    ,X
        STA    HDRFLAGS
        LEAX   1,X
        PSHU   X
        LDB    HDRFLAGS
        ANDB   #$1F
        CLRA
        PSHU   D
        JSR    TYPE
        JSR    SPACEW
        LDX    WWALK
        LEAX   1,X
        LDB    HDRFLAGS
        ANDB   #$1F
        CLRA
        LEAX   D,X
        LDD    ,X
        STD    WWALK
        BRA    WWLOOP
WWDONE:  JSR    CRW
        RTS

HEXDIGIT: PULU D
        CMPD   #10
        BLO    HDDIGIT
        ADDD   #'A'-10
        BRA    HDEMITE
HDDIGIT:  ADDD   #'0'
HDEMITE:  PSHU D
        JSR    EMIT
        RTS

HEXBYTE: PULU D
        STB    MSCR ; BUG FIX: STB writes ONE byte, at MSCR
        LDB    MSCR ; itself - both reads below used to read
        LSRB   ; MSCR+1 instead, a byte this routine never
        LSRB   ; writes at all, so both the high and low
        LSRB   ; nibble extraction ran on whatever stale
        LSRB   ; value happened to be sitting there, not
        CLRA    ; the byte actually passed in. Now reads

```

```

        PSHU D                ; from MSCR, where STB actually put it.
        JSR HEXDIGIT          ; Confirmed via MAME debugger: DUMP showed
        LDB MSCR              ; $03 instead of $7B for a memory fill
        ANDB #$0F             ; character, with the ASCII column (a
        CLRA                  ; separate code path) correctly showing
        PSHU D                ; "}".
        JSR HEXDIGIT
        RTS

; DUMP - includes the partial-final-line ASCII fix
DUMPW:  PULU D
        STD DUMPCNT
        PULU D
        STD DUMPADDR
        JSR CRW                ; FORMATTING: leading CR, so the first line
                                ; starts on its own fresh line - matches
                                ; DULEND's existing trailing CR after every
                                ; line (including the last), giving
                                ; consistent vertical alignment from the
                                ; first line to the last regardless of
                                ; where the cursor was when DUMP was called.

DULINE: LDD DUMPCNT
        LBEQ DUDONE            ; was BEQ - out of short-branch range
        LDD DUMPADDR
        STD HEXBUF
        CLR DUMPCOL
        LDA #16
        STA DUVALID
DUHEX:  LDB DUMPCOL
        CMPB #16
        BEQ DUASCII
        LDD DUMPCNT
        BNE DUHEXBYTE
        LDA DUMPCOL
        STA DUVALID
        BRA DUHEXPAD
DUHEXBYTE: LDX DUMPADDR
        LDB ,X
        CLRA
        PSHU D
        JSR HEXBYTE
        LDD #32
        PSHU D
        JSR EMIT
        LDX DUMPADDR
        LEAX 1,X
        STX DUMPADDR
        LDD DUMPCNT
        SUBD #1
        STD DUMPCNT
        INC DUMPCOL
        BRA DUHEX
DUHEXPAD: LDD #32
        PSHU D
        JSR EMIT
        PSHU D
        JSR EMIT
        PSHU D
        JSR EMIT
        INC DUMPCOL
        LDB DUMPCOL
        CMPB #16
        BNE DUHEXPAD
        BRA DUASCII
DUASCII: LDD #32
        PSHU D
        JSR EMIT
        CLR DUMPCOL
DUACHAR: LDB DUMPCOL
        CMPB #16
        BEQ DULEND

```

```

        CMPB DUVALID
        BHS DUABLANK
        LDX HEXBUF
        LDB DUMPCOL
        CLRA
        LEAX D,X
        LDA ,X
        CMPA #32
        BLO DUDOT
        CMPA #127
        BHS DUDOT
        BRA DUPRINT
DUDOT:   LDA #'.'
DUPRINT: TFR A,B
        CLRA
        PSHU D
        JSR EMIT
        INC DUMPCOL
        BRA DUACHAR
DUABLANK: LDD #32
        PSHU D
        JSR EMIT
        INC DUMPCOL
        BRA DUACHAR
DULEND:  JSR CRW
        LDD DUMPCNT
        LBNE DULINE          ; was BNE - out of short-branch range
DUDONE:  RTS

```

## 8.27 ABORT / QUIT Hand-Built Headers

```

; =====
; SECTION 26: TRUE / FALSE. This section used to also hold
; ABORT/QUIT's hand-built dictionary headers (ABORTHDR/QUITHDR) -
; moved into BASEDICT and renamed to H_ABORT/H_QUIT (see the end
; of SECTION 27) so every header in this ROM lives in one
; contiguous block, rather than header data sitting apart from
; the rest of the dictionary inside BASECODE.
; =====
; -----
; TRUE / FALSE - simple subroutines, not the CONSTANT/DODOES
; pattern these used before (JSR DODOES + FDB ATSIGN + a value
; cell, matching what interactive CONSTANT compiles). Each is now
; a direct LDD/PSHU/RTS, like every other ordinary ROM word's CFA
; - no indirection, no value cell.
;
; RESOLVED: the previous version of this code deliberately inverted
; TRUE/FALSE's values from the system's own convention, per an
; earlier explicit request - FALSE pushed $FFFF and TRUE pushed
; $0000, disagreeing with TRUEV/FALSEV ($FFFF/$0000) and every
; comparison operator (=, <, >, and the rest), which all still
; returned TRUEV for a true result and FALSEV for false. Restored
; to the standard convention now: TRUE pushes $FFFF, FALSE pushes
; $0000, matching TRUEV/FALSEV and every comparison operator again.
; Each is a direct LDD/PSHU/RTS, like every other ordinary ROM
; word's CFA - no indirection, no value cell, no DODOES trampoline.
; -----
TRUEBODY: LDD  #$FFFF
          PSHU D
          RTS

FALSEBODY: LDD  #$0000
          PSHU D
          RTS

; -----
; 1 / -1 / 2 / -2 - simple immediate-value words, same pattern as
; TRUE/FALSE above: direct LDD/PSHU/RTS, no indirection, no value
; cell. Headers are H_1/H_M1/H_2/H_M2 (SECTION 27, chained after

```

```

; H_FIND) - "1"/"-1"/"2"/"-2" aren't valid 6809 assembler labels.
; -----
ONEBODY:   LDD    #1
           PSHU   D
           RTS

MONEBODY:  LDD    #-1
           PSHU   D
           RTS

TWOBODY:   LDD    #2
           PSHU   D
           RTS

MTWOBODY:  LDD    #-2
           PSHU   D
           RTS

; Verify no collision with init code,
; value should match ORG INITCODE.
BASECODEEND EQU    *
BASECODESIZE EQU    BASECODEEND-BASECODE

; Prevent the assembler from extinguishing the gap between the
; BASECODE block and the INITCODE block when it generates the
; .bin file.
BASEND:
    FILL $FF,INITCODE-BASEND

```

## 8.28 Forth Dictionary (ROM Base Dictionary Headers)

```

; =====
; SECTION 27: FORTH DICTIONARY (ROM base dictionary headers)
; Every primitive word in the Glossary gets a real header here,
; chained via LINK, living in BASEDICT ($D83F-$E006, an exact fit
; for this dictionary's 1992 bytes - was $E000-$E7FF). CFA points
; directly at each primitive's own code label for every entry -
; these are all raw code entries, CFA = the label itself,
; including TRUE and FALSE (added in a later pass, chain's newest
; entries before H_ABORT/H_QUIT below), which originally used the
; DODOES-trampoline pattern matching interactive CONSTANT, but are
; now simple LDD/PSHU/RTS subroutines like everything else here -
; see TRUEBODY/FALSEBODY, section 26, for the current code and why
; they no longer need indirection.
;
; DOES> is included (H_DOESGT) - added in a follow-up pass after
; the original 214-entry generation flagged it as missing, then
; moved again so it sits immediately after CREATE in the chain
; (H_CREATE -> H_DOESGT -> H_VARIABLE) rather than at the chain's
; newest end, since the two words are tightly coupled and read
; better adjacent. SETDOES (the runtime it compiles a call to)
; lives beside DODOES/DOESRT0; DOESGT is DOES>'s code label,
; since a literal ">" is not valid in a 6809 assembler label.
;
; H_ABORT and H_QUIT (formerly ABORTHDR/QUITHDR, renamed to match
; this file's H_ naming convention) are hand-built, not produced
; by the same generation pass as the other 217 entries - see the
; note where they're defined, at the end of this section, for why
; HEADER/CREATE couldn't be used directly for these two. They used
; to live physically apart from the rest of the dictionary, inside
; BASECODE rather than BASEDICT, despite being header data rather
; than code - moved here so every header lives in one place and
; the chain (H_QUIT -> H_ABORT -> (newest entry below) -> ... ->
; (oldest entry) -> 0) is visible in one block. BASELATEST remains
; H_QUIT, the chain's true overall head.
;
; H_ABORT's LINK field, a placeholder 0 since it was first built,
; is resolved here: it now points to this chain's newest entry.
;
;

```

```
; Names containing a literal double-quote (S", .", ABORT") have
; that character split into a standalone FCB $22 rather than
; escaped inside FCC - see emit_name's comment for why.
; =====
```

```
      ORG    BASEDICT      ; BASEDICT is $D83F

H_KEY:
      FCB    $03
      FCC    "KEY"
      FDB    0
      FDB    KEY

H_KEYQ:
      FCB    $04
      FCC    "KEY?"
      FDB    H_KEY
      FDB    KEYQ

H_EMIT:
      FCB    $04
      FCC    "EMIT"
      FDB    H_KEYQ
      FDB    EMIT

H_ACCEPT:
      FCB    $06
      FCC    "ACCEPT"
      FDB    H_EMIT
      FDB    ACCEPT

H_EXPECTW:
      FCB    $06
      FCC    "EXPECT"
      FDB    H_ACCEPT
      FDB    EXPECTW

H_QUERY:
      FCB    $05
      FCC    "QUERY"
      FDB    H_EXPECTW
      FDB    QUERY

H_TYPE:
      FCB    $04
      FCC    "TYPE"
      FDB    H_QUERY
      FDB    TYPE

H_CRW:
      FCB    $02
      FCC    "CR"
      FDB    H_TYPE
      FDB    CRW

H_SPACEW:
      FCB    $05
      FCC    "SPACE"
      FDB    H_CRW
      FDB    SPACEW

H_SPACESW:
      FCB    $06
      FCC    "SPACES"
      FDB    H_SPACEW
      FDB    SPACESW

H_DUP:
      FCB    $03
      FCC    "DUP"
      FDB    H_SPACESW
      FDB    DUP

H_DROP:
      FCB    $04
      FCC    "DROP"
      FDB    H_DUP
      FDB    DROP

H_SWAP:
      FCB    $04
      FCC    "SWAP"
      FDB    H_DROP
```

```

H_OVER:  FDB  SWAP
          FCB  $04
          FCC  "OVER"
          FDB  H_SWAP
          FDB  OVER
H_ROT:   FCB  $03
          FCC  "ROT"
          FDB  H_OVER
          FDB  ROT
H_QDUP:  FCB  $04
          FCC  "?DUP"
          FDB  H_ROT
          FDB  QDUP
H_DEPTH: FCB  $05
          FCC  "DEPTH"
          FDB  H_QDUP
          FDB  DEPTH
H_DDUP:  FCB  $04
          FCC  "2DUP"
          FDB  H_DEPTH
          FDB  DDUP
H_DDROP: FCB  $05
          FCC  "2DROP"
          FDB  H_DDUP
          FDB  DDROP
H_DSWAP: FCB  $05
          FCC  "2SWAP"
          FDB  H_DDROP
          FDB  DSWAP
H_DOVER: FCB  $05
          FCC  "2OVER"
          FDB  H_DSWAP
          FDB  DOVER
H_NIP:   FCB  $03
          FCC  "NIP"
          FDB  H_DOVER
          FDB  NIP
H_TUCK:  FCB  $04
          FCC  "TUCK"
          FDB  H_NIP
          FDB  TUCK
H_PICK:  FCB  $04
          FCC  "PICK"
          FDB  H_TUCK
          FDB  PICK
H_ROLL:  FCB  $04
          FCC  "ROLL"
          FDB  H_PICK
          FDB  ROLL
H_DR0T:  FCB  $04
          FCC  "2R0T"
          FDB  H_ROLL
          FDB  DR0T
H_T0R:   FCB  $02
          FCC  ">R"
          FDB  H_DR0T
          FDB  T0R

```

```

H_FROMR:
    FCB    $02
    FCC    "R>"
    FDB    H_TOR
    FDB    FROMR
H_RFETCH:
    FCB    $02
    FCC    "R@"
    FDB    H_FROMR
    FDB    RFETCH
H_TWOTOR:
    FCB    $03
    FCC    "2>R"
    FDB    H_RFETCH
    FDB    TWOTOR
H_TWOFROMR:
    FCB    $03
    FCC    "2R>"
    FDB    H_TWOTOR
    FDB    TWOFROMR
H_TWORFETCH:
    FCB    $03
    FCC    "2R@"
    FDB    H_TWOFROMR
    FDB    TWORFETCH
H_PLUS:
    FCB    $01
    FCC    "+"
    FDB    H_TWORFETCH
    FDB    PLUS
H_MINUS:
    FCB    $01
    FCC    "-"
    FDB    H_PLUS
    FDB    MINUS
H_STAR:
    FCB    $01
    FCC    "*"
    FDB    H_MINUS
    FDB    STAR
H_SLASH:
    FCB    $01
    FCC    "/"
    FDB    H_STAR
    FDB    SLASH
H_MODW:
    FCB    $03
    FCC    "MOD"
    FDB    H_SLASH
    FDB    MODW
H_SLASHMOD:
    FCB    $04
    FCC    "/MOD"
    FDB    H_MODW
    FDB    SLASHMOD
H_NEGATE:
    FCB    $06
    FCC    "NEGATE"
    FDB    H_SLASHMOD
    FDB    NEGATE
H_ABSW:
    FCB    $03
    FCC    "ABS"
    FDB    H_NEGATE
    FDB    ABSW
H_MIN:
    FCB    $03
    FCC    "MIN"
    FDB    H_ABSW
    FDB    MIN
H_MAX:

```

```

        FCB    $03
        FCC    "MAX"
        FDB    H_MIN
        FDB    MAX
H_ONEPLUS:
        FCB    $02
        FCC    "1+"
        FDB    H_MAX
        FDB    ONEPLUS
H_ONEMINUS:
        FCB    $02
        FCC    "1-"
        FDB    H_ONEPLUS
        FDB    ONEMINUS
H_TWOPLUS:
        FCB    $02
        FCC    "2+"
        FDB    H_ONEMINUS
        FDB    TWOPLUS
H_TWOSTAR:
        FCB    $02
        FCC    "2*"
        FDB    H_TWOPLUS
        FDB    TWOSTAR
H_TWOSLASH:
        FCB    $02
        FCC    "2/"
        FDB    H_TWOSTAR
        FDB    TWOSLASH
H_STARSLASH:
        FCB    $02
        FCC    "*/"
        FDB    H_TWOSLASH
        FDB    STARSLASH
H_STARSLASHMOD:
        FCB    $05
        FCC    "*/MOD"
        FDB    H_STARSLASH
        FDB    STARSLASHMOD
H_UMSTAR:
        FCB    $03
        FCC    "UM*"
        FDB    H_STARSLASHMOD
        FDB    UMSTAR
H_UMSLASHMOD:
        FCB    $06
        FCC    "UM/MOD"
        FDB    H_UMSTAR
        FDB    UMSLASHMOD
H_MSTAR:
        FCB    $02
        FCC    "M*"
        FDB    H_UMSLASHMOD
        FDB    MSTAR
H_FMSLASHMOD:
        FCB    $06
        FCC    "FM/MOD"
        FDB    H_MSTAR
        FDB    FMSLASHMOD
H_SMSLASHREM:
        FCB    $06
        FCC    "SM/REM"
        FDB    H_FMSLASHMOD
        FDB    SMSLASHREM
H_DPLUS:
        FCB    $02
        FCC    "D+"
        FDB    H_SMSLASHREM
        FDB    DPLUS
H_DMINUS:
        FCB    $02

```

```

        FCC    "D- "
        FDB    H_DPLUS
        FDB    DMINUS
H_DNEGATEW:
        FCB    $07
        FCC    "DNEGATE"
        FDB    H_DMINUS
        FDB    DNEGATEW
H_DABSW:
        FCB    $04
        FCC    "DABS"
        FDB    H_DNEGATEW
        FDB    DABSW
H_MPLUS:
        FCB    $02
        FCC    "M+"
        FDB    H_DABSW
        FDB    MPLUS
H_STOD:
        FCB    $03
        FCC    "S>D"
        FDB    H_MPLUS
        FDB    STOD
H_DTOS:
        FCB    $03
        FCC    "D>S"
        FDB    H_STOD
        FDB    DTOS
H_DMAXW:
        FCB    $04
        FCC    "DMAX"
        FDB    H_DTOS
        FDB    DMAXW
H_DMINW:
        FCB    $04
        FCC    "DMIN"
        FDB    H_DMAXW
        FDB    DMINW
H_ANDW:
        FCB    $03
        FCC    "AND"
        FDB    H_DMINW
        FDB    ANDW
H_ORW:
        FCB    $02
        FCC    "OR"
        FDB    H_ANDW
        FDB    ORW
H_XORW:
        FCB    $03
        FCC    "XOR"
        FDB    H_ORW
        FDB    XORW
H_INVERT:
        FCB    $06
        FCC    "INVERT"
        FDB    H_XORW
        FDB    INVERT
H_LSHIFT:
        FCB    $06
        FCC    "LSHIFT"
        FDB    H_INVERT
        FDB    LSHIFT
H_RSHIFT:
        FCB    $06
        FCC    "RSHIFT"
        FDB    H_LSHIFT
        FDB    RSHIFT
H_CELLSW:
        FCB    $05
        FCC    "CELLS"

```

```

        FDB H_RSHIFT
        FDB CELLSW
H_CELLPLUS:
        FCB $05
        FCC "CELL+"
        FDB H_CELLSW
        FDB CELLPLUS
H_CHARSW:
        FCB $05
        FCC "CHARS"
        FDB H_CELLPLUS
        FDB CHARSW
H_CHARPLUS:
        FCB $05
        FCC "CHAR+"
        FDB H_CHARSW
        FDB CHARPLUS
H_ALIGNW:
        FCB $05
        FCC "ALIGN"
        FDB H_CHARPLUS
        FDB ALIGNW
H_ALIGNEDW:
        FCB $07
        FCC "ALIGNED"
        FDB H_ALIGNW
        FDB ALIGNEDW
H_EQUALW:
        FCB $01
        FCC "="
        FDB H_ALIGNEDW
        FDB EQUALW
H_LESSW:
        FCB $01
        FCC "<"
        FDB H_EQUALW
        FDB LESSW
H_GREATERW:
        FCB $01
        FCC ">"
        FDB H_LESSW
        FDB GREATERW
H_ZEROEQ:
        FCB $02
        FCC "0="
        FDB H_GREATERW
        FDB ZEROEQ
H_ZEROLT:
        FCB $02
        FCC "0<"
        FDB H_ZEROEQ
        FDB ZEROLT
H_ULESSW:
        FCB $02
        FCC "U<"
        FDB H_ZEROLT
        FDB ULESSW
H_NOTEQUAL:
        FCB $02
        FCC "<>"
        FDB H_ULESSW
        FDB NOTEQUAL
H_ZERONE:
        FCB $03
        FCC "0<>"
        FDB H_NOTEQUAL
        FDB ZERONE
H_ZEROGT:
        FCB $02
        FCC "0>"
        FDB H_ZERONE

```

```

      FDB  ZEROGT
H_UGREATER:
      FCB  $02
      FCC  "U>"
      FDB  H_ZEROGT
      FDB  UGREATER
H_WITHINW:
      FCB  $06
      FCC  "WITHIN"
      FDB  H_UGREATER
      FDB  WITHINW
H_DEQUAL:
      FCB  $02
      FCC  "D="
      FDB  H_WITHINW
      FDB  DEQUAL
H_DLESSW:
      FCB  $02
      FCC  "D<"
      FDB  H_DEQUAL
      FDB  DLESSW
H_DULESSW:
      FCB  $03
      FCC  "DU<"
      FDB  H_DLESSW
      FDB  DULESSW
H_IF:
      FCB  $82
      FCC  "IF"
      FDB  H_DULESSW
      FDB  IF
H_THEN:
      FCB  $84
      FCC  "THEN"
      FDB  H_IF
      FDB  THEN
H_ELSE:
      FCB  $84
      FCC  "ELSE"
      FDB  H_THEN
      FDB  ELSE
H_BEGIN:
      FCB  $85
      FCC  "BEGIN"
      FDB  H_ELSE
      FDB  BEGIN
H_UNTIL:
      FCB  $85
      FCC  "UNTIL"
      FDB  H_BEGIN
      FDB  UNTIL
H_AGAIN:
      FCB  $85
      FCC  "AGAIN"
      FDB  H_UNTIL
      FDB  AGAIN
H_WHILE:
      FCB  $85
      FCC  "WHILE"
      FDB  H_AGAIN
      FDB  WHILE
H_REPEAT:
      FCB  $86
      FCC  "REPEAT"
      FDB  H_WHILE
      FDB  REPEAT
H_RECURSE:
      FCB  $87
      FCC  "RECURSE"
      FDB  H_REPEAT
      FDB  RECURSE

```

```

H_DO:
    FCB    $82
    FCC    "DO"
    FDB    H_RECURSE
    FDB    DO

H_QDO:
    FCB    $83
    FCC    "?DO"
    FDB    H_DO
    FDB    QDO

H_LOOP:
    FCB    $84
    FCC    "LOOP"
    FDB    H_QDO
    FDB    LOOP

H_PLUSLOOP:
    FCB    $85
    FCC    "+LOOP"
    FDB    H_LOOP
    FDB    PLUSLOOP

H_IWORD:
    FCB    $01
    FCC    "I"
    FDB    H_PLUSLOOP
    FDB    IWORD

H_JWORD:
    FCB    $01
    FCC    "J"
    FDB    H_IWORD
    FDB    JWORD

H_LEAVE:
    FCB    $05
    FCC    "LEAVE"
    FDB    H_JWORD
    FDB    LEAVE

H_UNLOOP:
    FCB    $06
    FCC    "UNLOOP"
    FDB    H_LEAVE
    FDB    UNLOOP

H_EXIT:
    FCB    $84
    FCC    "EXIT"
    FDB    H_UNLOOP
    FDB    EXIT

H_CASEW:
    FCB    $84
    FCC    "CASE"
    FDB    H_EXIT
    FDB    CASEW

H_OF:
    FCB    $82
    FCC    "OF"
    FDB    H_CASEW
    FDB    OF

H_ENDOF:
    FCB    $85
    FCC    "ENDOF"
    FDB    H_OF
    FDB    ENDOF

H_ENDCASE:
    FCB    $87
    FCC    "ENDCASE"
    FDB    H_ENDOF
    FDB    ENDCASE

H_COLON:
    FCB    $01
    FCC    ":"
    FDB    H_ENDCASE
    FDB    COLON

H_SEMI:

```

```

        FCB    $81
        FCC    ","
        FDB    H_COLON
        FDB    SEMI
H_CREATE:
        FCB    $06
        FCC    "CREATE"
        FDB    H_SEMI
        FDB    CREATE
H_DOESGT:
        FCB    $85          ; $80 IMMEDIATE | 5 (length of "DOES>")
        FCC    "DOES>"
        FDB    H_CREATE
        FDB    DOESGT
H_VARIABLE:
        FCB    $08
        FCC    "VARIABLE"
        FDB    H_DOESGT
        FDB    VARIABLE
H_CONSTANT:
        FCB    $08
        FCC    "CONSTANT"
        FDB    H_VARIABLE
        FDB    CONSTANT
H_VALUEW:
        FCB    $05
        FCC    "VALUE"
        FDB    H_CONSTANT
        FDB    VALUEW
H_TOW:
        FCB    $82
        FCC    "TO"
        FDB    H_VALUEW
        FDB    TOW
H_TWOVARIABLE:
        FCB    $09
        FCC    "2VARIABLE"
        FDB    H_TOW
        FDB    TWOVARIABLE
H_TWOCONSTANT:
        FCB    $09
        FCC    "2CONSTANT"
        FDB    H_TWOVARIABLE
        FDB    TWOCONSTANT
H_BUFFERCOLON:
        FCB    $07
        FCC    "BUFFER: "
        FDB    H_TWOCONSTANT
        FDB    BUFFERCOLON
H_DEFERW:
        FCB    $05
        FCC    "DEFER"
        FDB    H_BUFFERCOLON
        FDB    DEFERW
H_DEFERFETCH:
        FCB    $06
        FCC    "DEFER@"
        FDB    H_DEFERW
        FDB    DEFERFETCH
H_DEFERSTORE:
        FCB    $06
        FCC    "DEFER!"
        FDB    H_DEFERFETCH
        FDB    DEFERSTORE
H_ISW:
        FCB    $82
        FCC    "IS"
        FDB    H_DEFERSTORE
        FDB    ISW
H_ACTIONOF:
        FCB    $89

```

```

        FCC  "ACTION-OF"
        FDB  H_ISW
        FDB  ACTIONOF
H_MARKERW:
        FCB  $06
        FCC  "MARKER"
        FDB  H_ACTIONOF
        FDB  MARKERW
H_IMMEDIATE:
        FCB  $09
        FCC  "IMMEDIATE"
        FDB  H_MARKERW
        FDB  IMMEDIATE
H_STATEW:
        FCB  $05
        FCC  "STATE"
        FDB  H_IMMEDIATE
        FDB  STATEW
H_LBRACKET:
        FCB  $81
        FCC  "["
        FDB  H_STATEW
        FDB  LBRACKET
H_RBRACKET:
        FCB  $81
        FCC  "]"
        FDB  H_LBRACKET
        FDB  RBRACKET
H_TICK:
        FCB  $01
        FCC  "'"
        FDB  H_RBRACKET
        FDB  TICK
H_COMPILECOMMA:
        FCB  $08
        FCC  "COMPILE, "
        FDB  H_TICK
        FDB  COMPILECOMMA
H_LITERALW:
        FCB  $87
        FCC  "LITERAL"
        FDB  H_COMPILECOMMA
        FDB  LITERALW
H_BRACKTICK:
        FCB  $83
        FCC  "[']"
        FDB  H_LITERALW
        FDB  BRACKTICK
H_POSTPONEW:
        FCB  $88
        FCC  "POSTPONE"
        FDB  H_BRACKTICK
        FDB  POSTPONEW
H_TOBODY:
        FCB  $05
        FCC  ">BODY"
        FDB  H_POSTPONEW
        FDB  TOBODY
H_EXECUTE:
        FCB  $07
        FCC  "EXECUTE"
        FDB  H_TOBODY
        FDB  EXECUTE
H_SLITERALW:
        FCB  $88
        FCC  "SLITERAL"
        FDB  H_EXECUTE
        FDB  SLITERALW
H_ABORTQUOTE:
        FCB  $86
        FCC  "ABORT"

```

```

        FCB $22          ; ''' - split out of FCC, not escaped within it
        FDB H_SLITERALW
        FDB ABORTQUOTE
H_ATSIGN:
        FCB $01
        FCC "@"
        FDB H_ABORTQUOTE
        FDB ATSIGN
H_STOREW:
        FCB $01
        FCC "!"
        FDB H_ATSIGN
        FDB STOREW
H_CFETCH:
        FCB $02
        FCC "C@"
        FDB H_STOREW
        FDB CFETCH
H_CSTOREW:
        FCB $02
        FCC "C!"
        FDB H_CFETCH
        FDB CSTOREW
H_PLUSSTORE:
        FCB $02
        FCC "+!"
        FDB H_CSTOREW
        FDB PLUSSTORE
H_DFETCH:
        FCB $02
        FCC "2@"
        FDB H_PLUSSTORE
        FDB DFETCH
H_DSTORE:
        FCB $02
        FCC "2!"
        FDB H_DFETCH
        FDB DSTORE
H_COMMA:
        FCB $01
        FCC ", "
        FDB H_DSTORE
        FDB COMMA
H_CCOMMA:
        FCB $02
        FCC "C, "
        FDB H_COMMA
        FDB CCOMMA
H_ALLOT:
        FCB $05
        FCC "ALLOT"
        FDB H_CCOMMA
        FDB ALLOT
H_HEREW:
        FCB $04
        FCC "HERE"
        FDB H_ALLOT
        FDB HEREW
H_VCOMMA:
        FCB $02
        FCC "V, "
        FDB H_HEREW
        FDB VCOMMA
H_VCCOMMA:
        FCB $03
        FCC "VC, "
        FDB H_VCOMMA
        FDB VCCOMMA
H_VALLOT:
        FCB $06
        FCC "VALLOT"

```

```

        FDB H_VCCOMMA
        FDB VALLLOT
H_VHEREW:
        FCB $05
        FCC "VHERE"
        FDB H_VALLLOT
        FDB VHEREW
H_PADW:
        FCB $03
        FCC "PAD"
        FDB H_VHEREW
        FDB PADW
H_UNUSEDW:
        FCB $06
        FCC "UNUSED"
        FDB H_PADW
        FDB UNUSEDW
H_VUNUSEDW:
        FCB $07
        FCC "VUNUSED"
        FDB H_UNUSEDW
        FDB VUNUSEDW
H_MOVEW:
        FCB $04
        FCC "MOVE"
        FDB H_VUNUSEDW
        FDB MOVEW
H_FILLW:
        FCB $04
        FCC "FILL"
        FDB H_MOVEW
        FDB FILLW
H_ERASEW:
        FCB $05
        FCC "ERASE"
        FDB H_FILLW
        FDB ERASEW
H_CMOVEW:
        FCB $05
        FCC "CMOVE"
        FDB H_ERASEW
        FDB CMOVEW
H_CMOVEGT:
        FCB $06
        FCC "CMOVE>"
        FDB H_CMOVEW
        FDB CMOVEGT
H_COUNT:
        FCB $05
        FCC "COUNT"
        FDB H_CMOVEGT
        FDB COUNT
H_WORD:
        FCB $04
        FCC "WORD"
        FDB H_COUNT
        FDB WORD
H_CHARW:
        FCB $04
        FCC "CHAR"
        FDB H_WORD
        FDB CHARW
H_BRACKCHAR:
        FCB $86
        FCC "[CHAR]"
        FDB H_CHARW
        FDB BRACKCHAR
H_PARSEW:
        FCB $05
        FCC "PARSE"
        FDB H_BRACKCHAR

```

```

      FDB  PARSEW
H_PARSENAME:
      FCB  $0A
      FCC  "PARSE-NAME"
      FDB  H_PARSEW
      FDB  PARSENAME
H_SQUOTE:
      FCB  $82
      FCC  "S"
      FCB  $22      ; ''' - split out of FCC, not escaped within it
      FDB  H_PARSENAME
      FDB  SQUOTE
H_DOTQUOTE:
      FCB  $82
      FCC  "."
      FCB  $22      ; ''' - split out of FCC, not escaped within it
      FDB  H_SQUOTE
      FDB  DOTQUOTE
H_COMPAREW:
      FCB  $07
      FCC  "COMPARE"
      FDB  H_DOTQUOTE
      FDB  COMPAREW
H_SEARCHW:
      FCB  $06
      FCC  "SEARCH"
      FDB  H_COMPAREW
      FDB  SEARCHW
H_DASHTRAILING:
      FCB  $09
      FCC  "-TRAILING"
      FDB  H_SEARCHW
      FDB  DASHTRAILING
H_SLASHSTRING:
      FCB  $07
      FCC  "/STRING"
      FDB  H_DASHTRAILING
      FDB  SLASHSTRING
H_REPLACESW:
      FCB  $08
      FCC  "REPLACES"
      FDB  H_SLASHSTRING
      FDB  REPLACESW
H_SUBSTITUTEW:
      FCB  $0A
      FCC  "SUBSTITUTE"
      FDB  H_REPLACESW
      FDB  SUBSTITUTEW
H_SNAMEW:
      FCB  $05
      FCC  "SNAME"
      FDB  H_SUBSTITUTEW
      FDB  SNAMEW
H_UNESCAPEW:
      FCB  $08
      FCC  "UNESCAPE"
      FDB  H_SNAMEW
      FDB  UNESCAPEW
H_LTNUM:
      FCB  $02
      FCC  "<#"
      FDB  H_UNESCAPEW
      FDB  LTNUM
H_NUMSIGN:
      FCB  $01
      FCC  "#"
      FDB  H_LTNUM
      FDB  NUMSIGN
H_NUMSIGNS:
      FCB  $02
      FCC  "#S"

```

```

      FDB H_NUMSIGN
      FDB NUMSIGNS
H_NUMGT:
      FCB $02
      FCC "#>"
      FDB H_NUMSIGNS
      FDB NUMGT
H_HOLD:
      FCB $04
      FCC "HOLD"
      FDB H_NUMGT
      FDB HOLD
H_HOLDS:
      FCB $05
      FCC "HOLDS"
      FDB H_HOLD
      FDB HOLDS
H_SIGN:
      FCB $04
      FCC "SIGN"
      FDB H_HOLDS
      FDB SIGN
H_DOT:
      FCB $01
      FCC "."
      FDB H_SIGN
      FDB DOT
H_UDOT:
      FCB $02
      FCC "U."
      FDB H_DOT
      FDB UDOT
H_DOTR:
      FCB $02
      FCC ".R"
      FDB H_UDOT
      FDB DOTR
H_UDOTR:
      FCB $03
      FCC "U.R"
      FDB H_DOTR
      FDB UDOTR
H_QMARK:
      FCB $01
      FCC "?"
      FDB H_UDOTR
      FDB QMARK
H_DDOT:
      FCB $02
      FCC "D."
      FDB H_QMARK
      FDB DDOT
H_DDOTR:
      FCB $03
      FCC "D.R"
      FDB H_DDOT
      FDB DDOTR
H_BASEW:
      FCB $04
      FCC "BASE"
      FDB H_DDOTR
      FDB BASEW
H_DECIMAL:
      FCB $07
      FCC "DECIMAL"
      FDB H_BASEW
      FDB DECIMAL
H_HEXW:
      FCB $03
      FCC "HEX"
      FDB H_DECIMAL

```

```

      FDB    HEXW
H_BINARYW:
      FCB    $06
      FCC    "BINARY"
      FDB    H_HEXW
      FDB    BINARYW
H_CATCH:
      FCB    $05
      FCC    "CATCH"
      FDB    H_BINARYW
      FDB    CATCH
H_THROW:
      FCB    $05
      FCC    "THROW"
      FDB    H_CATCH
      FDB    THROW
H_LPAREN:
      FCB    $81
      FCC    "("
      FDB    H_THROW
      FDB    LPAREN
H_BACKSLASH:
      FCB    $81
      FCC    "\"
      FDB    H_LPAREN
      FDB    BACKSLASH
H_ENVQUERY:
      FCB    $0C
      FCC    "ENVIRONMENT?"
      FDB    H_BACKSLASH
      FDB    ENVQUERY
H_SOURCEW:
      FCB    $06
      FCC    "SOURCE"
      FDB    H_ENVQUERY
      FDB    SOURCEW
H_SOURCEID:
      FCB    $09
      FCC    "SOURCE-ID"
      FDB    H_SOURCEW
      FDB    SOURCEID
H_REFILLW:
      FCB    $06
      FCC    "REFILL"
      FDB    H_SOURCEID
      FDB    REFILLW
H_EVALUATEW:
      FCB    $08
      FCC    "EVALUATE"
      FDB    H_REFILLW
      FDB    EVALUATEW
H_TIBW:
      FCB    $03
      FCC    "TIB"
      FDB    H_EVALUATEW
      FDB    TIBW
H_NTIBW:
      FCB    $04
      FCC    "#TIB"
      FDB    H_TIBW
      FDB    NTIBW
H_TOINW:
      FCB    $03
      FCC    ">IN"
      FDB    H_NTIBW
      FDB    TOINW
H_SPANW:
      FCB    $04
      FCC    "SPAN"
      FDB    H_TOINW
      FDB    SPANW

```

```

H_BLW:      FCB    $02
            FCC    "BL"
            FDB    H_SPANW
            FDB    BLW

H_DOTS:     FCB    $02
            FCC    ".S"
            FDB    H_BLW
            FDB    DOTS

H_WORDSW:   FCB    $05
            FCC    "WORDS"
            FDB    H_DOTS
            FDB    WORDSW

H_DUMPW:    FCB    $04
            FCC    "DUMP"
            FDB    H_WORDSW
            FDB    DUMPW

H_TRUE:     FCB    $04
            FCC    "TRUE"
            FDB    H_DUMPW
            FDB    TRUEBODY

H_FALSE:    FCB    $05
            FCC    "FALSE"
            FDB    H_TRUE
            FDB    FALSEBODY

```

```

; H_ABORT and H_QUIT are hand-built, not produced by the same
; generation pass as the 217 entries above - ABORT and QUIT need
; to be findable at the prompt, but HEADER/CREATE couldn't be used
; directly for these two (see the source conversation this file
; was derived from for why). Moved here from their own separate
; section so every header in this ROM lives in one contiguous
; block, with the chain visible end to end: H_M2 -> H_2 -> H_M1 ->
; H_1 -> H_FIND -> H_QUIT -> H_ABORT -> H_FALSE (the newest of the
; 217 generated entries) -> ... -> H_KEY -> 0. H_M2 is now the true
; head (see BASELATEST below) - H_FIND and the four number words
; were chained on after H_QUIT in a later turn.

```

```

H_ABORT:    FCB    5
            FCC    "ABORT"
            FDB    H_FALSE      ; resolved - was placeholder 0, then H_DUMPW,
                                   ; then H_DOESGT, then H_DUMPW again once
                                   ; DOES> moved out of the chain's newest slot;
                                   ; now H_FALSE, the chain's newest entry

            FDB    ABORT

```

```

H_QUIT:     FCB    4
            FCC    "QUIT"
            FDB    H_ABORT
            FDB    QUIT

```

```

; FIND had no dictionary entry at all - confirmed by checking for
; any H_FIND label or FCC "FIND" string anywhere in this file
; before adding this. Its actual implementation (verified before
; exposing it) already matches the standard stack effect exactly:
; success pushes xt then 1 (immediate) or -1 (normal); failure
; reconstructs and pushes the original c-addr, then 0.

```

```

H_FIND:     FCB    4
            FCC    "FIND"
            FDB    H_QUIT
            FDB    FIND

```

```

H_1:        FCB    1
            FCC    "1"
            FDB    H_FIND

```

```

        FDB    ONEBODY

H_M1:   FCB    2
        FCC    "-1"
        FDB    H_1
        FDB    MONEBODY

H_2:    FCB    1
        FCC    "2"
        FDB    H_M1
        FDB    TWOBODY

H_M2:   FCB    2
        FCC    "-2"
        FDB    H_2
        FDB    MTWOBODY

BASELATEST EQU  H_M2    ; the ROM dictionary's true head - referenced by
                        ; COLD to initialize LATEST. Was H_QUIT before
                        ; FIND and the four number words (1/-1/2/-2)
                        ; were chained on after it; before that,
                        ; QUITHDR before the H_QUIT rename; before
                        ; that, undefined entirely - a real bug, not
                        ; a placeholder, found and fixed several turns
                        ; before this one.

; Verify no collision with base code.
; Value should match ORG BASECODE
BASEDICTEND EQU    *
BASEDICTSIZE EQU    BASEDICTEND-BASEDICT

```

## 9. Alphabetical Index

Every word implemented, alphabetically, with the page in the Glossary (Section 3) where its full entry appears. Page numbers are live Word fields — update fields (Ctrl+A, F9) if they show as blank or “?” after edits.

|                 |    |
|-----------------|----|
| -.....          | 14 |
| - TRAILING..... | 28 |
| , .....         | 25 |
| ; .....         | 22 |
| : .....         | 21 |
| ! .....         | 25 |
| ? .....         | 30 |
| ?DO.....        | 20 |
| ?DUP.....       | 12 |
| . .....         | 29 |
| . " .....       | 27 |
| . R.....        | 29 |
| . S.....        | 32 |
| ' .....         | 23 |
| ( .....         | 31 |
| [ .....         | 23 |
| [ ' ] .....     | 24 |
| [ CHAR] .....   | 27 |

|           |    |
|-----------|----|
| ]         | 23 |
| @         | 24 |
| *         | 14 |
| */        | 15 |
| */MOD     | 15 |
| /         | 14 |
| /MOD      | 14 |
| /STRING   | 28 |
| \         | 31 |
| #         | 29 |
| #>        | 29 |
| #S        | 29 |
| #TIB      | 31 |
| +         | 14 |
| +!        | 25 |
| +LOOP     | 20 |
| <         | 18 |
| <#        | 29 |
| <>        | 19 |
| =         | 18 |
| >         | 18 |
| >BODY     | 24 |
| >IN       | 32 |
| >R        | 13 |
| 0<        | 18 |
| 0<>       | 19 |
| 0=        | 18 |
| 0>        | 19 |
| 1-        | 15 |
| 1+        | 15 |
| 2!        | 25 |
| 2@        | 25 |
| 2*        | 15 |
| 2/        | 15 |
| 2+        | 15 |
| 2>R       | 14 |
| 2CONSTANT | 22 |
| 2DROP     | 13 |
| 2DUP      | 12 |
| 2OVER     | 13 |
| 2R@       | 14 |

|                |    |
|----------------|----|
| 2R>.....       | 14 |
| 2ROT.....      | 13 |
| 2SWAP.....     | 13 |
| 2VARIABLE..... | 22 |
| ABORT.....     | 12 |
| ABORT".....    | 24 |
| ABS.....       | 15 |
| ACCEPT.....    | 11 |
| ACTION-OF..... | 23 |
| AGAIN.....     | 20 |
| ALIGN.....     | 18 |
| ALIGNED.....   | 18 |
| ALLOT.....     | 25 |
| AND.....       | 17 |
| BASE.....      | 30 |
| BEGIN.....     | 20 |
| BINARY.....    | 30 |
| BL.....        | 32 |
| BUFFER:.....   | 22 |
| C,.....        | 25 |
| C!.....        | 25 |
| C@.....        | 25 |
| CASE.....      | 21 |
| CATCH.....     | 30 |
| CELL+.....     | 18 |
| CELLS.....     | 17 |
| CHAR.....      | 27 |
| CHAR+.....     | 18 |
| CHARS.....     | 18 |
| CMOVE.....     | 26 |
| CMOVE>.....    | 26 |
| COMPARE.....   | 28 |
| COMPILE,.....  | 24 |
| CONSTANT.....  | 22 |
| COUNT.....     | 27 |
| CR.....        | 11 |
| CREATE.....    | 22 |
| D-.....        | 16 |
| D.....         | 30 |
| D.R.....       | 30 |
| D+.....        | 16 |

|                   |    |
|-------------------|----|
| D<.....           | 19 |
| D=.....           | 19 |
| D>S.....          | 17 |
| DABS.....         | 16 |
| DECIMAL.....      | 30 |
| DEFER.....        | 22 |
| DEFER!.....       | 23 |
| DEFER@.....       | 23 |
| DEPTH.....        | 12 |
| DMAX.....         | 17 |
| DMIN.....         | 17 |
| DNEGATE.....      | 16 |
| DO.....           | 20 |
| DOES>.....        | 22 |
| DROP.....         | 12 |
| DU<.....          | 19 |
| DUMP.....         | 32 |
| DUP.....          | 12 |
| ELSE.....         | 19 |
| EMIT.....         | 11 |
| ENDCASE.....      | 21 |
| ENDOF.....        | 21 |
| ENVIRONMENT?..... | 31 |
| ERASE.....        | 26 |
| EVALUATE.....     | 31 |
| EXECUTE.....      | 24 |
| EXIT.....         | 21 |
| EXPECT.....       | 11 |
| FILL.....         | 26 |
| FM/MOD.....       | 16 |
| HERE.....         | 25 |
| HEX.....          | 30 |
| HOLD.....         | 29 |
| HOLDS.....        | 29 |
| I.....            | 20 |
| IF.....           | 19 |
| IMMEDIATE.....    | 23 |
| INVERT.....       | 17 |
| IS.....           | 23 |
| J.....            | 21 |
| KEY.....          | 11 |

|              |    |
|--------------|----|
| KEY?         | 11 |
| LEAVE        | 21 |
| LITERAL      | 24 |
| LOOP         | 20 |
| LSHIFT       | 17 |
| M*           | 16 |
| M+           | 16 |
| MARKER       | 23 |
| MAX          | 15 |
| MIN          | 15 |
| MOD          | 14 |
| MOVE         | 26 |
| NEGATE       | 15 |
| NIP          | 13 |
| OF           | 21 |
| OR           | 17 |
| OVER         | 12 |
| PAD          | 26 |
| PARSE        | 27 |
| PARSE - NAME | 27 |
| PICK         | 13 |
| POSTPONE     | 24 |
| QUERY        | 11 |
| QUIT         | 12 |
| R@           | 14 |
| R>           | 14 |
| RECURSE      | 20 |
| REFILL       | 31 |
| REPEAT       | 20 |
| REPLACES     | 28 |
| ROLL         | 13 |
| ROT          | 12 |
| RSHIFT       | 17 |
| S"           | 27 |
| S>D          | 16 |
| SEARCH       | 28 |
| SIGN         | 29 |
| SLITERAL     | 24 |
| SM/REM       | 16 |
| SNAME        | 28 |
| SOURCE       | 31 |

|                 |    |
|-----------------|----|
| SOURCE-ID.....  | 31 |
| SPACE.....      | 11 |
| SPACES.....     | 11 |
| SPAN.....       | 32 |
| STATE.....      | 23 |
| SUBSTITUTE..... | 28 |
| SWAP.....       | 12 |
| THEN.....       | 19 |
| THROW.....      | 30 |
| TIB.....        | 31 |
| TO.....         | 22 |
| TUCK.....       | 13 |
| TYPE.....       | 11 |
| U.....          | 29 |
| U.R.....        | 30 |
| U<.....         | 18 |
| U>.....         | 19 |
| UM*.....        | 16 |
| UM/MOD.....     | 16 |
| UNESCAPE.....   | 28 |
| UNLOOP.....     | 21 |
| UNTIL.....      | 20 |
| UNUSED.....     | 26 |
| V,.....         | 25 |
| VALLLOT.....    | 26 |
| VALUE.....      | 22 |
| VARIABLE.....   | 22 |
| VC,.....        | 26 |
| VHERE.....      | 26 |
| VUNUSED.....    | 26 |
| WHILE.....      | 20 |
| WITHIN.....     | 19 |
| WORD.....       | 27 |
| WORDS.....      | 32 |
| XOR.....        | 17 |